MYC – PROJECT GREEN – WHITE CITY - 22.0107.000

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REGISTERED PROFESSIONAL ENGINEER
101132PE
DIGITALLY SIGNED
OREGON
SEP. 13, 2022
BRENTON C. BAUMER
EXPIRES: 12/31/2025

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SECTION 260500 – COMMON WORK RESULTS FOR ELECTRICAL SYSTEMS

1. GENERAL

1.1. RELATED DOCUMENTS

1.1.1. The general provisions of the Contract, including General and Supplementary Conditions and General Requirements, apply to the work specified in this section and other 26-Series Sections.

1.2. OWNER FURNISHED EQUIPMENT AND MATERIAL

1.2.1. Owner Furnished. Where specified herein, indicated in equipment list, or shown on the drawings, the Owner will furnish certain equipment and materials for installation, connection, and placement into satisfactory operating service by the Contractor. The Contractor shall review the drawings, specifications, manufacturer's recommendations, schedules, building structures and progress of other trades for clarification of his responsibility in the handling and installation of this equipment and material. Manufacturer's drawings and installation instructions shall be furnished with the equipment for use by the Contractor. Where applicable, and not in contradiction with the contract drawings and specifications, the Contractor shall install and connect the equipment in accordance with the manufacturer's recommendations and instructions.

1.2.2. Contractor's Responsibility. Electrical equipment furnished by the Owner for installation by the Contractor will be turned over to the Contractor. The Contractor shall be responsible for promptly checking said equipment and protecting it against mechanical and electrical damage. The Contractor shall determine the condition of the equipment at the time it is turned over to him and furnish the Owner's Representative a written report, giving the "megger" readings and general condition of the equipment. Should any damage or fault be found with the equipment, it shall be brought to the attention of the Owner's Representative in writing. The Contractor shall see that no marking for construction designations, or other, shall be made on the finished painted face of any equipment or apparatus. All painting touch-up, re-painting, drying out, repairs or other corrective actions due to Contactor negligence after release and prior to final acceptance shall be corrected by the Contractor and at his expense.

1.2.3. Protection of Equipment and Material. The Contractor shall be responsible for properly protecting the equipment and materials at all times from the weather or any other adverse environmental conditions. Such responsibility shall include adequate shelter and heat to protect the insulation of all electrical apparatus subjected to humid conditions, indoors or outdoors.

1.3. MATERIAL AND EQUIPMENT BY THE CONTRACTOR

1.3.1. Contractor Furnished. Unless otherwise noted on the drawings, equipment list, or specifications, the Contractor shall furnish and install all materials, devices and apparatus necessary for the complete electrical system. All materials and equipment shall be of types and manufacturer specified wherever practical. Should materials or equipment so specified be unobtainable, the Contractor shall submit a written request for deviation including description and manufacturer's literature of the replacement item along with the reason for the substitution of other material or equipment. This specification establishes performance requirements and the quality of equipment acceptable for use and shall in no way be construed to limit procurement from other manufacturers.

1.3.2. Equal or Equivalent. The term "or equal" and similar terms as used on the drawings or specifications shall be interpreted to mean "equal or equivalent" in the opinion of the Engineers.

1.3.3. Manufacturer's Prints. Where the Contractor furnishes equipment other than standard construction items, he shall furnish manufacturer's prints and reproducibles of all such equipment to the Engineers.

1.3.4. UL Listing. All equipment and materials shall be new and conform to the requirements of this specification. All equipment and materials shall be listed by the Underwriters Laboratories, Inc., and shall bear their label whenever standards have been established and label service is regularly furnished. All equipment and materials shall be of the best grade of their respective kind for the purpose.

1.4. ELECTRICAL ALTERNATES

1.4.1. Definition: An alternate is an amount proposed by Bidders and stated on the Bid Form that will be added to or deducted from Base Bid amount if the Owner decides to accept a corresponding change in either scope of work or in products, materials, equipment, systems or installation methods described in Contract Documents.

1.4.2. Coordination: Coordinate related work and modify or adjust adjacent work as required to ensure that work affected by each accepted alternate is complete and fully integrated into the project.

1.4.3. Notification: Immediately following award of Contract, prepare and distribute to each party involved, notification of the status of each alternate. Indicate

whether alternates have been accepted, rejected or deferred for consideration at a later date. Include a complete description of negotiated modifications to alternates, if any.

1.4.4. Schedule: A "Schedule of Alternates" is included in this section. Specification sections referenced in the Schedule contain requirements for materials and methods necessary to achieve the work described under each alternate.

1.4.5. Include as part of each alternate, miscellaneous devices and similar items incidental to, or required for a complete installation, whether or not mentioned as part of the alternate.

1.5. SITE VISITATION

1.5.1. Bidders are required to visit the site prior to bidding the project and inform themselves of all conditions under which this contract must be completed. Failure to do so will in no way relieve the successful bidder from the necessity of furnishing any material or performing any work that may be required to complete the work in accordance with the intent and meaning of the drawings and specifications without additional cost to the Owner. A pre-bid walkthrough with the Architect/Engineer and Owners' representative will be scheduled and all bidders will be notified.

1.6. COORDINATION OF NEW ELECTRICAL WORK

1.6.1. General: It is recognized that the contract documents are diagrammatic in showing certain physical relationships which must be established within the electrical work. The interface with other work including utilities and mechanical work is the exclusive responsibility of the Contractor, by the data and criteria given on the drawings, specifications and implied intent of construction objectives.

1.6.2. Arrange electrical work in a neat, well-organized manner with conduit and similar services running parallel with primary lines of the building construction.

1.6.3. Locate operating and control equipment properly to provide easy access and arrange entire electrical work with adequate access for operation and maintenance. All switchboards, panels, transformers, disconnects, motor starters shall be installed in accordance with Code requirements for clearances. This shall include providing a dedicated space 3'-0" in front of all electrical equipment from floor to ceiling. No mechanical piping, ductwork or other appurtenances will be allowed in this space; it shall be solely dedicated to electrical access and maintenance. Transformers shall be installed to provide ready access for inspection and maintenance as per NEC Article 450.

1.6.4. Contractor shall verify all clearances and shall be responsible for proper installation. Any changes in equipment location due to code violations will be done by the contractor, with no addition cost to the owner or the Engineer.

1.6.5. The minimum headroom of working spaces about electrical equipment shall be 6-1/2 feet as per NEC Article 110.

1.6.6. Adequate space shall be provided around cable trays to permit access for installation of cables and maintenance per NEC Article 318. A minimum of 2'-6" clear space in front of each cable tray section containing cables shall be provided, from floor to ceiling, for maintenance and access for installing cables under otherwise indicated on the plans.

1.6.7. Advise other trades of openings required in their work for the subsequent move-in of large units of electrical work (equipment).

1.7. COOPERATION AND PROGRESS

1.7.1. The Contractor shall be informed about the work of all other trades engaged in the project and shall execute his work in such a manner as not to delay or interfere with the progress of other crafts. This contractor shall schedule his work so that no other craft is delayed by the execution of his work. Complete cooperation with all trades is expected.

1.7.2. Schedule and coordinate the work of this Division with the schedule of the general contractor so the work progresses expeditiously. Avoid unnecessary delays.

1.7.3. Examine fully the drawings and specifications for work required by other contractors and coordinate the installation of electrical work with that of the other crafts. Consult and cooperate with other crafts in determining space requirements and adequate clearances with respect to his equipment, other equipment, and the building. The Owner's representative reserves the right to determine space priority in the event of interference between piping, conduit, ducts, lighting and equipment of the various contractors.

1.7.4. Conflicts between the drawings and the specification shall be called to the attention of the Owner's representative. If clarification is not asked for prior to the opening of bids, it will be assumed that none is required and that the contractor is in agreement with the drawings and specifications as issued. If clarification is required after the Contract is awarded, such clarification will be made by the Owner's Representative and his decision will be final.

1.7.5. Coordinate the installation of all electrical system components with all other trades, including structural components and mechanical trades. Allocate space in all areas to allow for the installation of lighting, speakers, detectors, disconnects, panels, and electrical equipment above ceilings and in equipment spaces. Rerouting, resizing or relocation of electrical components, so all trades can install their systems in the space allotted shall be the responsibility of this Contractor. Any proposed changes from the systems layout on the drawings, shall be done in accordance with the design criteria

specified in the applicable codes and shall be subject to the review and acceptance of the Owner's Representative.

1.8. SUBMITTALS

1.8.1. General: Shop drawings, product data and O&M manuals shall be submitted for review in electronic, PDF format, via the project management software. On this project that software is ProCore.

1.8.1.1. Product Data: Submit product data for all materials proposed for use.

1.8.1.2. Warranties (Guarantees): Refer to individual sections for specific warranty requirements. If there is no warranty specified for a particular device or material, the warranty shall be for a period of a minimum of one year after project substantial completion. Submit statement of warranty with the O&M manuals.

1.8.1.3. Operation & Maintenance Manuals: Include wiring diagrams, maintenance and operating instruction, parts listings, and copies of other submittals indicated for inclusion.

1.8.1.4. Organize each operation & maintenance manual per specification section plus a warranty section.

1.9. CHANGE ORDERS

1.9.1. General: Refer to the General Conditions and Supplementary General Conditions for general requirements concerning change orders. For electrical work, the following is also required:

1.9.1.1. Change Order Format: Any changes in the work which result in additional cost or reduction in cost shall be covered by an appropriate addition to or deduction from the contract sum. In submitting a breakdown for an addition or reduction from the contract sum, for any price exceeding one hundred dollars, the Contractor shall submit a breakdown showing unit material price and unit labor price. This shall include total feet of conduit, fittings, wire, etc. with unit price.

1.10. SUBSTITUTIONS AND APPROVALS

1.10.1. It is the intent of this specification and the accompanying contract drawings to secure an electrical installation consistent with the level of quality and performance desired by the Owner.

1.10.2. Reference to any article, device, product, material, fixture, form or type of construction, either by name, make, or catalog number, shall be interpreted as establishing a standard of quality level and form of construction required.

1.10.3. Where only one manufacturer is named for a particular product, the Contractor shall base his proposal on that manufacturer. However, when only one manufacturer is suggested in the contract documents provisions for products

of other competing manufacturers will be considered under alternate proposals.

1.10.4. Where two or more manufacturers are named, the Contractor shall have the option of selecting one of those listed.

1.10.5. Where no manufacturer is listed, provide a standard product meeting the requirements of the drawings and specifications and manufactured by a firm regularly engaged in the manufacture of such products.

1.10.6. Manufacturers listed for luminaires on the fixture schedule on the drawings are the Basis of Design. Requests for substitutions will be accepted for consideration as equal substitutes for those specified, providing they meet or exceed the quality of those listed in the Luminaire Schedule. It shall be considered mandatory that all substitutions contemplated on any basis shall be on an equality basis as to material, type, design, appearance and function, and not on price. Requests for substitution are not to be construed as approval. Only once the Engineer has reviewed and approved the request will the substitution be allowed.

1.11. ERECTION OF EQUIPMENT

1.11.1. Comply with the IBC and any and all other applicable codes for seismic restraint requirements for supporting and bracing equipment, fixtures and raceways. Utilize anchoring materials and methods suitable for use in project seismic zone. Anchor switchboard and MCC's to floor slabs or equipment pads and brace to structural wall or ceiling to meet IBC Seismic Design Category D and ASCE 7-16 Ch. 13. Anchor transformers to floor slabs or equipment pads to meet IBC Seismic Design Category D and ASCE 7-16 Ch. 13. Tie light fixtures to grids and support each corner of fixtures to structure above with wires or safety chains. Brace suspended raceways to meet seismic requirements.

1.11.2. Motors. Normally motors shipped separately from driven equipment shall be set, coupled and aligned in the field by others and will require electrical connection by the Contractor.

1.11.3. Manufacturer's Installation Instructions. Where furnished or called for by the manufacturer's representative, equipment manufacturer's installation instructions shall be considered a part of this specification and fully complied with. Where Contractor damages the finishing coat of paint in existing or completed areas by welding, or other means, he shall refinish with matching paint. Manufacturer's installation instructions shall be strictly adhered to by

the contractor for installation of substations, motor control centers, contactors, panels, and similar equipment.

1.11.4. Mounting Heights. Individual safety switches and buttons and devices shall be installed on columns and walls at the following mounting heights, to center unless otherwise noted:

1.11.4.1.	Safety Switches	5’-0” (to bottom)
1.11.4.2.	Push Buttons	4’-0”
1.11.4.3.	Manual Switches	4’-6”
1.11.4.4.	Toggle Switches	4’-0”
1.11.4.5.	Receptacles	44” (process and warehouse areas)
1.11.4.6.	Receptacles	1’-6” (office area)

1.11.5. Mounting. Equipment and control devices shall be supported independent of conduit connections. Panels or cabinets shall not be mounted directly in contact with structure. Except where mounted on machine or pedestal, control devices and metal enclosures shall be bolted to galvanized steel channel (Unistrut) or galvanized steel plate supports which are attached to walls with expansion bolts or to columns or steel by welding.

1.12. PRODUCT, ELECTRICAL WORK

1.12.1. Compatibility: Provide products which are compatible with other products of the electrical work, and with other work requiring interface to the electrical work, including electrical connections and control devices. For exposed electrical work, coordinate colors and finishes with adjacent materials and colors.

1.12.2. Material and equipment shall be new and free from defects. All materials used in this work, which is included in Underwriters Label Service, shall be new and bear the National Board of Fire Underwriters Inspection Label. Materials not included in Underwriters Label Service shall be new and conform to NEMA or other applicable industry standards.

1.12.3. Install all materials and equipment in accordance with the manufacturer’s current published recommendations.

1.12.4. Certain materials and equipment are specified by manufacturer and model or catalog number. Such specified items are the basis of design and establish a degree of quality, performance and physical configuration. The contractor shall be responsible for the quality, performance and physical configuration of the work should he use products of a listed acceptable manufacturer in lieu of the specific product upon which the design is based.

1.13. ACCESS: CUTTING AND PATCHING

1.13.1. Access Units:

1.13.1.1. General: The work of this article is limited to the provisions for access through work of other trades for access to electrical work. This does not include internal access provisions (within the electrical equipment). In general, and where possible, furnish or furnish-and-mount required access units where required in the work of other trades prior to their work, so that cutting and patching for the subsequent installation of such access units will not be required. In occupied spaces, provide finished access units of the maximum concealment type, including locks where appropriate, and matching other access units provided in the same expanse of finish (which may be installed for nonelectrical access, if any).

1.13.1.2. Scope: The scope of access units to be furnished or provided as required to serve as electrical work. Includes those units indicated on the drawings or specified on the drawings and those additional units required for adequate access to electrical work and not shown or specified individually.

1.13.1.3. Access Doors: Standard welded-steel construction, 16 gage frames and 14 gage door panels, 175 degree concealed spring hinges, rust-inhibitive prime coat, flush cam lock (for screwdriver operation where keyed lock in not required), recessed to receive applied finish where applicable, 5-pin/disk tumbler lock where indicated.

1.13.1.4. Removable Access Plates: Where only hand access is sufficient, provide removable plate-type access units, or minimum size which will facilitate the required access. Provide units of the type, style, design, material and finish appropriate for the location and exposure in each instance. In exposed surfaces of occupied spaces provide round plate units, flush floor units and frameless low-profile wall units, primed for paint in painted surfaces and polished chrome or stainless steel finish in other surfaces.

1.14. Cutting and Patching:

1.14.1. General: The Contractor shall do all cutting of floors, walls, and partitions, and do all fitting necessary for the proper installation of his work. Before doing any cutting, the Electrical Contractor shall obtain permission from the Engineer and follow his instructions as to the method of cutting. All holes shall be cut as small as possible to admit conduit and with as much care as possible. This shall include cutting of all floors that are poured where sleeves or openings are omitted. Repairing around such holes, if required, shall be the responsibility of the Electrical Contractor.

1.14.2. Except as individually authorized by the Architect/Engineer, cutting and patching of electrical work to accommodate the installation of other work is not permitted.

1.14.3. Avoid cutting of walls and floors insofar as possible by setting sleeves, frames, etc., or requesting opening in advance.

1.14.4. If cutting of newly finished work is required by the Contractor, he is responsible for the repair of work to its original condition.

1.14.5. In buildings where cutting of holes through reinforced concrete is necessary for the installation of conduit, such holes shall be cut by rotary type drill. The use of pneumatic hammer type drills, impact type electric drills, and hand or manual hammer type drills will not be allowed. All holes shall be located so as not to affect structural sections such as ribs or beams. Holes shall be laid out in advance and drilling shall be done only after approval by the Engineer.

1.15. SCAFFOLDING, RIGGING AND HOISTING

1.15.1. Furnish necessary scaffolding, rigging equipment and hoisting as required to install the referenced electrical equipment. Measure all door openings prior to installation and compare with physical equipment dimensions of all major items of electrical equipment to be installed prior to installation of doors or door openings.

1.15.2. Secure all equipment. The Owner will not be responsible for replacing damaged, missing or stolen equipment. Remove equipment from premises when no longer needed. Provide necessary services to deliver, erect, place and install all equipment and apparatus furnished.

1.16. OPERATING INSTRUCTIONS

1.16.1. Upon completion of work and prior to final acceptance, equipment and system suppliers shall instruct the Owner on operation and maintenance of all equipment and systems. Use the operation and maintenance manual as a guide. Provide a total of one 8-hour day of instruction which may be split into 2 to 4 sessions, with 2 hour duration minimum, or as specified in the individual specification section. Advise the Owner of the time scheduled for the instruction period at least 7 days in advance. Use the instruction period to demonstrate the proper operation of all equipment and systems. The project foreman shall be present for the training in addition to factory personnel. Training shall be conducted by qualified individuals familiar with the specific installation on the specific project. Training conducted by sales staff and/or others unfamiliar with the specific project installation and equipment or system shall not be acceptable.

1.17. PROJECT CLOSEOUT

1.17.1. Upon written notice from Contractor stating that electrical work is complete, the Owner's representative will prepare a punchlist of items determined to be incomplete or otherwise not in compliance with the intent of the Contract Documents. This list shall be presented to the Contractor who shall schedule repairs or modifications promptly.

1.17.2. When required, subsequent visits to review completion of punchlist work will be made after receipt of written statement from Contractor indicating punchlist work is complete. Include copies of intermediate observation reports and final punchlists with individual items initialed by Contractor to attest that individual work items are completed.

1.18. DELIVERY AND STORAGE OF EQUIPMENT AND MATERIAL

1.18.1. Make provisions for delivery and safe storage of equipment and materials.

1.18.2. Make arrangements for the introduction into the building of equipment too large to pass through finished openings and spaces. Arrange with Owner for on-site storage space for equipment.

1.19. PROTECTION

1.19.1. Protect all new equipment and material from damage. Protect all existing Owner materials, equipment and appurtenances from damage. Repair or replace damaged items as necessary. Protect equipment, material and completed work until acceptance by Owner's Representative as Substantially Complete or until Owner assumes occupancy and responsibility for operation and maintenance of electrical system. The contractor shall assume responsibility for operation and maintenance of the electrical system throughout the course of construction.

1.19.2. Close open ends of work and stored equipment and materials with temporary covers or plugs to prevent entry of foreign objects, dirt, water or debris.

1.20. WARRANTY

1.20.1. Warranty all equipment, materials, workmanship and proper operation of equipment and apparatus for a period of one (1) years from date of project substantial completion. If other provisions of the Contract Documents specifically require more stringent warranty requirements, then the more stringent requirements shall apply.

1.20.2. Repair or replace at no cost to the Owner, work which is judged defective by the Owner's Representative during the warranty period.

1.21. CLEANUP

1.21.1. Maintain project site free and clear of rubbish and debris on a daily basis. At completion of the project, check and thoroughly clean all equipment, in and around all electrical equipment, panelboards. Clean exposed surfaces of conduit, equipment, hangers, cable trays and interior cabinets. All equipment and materials shall be thoroughly cleaned before a final inspection will be performed.

1.22. RECORD DRAWINGS

1.22.1. Maintain a complete set of project construction drawings indicating all changes in red pencil, and all additions and deletions in green pencil. Record Construction (As-Built) drawings shall be maintained and submitted. Include changes such as locations of equipment, system layout, and conduit and wire sizes.

1.22.2. During construction the contractor shall maintain, at the job site, a set of construction documents for the purpose of recording any revisions due to addenda, construction bulletins, change orders, field orders or other deviations from the construction drawings and specifications.

1.22.3. Drawings shall be kept up-to-date on a daily basis and shall indicate accurate routing of wiring and point-to-point wiring connection diagrams.

1.22.4. All drawings generated by the Contractor shall be 24x36 inch format. Include these and other drawings, such as schematics and wiring diagrams, as part of the project record drawings. Contractor shall submit reproducibles of all drawings made by him during the course of this work.

1.22.5. At the completion of the construction, submit these drawings to the Owner's Representative.

1.22.6. After the Owner, or his designee has reviewed and accepted the drawings as reasonably representing the constructed conditions, the contractor will transfer the marked-up information in pencil onto a copy of the original contract drawings, label them as "RECORD CONSTRUCTION DRAWINGS" and give them to the Owner.

1.23. PAINTING ELECTRICAL EQUIPMENT

1.23.1. General: The work of this article is defined to include general painting of electrical equipment at the project site. Coordinate the painting of equipment with the painting of other work of a similar nature and comply with indicated color matching requirements.

1.23.2. Painting Requirements:

1.23.3. If the factory finish on any equipment furnished by the Contractor is damaged in shipment or during construction of the equipment shall be refinished by the Contractor to the satisfaction of the Owner.

1.23.4. Do not paint over name plates on equipment, equipment grounding points or surfaces, sliding/rotating shaft surfaces, non-ferrous hardware/accessories/trim, and similar items where painting would normally be omitted.

1.24. ELECTRICAL WORK CLOSEOUT

1.24.1. General: Maintain a daily log of operational data on electrical equipment and systems through the closeout period; record hours of operation, assigned personnel, power consumption and similar information pertinent to the final acceptance of the installation; submit copy to Owner.

1.24.2. Coordination with Mechanical: Coordinate closeout operations with closeout of process equipment, mechanical system, elevators, and other power consuming equipment. Test run electrical equipment in coordination with test runs of mechanical systems. Clean and lubricate operational equipment. Instruct Owner's operating personnel thoroughly in the operation, sequencing, maintenance, safety and emergency provisions of the electrical systems. Turn over the operations to the Owner's personnel at the time of substantial completion. Until the time of final acceptance of the total work of the Contract, respond promptly with consultation and services to assist the Owner's personnel with operation of electrical systems.

1.24.3. Operating Instruction: Conduct a two-day walk-through instruction seminar for the Owner's personnel to be involved in the continued operation and maintenance of electrical equipment and systems. Explain the identification system, operational diagrams, emergency and safety, efficiency and similar features of the systems.

1.24.4. Turn-Over of Operation: At the time of substantial completion, turn over the prime responsibility for operation of the electrical equipment and systems to the Owner's operating personnel. However, until the time of final acceptance, provide one full-time operating engineer, who is completely familiar with the work, to consult with and continue training the Owner's personnel.

END OF SECTION

SECTION 260519 - LOW-VOLTAGE POWER CONDUCTORS AND CABLES

1. PART 1 - GENERAL

1.1. DESCRIPTION OF WORK

1.1.1. The requirements of this section apply to the conductors, cable and wire to be furnished and installed on this project.

1.1.2. The extent of electrical conductors, cable and electrical wiring is indicated by drawings and schedules, and by the requirements of this section.

1.1.3. The types of cable, wire and connectors required for the job include the following:

1.1.3.1. Copper conductors.

1.1.3.2. Tap Type Connectors

1.1.3.3. Split-Bolt Connectors

1.1.4. This specification covers the requirements for all power and lighting wire and cable to be used in the installation of the electrical systems for this project.

1.1.5. Wire and cable shall be furnished and installed by Contractor:

1.1.5.1. The Contractor shall submit for approval by the Owner any deviations anticipated or proposed in respect to the cable manufacturer, cable type, or specification contained herein.

1.1.5.2. It shall be the contractor's responsibility to make an accurate take – off of cable for each area on receipt of drawings and/or revisions so the cable can be purchased and delivered in a timely manner.

1.1.5.3. The take-offs shall include quantity, type, preferred cutting lengths (reel size) and latest acceptable delivery date.

1.2. QUALITY ASSURANCE

1.2.1. Installer: A firm with at least 5 years of successful installation experience on projects with electrical wiring installation work similar to that required for the project.

1.2.2. NEC Compliance: Comply with National Electrical Code (NFPA 70) Article 300 as applicable to construction and installation of electrical conductors, cable, wire and connectors.

1.2.3. UL Labels: Provide electrical conductors, cable, wire and connectors which have been listed and labeled by Underwriters Laboratories (UL) and comply with NEC Article 310.

1.2.4. NEMA/IPCEA Compliance: Comply with National Electrical Manufacturers Association/Insulated Power Cable Engineers Association Standards publications pertaining to materials, construction and testing of wire cable, where applicable.

1.2.5. ANSI/ASTM: Comply with applicable portions of ANSI/ASTM standards pertaining to construction of wire and cable.

1.3. SUBMITTALS

1.3.1. Manufacturer's Data; Conductors, Cable, Wire and Connectors: Submit manufacturer's data on electrical cable, wire and connectors.

1.4. PRODUCT DELIVERY, STORAGE AND HANDLING

1.4.1. Provide factory-wrapped waterproof flexible barrier material for covering wire and cable on wood reels, where applicable; and weather resistant fiberboard containers for factory-packaging of cable, wire and connectors, to protect against physical damage in transit. Do not install damaged cable, wire or connectors; remove from project site. Inspect reels for protruding nails of staples which could damage wire during installation.

1.4.2. Store cable, wire and connectors in factory-installed coverings and a clean, dry indoor space which provides protection against the weather.

2. PART 2 - PRODUCTS

2.1. GENERAL REQUIREMENTS

2.1.1. All wire and cable shall be Underwriters' Laboratories (UL) listed. In addition to other standard labeling, all wire and cable shall be marked UL on the outer surface indicating Underwriters' Laboratories, Inc. certification.

2.1.2. All insulated wire and cable shall conform to the minimum requirements of Insulated Power Cable Engineers Association (IPCEA) Standards for cable installed in wet locations, with the cable being subjected to all degrees of moisture conditions. Wire and cable shall comply with the applicable requirements of the National Electrical Code (NEC), in regards to cable construction and usage.

2.1.3. The conductors of insulated wire and cable shall be of copper (tinned where specified), having conductivity in accordance with the standardization rules of the Institute of Electrical and Electronic Engineers, Inc. (IEEE). The conductor and each strand shall be round and free of kinks and defects.

2.1.4. Grounding conductors, where insulated, shall be colored solid green. Conductors intended as a neutral shall be colored white (208Y/120V) or gray (480Y/277V).

2.1.5. All cable specified for use in tray shall be UL and NEC approved for tray installation.

2.1.6. Conductors for power, lighting, control and signal circuits shall have copper conductors of not less than 98% conductivity. Power, lighting, control wire and cable shall be insulated to 600 volts and shall be stranded. Solid or stranded conductors can be used for sizes up to NO. 8 AWG, but shall not be used in applications subject to vibration bending or isolated panel systems. Aluminum

conductors are not acceptable. Conductors larger than No. 8 AWG shall be Class B stranded.

2.1.7. Motor circuits and feeders shall utilize stranded conductors. Connectors shall be as specified in Paragraphs herein.

2.1.8. Minimum size conductors shall be No. 12 AWG for power and lighting.

2.2. CABLE, WIRE AND CONNECTORS

2.2.1. Manufacturers: Firms regularly engaged in manufacturer of electrical wire and cable products of types and ratings required, whose products have been in satisfactory use in similar service, for not less than five years. Provide products produced by one of the following (for each type of cable, wire and connectors):

2.2.1.1. Conductors, Cable and Wire:

2.2.1.1.1. Southwire Co.

2.2.1.1.2. Cerro.

2.2.1.1.3. Republic.

2.2.1.1.4. General Cable

2.2.1.2. Connectors:

2.2.1.2.1. Burndy Corporation

2.2.1.2.2. General Electric Company

2.2.1.2.3. Ideal Industries, Inc.

2.2.1.2.4. O. Z./Gedney Company

2.2.1.2.5. Thomas & Betts Company

2.3. 600 VOLT, MULTI-CONDUCTOR POWER CABLE

2.3.1. Tray Cable Type TC.- Non Shielded (Shall be used for non-VFD feeder cables to motors, distribution panels, and other equipment, where installed in cable tray.)

2.3.1.1. All 600 volt power cable larger than 4/0 AWG may be single or multi-conductor but shall be approved for tray installation.

2.3.1.2. “Type TC” Power Cable with Ground meets or exceeds all applicable ASTM specifications, UL standard 1581 for CT use, and UL standard 1277 for direct burial and sunlight resistance. Passes IEEE 1202 (70,000 Btu/hr) and ICEA T-29-520 (210,000 Btu/hr) vertical cable tray flame tests. The conductors are color coded per ICEA S-61-402, Method 4.

2.3.1.3. When specified, the current carrying capacity of the bare ground wires shall not be less than the current carrying capacity of one of the phase conductors. Ground wire sizes which meet NEC and UL Standards shall normally be acceptable.

2.3.1.4. Color code for 3-conductor cable shall be brown, orange and yellow. Color code for 4-conductor cable shall be brown, orange, yellow and white. Colored hash marks are acceptable.

2.3.1.5. Tray Cable Type TC. – Shielded (VFD feeder cables to motors where installed in cable trays).

2.3.1.5.1. Variable Frequency Drive branch circuit cables shall be AmerCable Type 37-108 Flexible TC-ER VFD 600Volt.

2.4. SINGLE CONDUCTOR LIGHTING AND POWER

2.4.1. 600 Volt Single Conductor Lighting and Power

2.4.1.1. For 600 volt power Lighting and Power, single conductor cable shall normally be Type THWN-2/THHN insulated wire rated to withstand conductor temperature of 75 degrees C. (wet) and 90 degrees C. (dry) without deterioration.

2.4.2. Type THW, THWN, or XHHW (75 deg. C): Use for branch circuits, panel and equipment feeders in wet and dry locations.

2.4.3. Type RHH, THHN, or XHHW (90 deg. C): Use for branch circuits, panel and equipment feeders in dry location only. Use where lighting branch circuit conductors are routed through fluorescent fixture channels. RHW cable shall be low-pass ERP insulation with Hypalon jacket. XHHW cable shall be cross-linked polyethylene insulation. Wire ampacity will be limited to the 75 deg. C wire rating.

2.4.4. Provide factory-fabricated cable of sizes, ratings, materials, and jacketing/sheathing as indicated for each type service. Where not indicated, provide proper selection as determined by installer to comply with installation requirements and NEC standards. Select from the following types:

2.5. METAL CLAD “”MC” CABLE

2.5.1. Metal clad type “MC” cable may be used <u>in the office area only</u> for branch lighting and power circuits 20A and smaller. MC cable may also be used for lighting fixture whips in lengths no greater than 6’0”.

2.5.2. Conductors shall be copper and #12 AWG with THWN/THHN insulation.

2.5.3. Terminate MC cables at boxes and panelboard cabinets with appropriate anti-short insulators and connectors listed for use with the cable.

2.5.4. MC cables that contain conductors for 0-10V dimming lighting circuits are acceptable.

2.6. CONDUCTOR CONNECTIONS AND SUPPORTS

2.6.1. Make necessary joints in circuit work at outlets with twist-on “wirenut” or push-in “Wago” style connectors rated for the number and size of conductors being spliced.

2.6.2. Leave at each fixture outlet a loop or end wire not less than 8” long for connections to fixtures.

2.6.3. All circuit wiring in cabinets, panels, pull boxes, and junction boxes shall be neatly trained and held with nylon cable ties; T&B “Ty-Rap”, Holub Industries “Quick-Wrap”, or approved equal.

2.6.4. For stranded wire size #8 and larger. Ilsco Type CSW Series, or “Burndy Hydent” or other approved hydraulic compression type, 600V insulated.

2.6.5. Wire #6 and larger shall be connected as specified herein and thoroughly taped with “Scotch #33 or approved equal electrical tape.

2.6.6. Lugs.

2.6.6.1. All lugs shall be furnished and installed by the Contractor.

2.6.6.2. Lugs for copper power wiring, Sizes No. 12 and No. 10 AWG, shall be T&B “Sta-Kon” uninsulated ring type lugs. Lugs for copper power wiring form No. 10 AWG to size No. 1.0 AWG shall be Ilsco 1-hole type CSWS Series. Size No. 2/0 AWG and larger lugs shall be 2-hole Type CSWD Series (except where 1-hole is required to match motor lead lugs). Size above No. 1/0 are to be applied using hydraulic crimp tool.

2.6.6.3. Where motor leads are furnished without lugs, Ilsco CT series 2 way-connectors (splicing sleeves) shall be used. Splice sleeves may be desirable where limited space for termination exists.

2.6.6.4. The proper lugs shall be furnished by Contractor in all Owner furnished equipment. All other lugs shall be furnished by Contractor. If any mechanical type lugs are furnished with Owner furnished equipment, the Contractor shall replace them with proper compression type lugs where practical.

3. PART 3 - EXECUTION

3.1. GENERAL

3.1.1. Separation of Usage. Motor and control wiring shall be routed in conduits, or other raceways as shown on the drawings. Power and control wiring shall not be routed in a common raceway except where shown on drawings.

3.1.2. Cables shall be installed in accordance with installation instructions and recommendations of the cable manufacturer. Maximum pulling tension shall not be exceeded. Cable pulling rig must positively limit this tension or a suitable dynamometer which continuously measures the pulling tension shall be used.

3.1.3. Cable termination shall be made by qualified personnel, using proper tools, equipment, methods and precautions recommended by the cable manufacturer to ensure that the cable will be installed without damage.

3.1.4. Cables at tray elevation/directional changes shall be bundled and tie-wrapped together. Cables shall be tie-wrapped to the first rung at both elevation/direction

changes. Cables shall be tie-wrapped to tray rungs at 6'-0" intervals. Tie-wraps shall be Panduit Corp. PAN-TY cable ties, type PRT-EH-100 (Releasable) weather-resistant polypropylene (black).

3.1.5. Immediately after each cable installation, the cable must be permanently tagged at both ends and at location where cable exits tray into conduits. Tagging shall indicate the assigned equipment I.D. No. as shown on the construction drawings.

3.1.6. A sufficient number of pulling blocks shall be used so that undue stress is not exerted on the cable during installation. Manufacturer recommendations regarding type of roller, rollers spacing, sheeve sizes and location, pulling equipment manufacturers, sheeve radii, direction of pull etc., shall be followed for routing through cable tray system.

3.1.7. After single or multiple pulls cables shall be laid neat and straight into the tray and secured.

3.1.8. AC and DC circuits wires shall not be run in the same conduit.

3.2. INSTALLATION

3.2.1. Conductors in conduit shall be continuous with splices only at outlet or junction boxes.

3.2.2. Circuit feeders and branch circuits are shown diagrammatically on drawings. Suggested deviations must be approved by Engineer.

3.2.3. Wire shall not be installed in the conduit system until the building is enclosed and masonry work is completed.

3.2.4. Conduit shall be swabbed free of moisture and debris prior to pulling in wire.

3.2.5. Cable for all feeders shall be continuous from origin to termination. Splices will not be allowed.

3.2.6. Three phase wiring. Install separate neutral conductors for each branch circuit. Do not share neutrals. Derating factors for additional number of conductors in conduits shall be applied according to NEC.

3.2.7. Splices in branch circuit wires shall be made only in accessible junction boxes.

3.2.8. Install all circuit conductors in a single raceway at one time, ensuring conductors do not cross one another while being pulled into raceway. Conductors must be pulled within allowable tension and minimum bending radius.

3.2.9. Conductor Support. Provide conductor vertical supports as required by codes and recommended by cable manufacturer. Where required, provide cable supports in vertical conduits similar to OZ to C.M.T. and provide lower end of conduit with OZ type KVF ventilators.

3.2.10. Make conductor length for parallel feeders identical.

3.2.11. No wire shall be pulled until the conduit system is complete.

3.2.12. Pulling. Where mechanical assistance is used for pulling conductors, contractor may use pulling compounds appropriate for the type of wire being pulled and shall not damage or degrade the wire insulation. Interior of all raceways shall be free from grease, filings or foreign matter before conductors are pulled in.

3.2.13. Use wire-pulling lubricant for pulling No. 4 AWG and larger wire. No grease, oil, or lubricant other than powdered soapstone, or approved pulling compound shall be used to facilitate the pulling of wires. Wire-pulling lubricant shall be equal to Ideal “Yellow” or Dow-Corning compound #4.

3.2.14. Splice shall utilize preinsulated connectors, 3M Company "Scotchlock" or Ideal Industries, Inc. "Super Nut", for splices and taps in conductors No. 10 AWG and smaller. All other twist-on connectors must be approved by Architect prior to installation. Utilize this type of connector for approval.

3.2.15. Pressure indent type connectors such as “Wago” style may be used in lieu of twist-on connectors at contractors option.

3.2.16. Tape all bolted splices and joints with vinyl plastic tape manufactured by 3M Company, St. Paul, Minnesota. Use sufficient tape to secure insulation strength equal to that of conductors joined.

3.2.17. Conductor Termination. Provide all power and control conductors, that terminate on equipment or terminal strips, with solderless lugs or fork and flanged tongue terminals. Provide T. and B. "Sta-Kon" tongue terminal. (This type conductor termination is not required when equipment is provided with solderless connectors.)

3.2.18. Conductor Identification. All secondary branch circuit conductors (No. 10 AWG and smaller) throughout project shall be provided with color coded insulation as follows:

3.2.18.1.1.1.	208Y/120V	Phase	480Y/277V
3.2.18.1.1.2.	Black	A	Brown
3.2.18.1.1.3.	Red	B	Orange
3.2.18.1.1.4.	Blue	C	Yellow
3.2.18.1.1.5.	White	Neutral	Gray
3.2.18.1.1.6.	Green	Ground	Green
3.2.18.1.1.7.	Isolated Grd	Green with Yellow Trace	

3.2.19. On insulated grounding or grounded conductors No. 6 AWG and larger ¾” colored marking tape manufactured for the purpose may be applied at 6” intervals the entire length exposed at junction, pull or outlet boxes, (except LB’s and LBD’s) and at 5’-0” intervals the entire length where direct buried, or run underground in PVC conduit or duct. This applies only to black colored insulation.

3.2.20. Size wire for 120-volt branch circuits for 3% maximum voltage drop. Size feeder circuits for 2% maximum voltage drop. Combined voltage drop of feeder and branch circuits shall not exceed 5% maximum.

3.2.21. Branch circuits whose length from panel to first outlet exceeds 75 feet for 120 volt circuits shall be #10 or larger, as required to comply with the NEC.

3.2.22. Conduit and conductor sizes shown on the drawings are based on use of copper conductors with THW insulation and rigid steel conduit.

3.2.23. Lighting Conductors Splices in lighting conductors shall be made with splicing caps with metal inserts only, such as 3M Company's "Scotchlock" spring connectors. The splices shall be firmly and neatly taped to prevent entry of moisture.

3.2.24. Wire identification numbers shall be placed on each end of the conductor involved by using self-laminating marker tags, T&B Company E-Z Code Type WSL or equal.

3.3. MISCELLANEOUS WIRING MATERIAL

3.3.1. Miscellaneous Connecting and Splicing Devices. Miscellaneous products, such as heat shrink tubing, electrical insulation, plug cabs, splices and kits, tapes, terminal blocks, and terminations, shall be approved for the specific application.

3.3.2. All feeder connectors shall be made with joint compounds similar to NO-A-LOX or as required for the specific type metal joint to be prepared.

3.3.3. Cable ties, clamps, and identification shall be nylon, self-locking.

3.3.4. Fire seal fittings, certified by NEL-PIA, shall be used for installations where sleeves penetrate fire-rated walls, floors, etc., as required by NEC. Size fire seals for the application.

3.3.5. Where cable tray penetrates fire walls, provide fire stopping per the Raceways specification section.

3.4. INSULATING

3.4.1. All voids, sharp corners and bolt projections shall be made smooth before applying insulation. All loose strands of wire shall be removed before insulating. Duxseal will not be permitted.

3.4.2. Provide heat shrink boots on all motor lead terminations. Heat-shrinkable tubing, boots, and end caps made of thermally stabilized cross-linked polyolefin, heat-shrinkable insulators shall be used.

3.4.2.1. Provide T&B Shrink-Kon motor lead kits or approved equal.

3.4.3. All insulating, splicing and termination materials shall be furnished by the Contractor.

3.5. FIELD QUALITY CONTROL

3.5.1. Prior to energization, check cable and wire for continuity of circuitry, and for short circuits, correct malfunction when detected.

3.5.2. Subsequent to wire and cable hook-ups, energize circuitry and demonstrate functioning in accordance with requirements.

END OF SECTION

SECTION 260526 – GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS

1. PART 1 - GENERAL

1.1. DESCRIPTION OF WORK

1.1.1. Extent of grounding is indicated by drawings and schedules and is further delineated as follows:

1.1.1.1. Provide Building and equipment ground as indicated on the plans.

1.2. SYSTEM DESCRIPTION

1.2.1. Ground each separately-derived system neutral as required to building steel and to plant underground ground grid system.

1.2.2. Provide main electric feeder ground buss and grounding conductor at point of power entrance.

1.2.3. Bond together system neutrals, service equipment enclosures, exposed non-current carrying metal parts of electrical equipment, metal raceway systems, grounding conductor in raceways and cables and receptacle ground connections.

1.2.4. As Required by the NEC. In general, fixtures, outlets, the enclosing cases, mounting frames, etc., of all switches, circuit breakers, control panels, motors and any other raceways shall be effectively and permanently grounded with a separate copper grounding conductor of cross-section as required by the National Electrical Code and drawings. It shall be of capacity sufficient to ensure continuity and continued effectiveness of the ground connections to carry fault currents. Ground conductors must be as short and straight as possible, protected from mechanical injury and if practicable without splice or joint. The grounding conductor shall be run from a ground established at the source of supply to the equipment to be grounded. Ground wires from below grade shall be protected by galvanized conduit and the conductor shall be brazed to conduit sleeve on each end. All grounding conductors shall be copper.

1.2.5. Types of grounding and ground-fault protection in this section include the following Grounding:

1.2.5.1. Underground metal structures.

1.2.5.2. Metal building frames.

1.2.5.3. Grounding electrodes.

1.2.5.4. Grounding plates.

1.2.5.5. Grounding rods.

1.2.5.6. Separately derived systems.

1.2.5.7. Service equipment.

1.2.5.8. Enclosures.

1.2.5.9. Systems.

1.2.5.10. Equipment.

1.3. QUALITY ASSURANCE

1.3.1. To be considered for approval, the manufacturer shall furnish products listed by Underwriters Laboratories Incorporated (UL), or testing firm acceptable to the authority having jurisdiction as suitable for application specified.

1.3.2. The ground-fault protection device manufacturing facility shall be Registered by Underwriters Laboratories Inc. to the International Organization for Standardization ISO 9000 Series Standards for quality.

1.3.3. Installer: Qualified with at least three years of successful installation experience on projects with electrical grounding and ground-fault protection work similar to that required for project.

1.4. REFERENCE STANDARDS

1.4.1. NEMA AB 1 Current Edition - (National Electrical Manufacturers Association) Molded Case Circuit Breakers and Molded Case Switches

1.4.2. UL 489 - (Underwriters Laboratories Inc.) Molded Case Circuit Breakers and Circuit Breaker Enclosures.

1.4.3. UL 943 - Standard for Ground Fault Circuit Interrupters

1.4.4. IEC 157-1 - (International Electrotechnical Commission) Low-Voltage Switchgear and Control Gear Part 1: Circuit Breakers)

1.4.5. Federal Specification W-C-375B/GEN - Circuit Breakers, Molded Case; Branch Circuit and Service.

1.4.6. NFPA 70 - (National Fire Protection Association) NEC (National Electrical Code).

1.4.7. ANSI/IEEE Compliance: Comply with C114.1 (IEEE Std. 142) and IEEE Stds Nos. 241 and 242 pertaining to grounding and ground-fault protection of power systems.

1.4.8. ANSI/UL Compliance: Comply with requirements of ANSI/UL and UL standards pertaining to grounding and ground-fault protection equipment and devices. Provide products which have been UL-listed and labeled.

1.5. SUBMITTALS

1.5.1. Manufacturers Installation Instructions for Electronic Trip Circuit Breakers:

1.1 Includes instructions for identification and receiving inspection, mounting, functions and adjustments, trouble shooting, accessories, wiring diagrams and dimensional drawings.

2. PART 2 - PRODUCTS

2.1. GROUNDING

2.1.1. Materials and Components:

2.1.1.1. General: Except as otherwise indicated, provide each electrical grounding system indicated, with assembly of materials including, but not necessarily limited to, cables/wires, connectors, terminals (solderless lugs), grounding rods/electrodes and plate electrodes, bonding jumper braid, and other items and accessories needed for complete installation. Where more than one type meets indicated requirements, selection is Installer's option. Where materials or components are not otherwise indicated, comply with NEC, NEMA and established industry standards for applications indicated.

2.1.2. Electrical Bonding Jumpers:

2.1.2.1. Bonding Jumper Braid: Copper braided tape, constructed of 30-gage bare copper wires and properly sized for indicated applications.

2.1.2.2. Flexible Jumper Strap: Flexible flat conductor, 480 strands of 30-gage bare copper wire; 3/4" wide, 9-1/2" long; 48,250 CM. Protect braid with copper bolt hole ends with holes sized for 3/8" dia. bolts.

2.1.2.3. Electrical Grounding Conductors: Unless otherwise indicated, provide electrical grounding conductors for grounding connections matching power supply wiring materials and sized according to NEC.

2.1.2.4. Equipment Grounding Conductors: Provide an equipment ground conductor in every conduit sized per table 250-122 of the NEC. Provide a grounding conductor the same size as the receptacle feeder conductor for all receptacles or as noted on the plans a #10 ground may be required for receptacles in some areas.

2.1.2.5. Bonding Plates, Connectors, Terminals and Clamps: Provide electrical bonding plates, connectors, terminals and clamps as recommended by bonding plate, connector, terminal and clamp manufacturers for indicated applications.

2.1.2.6. Electrical Grounding Connection Accessories: Provide electrical insulating tape, heat-shrinkable insulating tubing, solder, soldering flux, bonding straps, as recommended by accessories manufacturers for type services indicated.

2.1.3. Mechanical Connectors

2.1.3.1. Material: Bronze.

2.1.3.2. Grounding Clamps: Thomas and Betts No 390 BU for connection to water pipes (add number as required for pipe size). Thomas and Betts No 3846 for connection to grounding electrodes.

2.1.3.3. Wire

2.1.3.3.1. Material: Stranded copper. Bare for below grade grounding conductors and insulated for above grade, exposed or equipment grounding conductors routed in raceway with feeders and branch circuit conductors. Insulation type to be identical to the circuit conductors serving the load.

2.1.3.3.2. Insulation shall be colored green for #10 AWG and smaller. Use color coded marking tape for #8 AWG and larger.

2.1.3.3.3. Grounding Electrode Conductor: Size to meet NFPA 70 requirements.

2.2. GROUND-FAULT PROTECTION DEVICES

2.2.1. Circuit Interrupters/Duplex Receptacles: Provide specification grade, duplex receptacles, with integral ground fault circuit interrupters, UL-rated Class A, Group 1; select with feed-thru features capable of protecting connected downstream receptacles on single circuit; grounding types, 20-amperes ratings, 120-volts, 60 hertz. Provide with solid-state ground fault sensing and signaling, with 5 milliamperes ground fault trip level, plus or minus 1 milliampere; equip with 15-amperes NEMA receptacle configurations 5-15R.

2.2.2. Manufacturer: Subject to compliance with requirements, provide ground fault circuit interrupters of one of the following:

2.2.2.1. Eaton/Arrow Hart/Cooper

2.2.2.2. Leviton Mfg.

2.2.2.3. Hubbell/Bryant Electric Company

2.2.2.4. Pass and Seymour Inc.

3. PART 3 - EXECUTION

3.1. INSTALLATION OF GROUNDING AND GROUND-FAULT PROTECTION SYSTEMS

3.1.1. Install electrical grounding systems and ground-fault protection devices as indicated, in accordance with manufacturer's written instructions and with recognized industry practices to ensure grounding and ground-fault protection devices comply with requirements. Comply with requirements of NEC, NESC, and NEMA standards for installation of grounding and ground-fault protection systems and devices.

3.1.2. Bond together each metallic raceway, pipe, duct and other metal object.

3.1.3. Install grounding bushings, grounding studs, and grounding jumpers at distribution centers, pull boxes, motor control centers and panelboards.

3.1.4. All power transformers shall have the neutral ground connection made to the ground conductors at the transformer.

3.1.5. Provide green insulated copper bonding jumpers, size correlated with over-current device protecting the branch circuit or feeder. Attach bonding jumpers to grounding circuit or feeder. Attach bonding jumpers to grounding bushings on conduit, lugs on boxes, and other enclosures.

3.1.6. All ground connection hardware, bolts, and nuts shall be high strength, high conductivity copper alloy.

3.1.7. Underground connections or connections exposed to weather in the grounding system shall be made with an exothermic welding (CADWELD) process.

3.1.8. Connect the ground terminal on each outlet receptacle to the branch circuit grounding conductor and to the metallic raceway system with bonding jumpers.

3.1.9. Connect the grounding stud on each luminaire to the branch circuit grounding conductor and to the metallic raceway system with bonding jumpers.

3.1.10. Ground connections to equipment and ground buses shall be by ground lugs or clamps. Connections to enclosures not provided with ground buses or ground terminals shall be by clamp type lugs added under permanent assembly bolts or under new bolts drilled and added through enclosures or by grounding locknuts or bushings. Ground cable connections to anchor bolts; against gaskets, paint, or varnish; or on bolts holding removable access covers will not be permitted.

3.1.11. Ground the secondary electrical systems in accordance with the latest edition of the National Electrical Code.

3.1.12. To establish the grounding electrical system, connect grounding electrode conductors to each of the following:

3.1.12.1. Underground grounding grid system.

3.1.12.2. Metal water piping systems

3.1.13. Grounding conductors on equipment shall be formed to the contour or the equipment and firmly supported.

3.1.14. Power Conductors Supplying Equipment. A copper grounding conductor must be run inside the conduit or raceway, enclosing the power conductors supplying the equipment, or in case of a multi-conductor power cable, must be located within the sheath.

3.1.15. Connect at Source. Ground conductors in power cable or ground wire in conduits shall always be connected directly to station ground at the source end, and to motor frame or equipment enclosure and/or equipment ground bar.

3.1.16. Motors. The contractor shall drill and tap holes in motor frames and install a servit post within the terminal box enclosure for connection of ground wires furnished in power cables or ground wires in conduits. The terminal box screws shall not be used for this purpose.

3.1.17. System Neutral. The equipment ground conductor in all circuits shall be connected to the frame and ground lug in the load centers and not the neutral bus. Equipment ground connections to a system neutral are not permitted.

3.1.18. Fuses. In all cases of grounded circuits, fuses must be omitted from the grounded neutral conductor throughout the entire installation.

3.1.19. Equipment Frames. Frames of transformers, motor control centers, and other electrical apparatus will be connected to the grounding system. Neutrals of service transformers shall be connected to the grounding system as shown by typical details.

3.1.20. Metallic Raceways. All metallic conduits and wiring channels must be connected at each end to the grounding conductor with a good electrical contact.

3.1.21. Identification. The grounding conductor shall be stranded and covered with a green jacket or bare wire.

3.1.22. In all power wiring cases the white wire should be used for the current-carrying neutral only and never as a grounding conductor, or other purpose.

3.1.23. Below Grade. All ground connections below grade and to column steel shall be exothermic welded (cadwelded). All grounding conductors below grade may be green, PVC, insulated. This may be specified where necessary for cathodic type protection, where ground wires follow underground pipe, in highly corrosive areas, or in the interconnection of various building loops.

3.1.24. Test. Ground tests shall be performed on all new major equipment grounds, loops and rods. Results shall be reported to the Owner's Representative on proper forms for approval and acceptance. See "QUALITY CONTROL and "TEST REPORTS" sections of this spec for acceptable method of tests and checks.

3.1.25. Install ground and bonding conductors with sufficient slack to avoid breakage due to settlement and movement of conductors at attached points.

3.1.26. All grounding conductors above grade, and not subject to mechanical damage, shall be #4/0 AWG stranded copper with THHN/THWN green insulation or as shown on the drawings. All grounding conductors below grade shall be #4/0 AWG, solid, bare tinned copper. All grounding conductor connections from existing below grade ground grid to above grade equipment, and subject to mechanical damage, shall be # 4/0 AWG, solid, bare, tinned copper in ½" conduit.

3.1.27. Provide one time hit 1 lb. hammer test on all cad weld connections.

3.1.28. All below grade grounding connections shall be made by the cadweld process. All materials used (molds, welding metal, tools etc.) shall be as manufactured by "cadweld" or approved manufacturer and shall be installed per the manufacturer's recommendations and procedures.

3.1.29. All above grade grounding connections shall be made using two hole crimp type compression connections. Mechanical connections, fittings or connections that depend solely on solder shall not be used.

3.1.30. After all grounding connections have been made, the contractor shall test the ground resistance and submit a test report to the owner's representative confirming that the maximum allowable ground resistance of 5 OHMS is achieved. This shall be done before and after tie-ins to the existing ground ring to ensure that the integrity of the existing ground ring is maintained.

3.2. FIELD QUALITY CONTROL

3.2.1. Inspect grounding and bonding system conductors and connections for tightness and proper installation.

3.2.2. Use suitable test instrument to measure resistance to ground of system. Perform testing in accordance with test instrument manufacturer's recommendations using the fall-of-potential method.

3.3. TEST REPORTS

3.3.1. Grounding systems shall be tested and test reports shall be submitted prior to energizing the system.

3.3.2. The system shall, when tested, yield a maximum 5 OHM ground resistance. If this value is not obtained, modifications shall be made to obtain this value without additional cost to the project.

3.3.3. Make three point megger ground resistance tests after grounding system is complete. Make additional tests as required after modifications to verify ground resistance as specified.

3.3.4. Before energizing any circuits, make ground continuity tests on all circuits to ensure proper grounding.

3.3.5. Results of all tests shall be recorded and submitted to the Owner's Representative's representative before the system is energized.

3.3.6. Upon completion of installation of electrical grounding system, test ground resistance with ground resistance tester. Where tests show resistance-to- ground is over 5 ohms, take appropriate action to reduce resistance to 5 ohms or less by driving additional ground rods and/or by chemically treating soil encircling ground rods with sodium chloride, calcium chloride, copper sulphate, or magnesium. Then retest to demonstrate compliance.

3.3.7. Perform resistance-to-ground tests of electrical grounding system one year following time of substantial completion and submit written report indicating results.

Where resistance is greater than 3 ohms, chemically treat soil to reduce resistance to 3 ohms or less. Then retest to demonstrate compliance.

END OF SECTION

SECTION 260533 – RACEWAYS AND BOXES FOR ELECTRICAL SYSTEMS

1. PART 1 - GENERAL

1.1. DESCRIPTION OF WORK

1.1.1. Work specified in this section. Installation of all conduits, raceways and fittings furnished and installed on this project shall comply with State and Local Codes which have jurisdiction and Article 300 of the National Electric Code.

1.1.2. The extent of electrical raceway work is indicated by drawings and schedules, and by the requirements of this section.

1.1.3. The types of electrical raceways required for the project include the following:

1.1.1.1. Rigid metal conduit (GRC, RGS).

1.1.1.2. Rigid nonmetallic conduit (polyvinyl chloride) PVC.

1.1.1.3. Electrical Metallic Tubing (EMT).

1.1.1.4. Liquid-tight flexible metal conduit.

1.1.1.5. Flexible metal conduit

1.1.1.6. Cable tray (where shown on the drawings only).

1.1.4. All materials shall be Underwriters Laboratories (U.L.) labeled.

1.2. QUALITY ASSURANCE

1.2.1. Manufacturers: Firms regularly engaged in the manufacture of electrical raceways of the types and capacities required, whose products have been in satisfactory use in similar service for not less than 5 years.

1.2.2. Installer: A firm or individual with not less than 3 years of successful experience in installation of raceways similar to those required for this project.

1.2.3. NEMA Compliance: Comply with applicable portions of NEMA standards pertaining to raceways.

1.2.4. NEC Compliance: All raceway installations shall comply with Article 300 of the National Electric Code and State and Local Codes in effect and in accordance with authorities having jurisdiction.

1.3. SUBMITTALS

1.3.1. Product Data: Submit manufacturer's data including specifications, installation instructions and general recommendations, for each type of raceway required.

1.3.2. Samples: Where specified on drawings, submit 6" length of exposed type surface raceways with finish required. Prepare samples on material to be used for the work.

1.4. PRODUCT DELIVERY, STORAGE AND HANDLING

1.4.1. Provide thread protectors on the exposed threads of threaded rigid conduit.

1.4.2. Handle conduit and tubing carefully to prevent end damage and to avoid scoring the finish.

1.4.3. Store pipe and tubing inside and protect from weather. When necessary to store outdoors, elevate well above grade and enclose with durable, waterproof wrapping. PVC conduit shall not be stored in areas exposed to sunlight.

2. PART 2 - PRODUCTS

2.1. MATERIALS AND COMPONENTS

2.1.1. General:

2.1.1.1. For each electrical raceway system indicated, provide a complete assembly of conduit, tubing or duct with fittings, including, but not necessarily limited to connectors, nipples, couplings, expansion fittings, other components and accessories as needed to form a complete system of the type indicated.

2.1.1.2. The materials used in the fabrication of the conduit system shall be products of a manufacturer regularly engaged in the manufacturing of the specified material. Where one manufacturer is specified for a particular material, the material of other manufacturers will be considered equal provided the material meets all requirements of this specification.

2.1.1.3. Circular Raceways. Circular raceways shall be 3/4" minimum diameter for branch circuits for exterior work and interior work where NEC conductor fill requirements are met or, unless shown or specified otherwise.

2.1.1.4. Support all conduits and cable trays per IBC and any and all other applicable codes seismic requirements for the project area. Utilize only anchoring methods allowed by applicable codes.

2.2. METAL CONDUIT, TUBING AND FITTINGS

2.2.1. General:

2.2.1.1. Provide metal conduit, and fittings of the type, grade, and size indicated for each service. Where type and grade are not indicated, provide proper selection determined by Installer to fulfill the wiring requirements, and complying with the National Electrical Code for electrical raceways.

2.3. RACEWAYS

2.3.1. Rigid Conduit: In all areas where other raceway systems are not specifically allowed, and specifically in all process, production, and grow rooms and in all above ground outdoor areas, furnish and install rigid galvanized steel (RGS) (also known as Galvanized Rigid Conduit – GRC) conduits, galvanized inside

and outside by hot-dip or electro-galvanized steel (RGS) conduits, galvanized inside and outside by hot-dip or electro-galvanizing process. Conduits shall conform to Federal Spec WW-C-581, W-F-408 and ANSI C80.1 requirements. All couplings and fittings shall be threaded type. Terminate conduits in raintight hubs where connecting to raintight enclosures. Minimum size of 3/4".

2.3.2. Liquid Tight Flexible Metal Conduit (3/4" min. size, 36" max. length): Conduit shall be one continuous length interlocked steel construction with PVC jacket i.e. spirally wound steel strip, with water and oil tight neoprene jacket resistant to sunlight. Use for final connections to vibrating equipment such as motors located in wet areas and at all exterior locations.

2.3.3. Electrical Metallic Tubing (EMT): For use in interior, dry, above grade areas not subject to physical damage sizes 4" and less in the office area, Lab, Maintenance, Electrical, MCC and Utility rooms only. Conceal inside walls and above ceilings in finished areas. EMT may be exposed in electrical and utility rooms. EMT is not allowed in any Process, Production or Grow Room areas. Minimum size of 3/4". EMT fittings shall be steel.

2.3.4. Flexible Metal Conduit (3/4" min. size, 36" max. length): Conduit shall be one continuous length interlocked steel construction i.e. spirally wound steel strip. Use for final connections to vibrating equipment such as motors and transformers located in dry interior areas.

2.3.5. Rigid Non-metallic Conduit (PVC): Allowed for use for exterior underground branch circuits and feeders and interior under slab panel and transformer feeders only. Do not use for interior above or below slab branch circuits or motor or process equipment feeders. Provide Schedule 80 PVC (or GRC) for exterior underground or interior under slab switchboard, panelboard, MCC or transformer feeders not encased in concrete. Schedule 40 PVC may be used for underground exterior branch circuits only.

2.4. CONDUIT FITTINGS

2.4.1. Rigid Conduit Fittings shall be threaded, galvanized malleable iron or heavy steel, water and concrete tight. Provide grounding type fittings where required for connectors at cabinets, boxes and gutters. Use T & B or O.Z. Type “B”, steel, insulated bushings on all RSC entering panel, cabinets, outlet boxes, etc. Use O.Z. “BLG” where required for grounding. Connections exposed to the weather shall be made with T&B “Bullet” weathertight hubs.

2.4.2. Liquid Flexible Conduit Fittings shall be galvanized malleable iron or steel with watertight gaskets, "O" ring and retainer.

2.4.3. Condulet Fittings. For exposed conduit, conduit fittings shall be condulet type for sharp turns and tees. Minimum radius shall be as necessary to accept cable to be installed.

2.4.4. EMT fittings shall be steel. Pot metal or die cast fittings are not acceptable. Both steel set screw and steel compression fittings are acceptable.

2.4.5. Expansion fittings across structural joints shall be of a design to compensate for expansion and contraction and shall be OZ Type “DO” and shall be sealed to prevent entrance of water or moisture. Expansion fittings shall be approved for grounding duty.

2.5. CABLE TRAY

2.5.1. Manufacturer: Subject to compliance with these specifications, cable tray, systems to be installed shall be as manufactured by Cooper B-Line, Inc. or engineer approved equal.

2.5.2. General:

2.5.2.1. Except as otherwise indicated, provide metal cable trays, of types, classes and sizes indicated; with splice plates, bolts, nuts and washers for connecting units. Construct units with rounded edges and smooth surfaces; in compliance with applicable standards; and with the following additional construction features.

2.5.2.2. Materials and Finish: Material and finish specifications for each tray type are as follows:

2.5.2.2.1. Aluminum: Straight section and fitting side rails and rungs shall be extruded from Aluminum Association Alloy 6063. All fabricated parts shall be made from Aluminum Association Alloy 5052.

2.5.3. Type of tray system:

2.5.3.1. Ladder type trays shall consist of two longitudinal members (side rails) with transverse members (rungs) welded to the side rails. Rungs shall be spaced 9 inches on center. Spacing in radiused fittings shall be 9 inches and measured at the center of the tray's width. Rungs shall have a minimum cable bearing surface of 7/8" with radiused edges. No portion of the rungs shall protrude below the bottom plane of the side rails. Each rung must be capable of supporting the cable load, with a safety factor of 1.5, and a 200 lb. concentrated load when tested in accordance with NEMA VE 1, section 5.4.

2.5.3.2. Tray sizes shall have minimum 6” depth or as noted on the drawing.

2.5.3.3. Straight tray sections shall have side rails fabricated as I-Beams. All straight sections shall be supplied in standard 20 foot lengths, except where shorter lengths are permitted to facilitate tray assembly lengths as shown on drawings.

2.5.3.4. Tray widths shall be as shown on drawings.

2.5.3.5. All fittings must have a three inch tangent and a minimum radius of 24 inches.

2.5.3.6. Splice plates shall be the bolted type made as indicated below for each tray type. The resistance of fixed splice connections between an adjacent

section of tray shall not exceed .00033 ohm. Splice plate construction shall be such that a splice may be located anywhere within a continuously supported span without diminishing rated loading capacity of the cable tray.

2.5.3.6.1. Aluminum Tray - Splice plates shall be made of 6063-T6 aluminum, using four square neck carriage bolts and serrated flange locknuts. Hardware shall be zinc plated in accordance with ASTM B633, SC1. If aluminum cable tray is to be used outdoors, then hardware shall be Type 316 stainless steel.

2.5.3.7. Splice plates shall be furnished with straight sections and fittings.

2.5.3.8. Cable Tray Supports: Shall be placed so that the support spans do not exceed the span indicated for the load. Supports shall be constructed from 12 gauge steel formed shape channel members 15/8" x 15/8" with necessary hardware such as Trapeze Support Kits (9G-55XX-22SH) as manufactured by Cooper B-Line, Inc. or engineer approved equal. Cable trays installed adjacent to walls shall be supported on wall mounted brackets such as B409 as manufactured by Cooper B-Line, Inc. or engineer-approved equal.

2.5.3.9. Trapeze hangers and center hung supports shall be supported by 1/2" (minimum) diameter rods.

2.5.3.10. Barrier Strips: Shall be placed as specified on drawings and be fastened into the tray with self drilling screws.

2.5.3.11. Accessories: Special accessories shall be furnished as required to protect, support, and install a cable tray system. Accessories shall consist of, but are not limited to; section splice plates, expansion plates, blind-end plates, specially-designed ladder drop-outs, barriers, etc.

2.5.4. Loading capacities:

2.5.4.1. Cable tray shall be capable of carrying a uniformly distributed load of 100 lbs./ft. on a 20 ft. support span with a safety factor of 1.5 when supported as a simple span and tested per NEMA VE 1, section 5.2. In addition to the uniformly distributed load the cable tray shall support 200 lbs. concentrated load at mid-point of span. Load and safety factors specified are applicable to both the side rails and rung capacities. Cable tray shall be made to manufacturing tolerances as specified by NEMA.

2.5.5. Peak Flanged Covers:

2.5.5.1. Outdoor trays shall be furnished with peak flanged covers.

2.6. OUTLET AND JUNCTION BOXES

2.6.1.1. Outlets installed outdoors or in process locations: Use Crouse-Hinds Type FS or FD box with No. WLRS-1 or No. WLRD-1 cover plate for receptacles, and NO. DS185 cover plates for switches.

2.6.1.2. Contractor Furnished. The Contractor shall furnish and install all electrical boxes required for the proper installation of the electrical systems.

2.6.1.3. Surface Mounted. Fixture, outlet and switch boxes that are surface mounted in plant, shop and operating areas shall be threaded, cast alloy iron or malleable iron. Iron type shall have a cadmium/zinc electroplate, or galvanized finish with appropriate lacquer Boxes shall be of the approved type for the outlets, switches, and fixtures served and shall be made of the material and finish compatible with the conduit system and location. Boxes shall be provided with the appropriate cast covers and gaskets.

2.6.1.4. Stamped steel boxes for devices with appropriate plaster rings are allowed where concealed in walls withing finished areas (office area). Stamped steel pull and junction boxes are allowed above ceilings in finished areas.

2.7. SPLICE AND TAP BOXES

2.7.1.1. Splice and tap boxes for power circuits shall be used only where designated on the drawings and shall be of the type and sizes indicated. Otherwise all power wiring shall be continuous, splice and tap free, between equipment. On lighting and convenience receptacle circuitry, wiring may be spliced and boxes shall be provided for concealed or surface mounting as previously specified or may be JIC oil-tight minimum size as specified in the National Electric Code.

2.8. PULL AND JUNCTION BOXES

2.8.1.1. Construction, sizes and installation of pull and junction boxes shall comply with NEC, Article 314.

2.8.1.2. Pull and junction boxes not specifically described otherwise in NEC, Article 370, shall be fabricated of minimum 12 gauge galvanized steel with screw covers, brass screws and hardware with enamel finish.

2.8.1.3. Pull and junction boxes for outdoor installations and process areas shall be NEMA 4 rated.

2.8.1.4. Pull boxes for interior or outdoor exposed power wiring shall be provided where required to facilitate the installation of the wiring. Pull boxes shall not be located in finished rooms and shall be accessible for maintenance use. For conduit sizes 3/4 and 1 inch, conduit fittings of the "C", "LB", "TB" and similar types may be used for "Pulling In". Unless designated

otherwise, all pull boxes shall be the straight-through type and changes in direction shall not be made in the box. Where pull boxes are required, each power circuit shall have a separate galvanized steel threaded hub type box with a gasketed cover installed.

2.8.1.5. For exterior and Process Areas, exposed work pull boxes shall be of NEMA 4 construction and shall be threaded hub type with gasketed cover.

2.8.1.6. Outdoor or indoor junction and terminal boxes are required to be NEMA 4 and shall be compression molded fiberglass reinforced polyester material such as Crouse-Hinds Co. "Krydon" Type NAB (bonding jumpers required for ground continuity). All and Process Area and areas below 12'-0" AFF in the plant area are subject to hose down and shall be classified as NEMA 4.

3. PART 3 - EXECUTION

3.1. GENERAL

3.1.1. In general all lighting circuits and 120 volt power circuits shall be routed in conduit unless otherwise specifically allowed per the specification for specific areas. All circuits shall be routed to underside of respective floor level, surface mounted to lighting fixtures then surface mounted at underside of floor, down columns etc.

3.1.2. All feeders to process equipment motors shall be 3/C type "TC" (w/ground) cables routed via cable tray where shown on the drawings. All power feeders dropouts from cable trays to motor shall be installed in conduit.

3.2. RACEWAY SUPPORT

3.2.1. Support and brace raceways to comply IBC Seismic Design Category D and ASCE 7-16 Ch. 13 requirements.

3.2.2. Provide raceway straps, hangers and support devices of approved type to rigidly support raceways. This includes: Structural steel members, suspension rods, conduit clamps, concrete inserts, expansion shields, and beam clips. All devices shall have galvanized finish or corrosion resistant finish. Hanger straps and supports shall be installed as follows:

3.2.2.1. Where one or more runs of conduit is routed on surfaces of structure, use conduit clamps mounted on UNISTRUT channel to maintain 1" (minimum) clearance between conduit and structure.

3.2.2.2. Where single run of conduit is suspended, use split ring conduit clamp and rigid suspension support.

3.2.2.3. Where multiple parallel runs of conduit are suspended from overhead, use split ring conduit clamps, uniformly spaced and supported on trapeze hangers fabricated of UNISTRUT channels, suspended by steel drop rod.

3.2.2.4. Maximum hanger and support spacing shall be in accordance with NEC, Sections 344.30. Regardless of listed spacing provide additional hangers or supports at not more than 2'-0" from each change of direction and at each side of any box or fitting.

3.2.2.5. Hanger straps and supports shall be anchored to poured concrete structure with malleable iron or steel concrete inserts attached to concrete forms.

3.2.2.6. Hanger straps or supports anchored to structural steel shall be beam clamps and/or steel channels as required by the structural system.

3.2.2.7. Hanger straps or supports anchored to metal deck shall be attached with drill-in expansion anchors.

3.3. GENERAL

3.3.1. All wiring outside of cable trays shall be run in rigid steel conduit except as specifically noted on the drawings or specified otherwise herein. Conduit size where not specified on the drawings, shall be of size required by code for number of wires indicated or required but in no case shall it be less than ¾". Conduit sizes indicated or required by code may be increased if desired to facilitate the pulling of cables.

3.3.2. Conduits shall be sized in accordance with NEC except where larger conduits are called for on drawings. Conductor and conduit sizes shown on drawings are based on the use of copper conductors.

3.4. INSTALLATION

3.4.1. Conduit for convenience outlets, wall mounted fixtures and other wall outlets shall be routed overhead and dropped to the outlet. Conduits shall not be installed within concrete floor slabs. Conduits shall be supported independently from mechanical system components.

3.4.2. Conduit shall be routed in lines parallel to building construction.

3.4.3. Conduit shall be installed so that it is at least 6" from steam or hot fluid lines and a minimum of 3 inches from other piping systems.

3.4.4. Conduits shall not be embedded in structural concrete work.

3.4.5. Rigid Galvanized. Rigid Galvanized conduit, elbows, couplings, and conduit bodies shall be used where conduit is run exposed.

3.4.6. Minimum Size. Minimum conduit and tubing size shall be ¾ inch. Where vertical drops are made to equipment in open space, the vertical conduit shall be rigidly supported from the equipment and a minimum of 7 feet above floor using not less than 1 inch conduit regardless if a smaller size is indicated on the drawings.

3.4.7. Minimum size allowable for liquid tight flexible metal conduit shall be 3/4 inch.

3.4.8. Joints in conduit must be drawn up tight. Contractor shall square cut all conduit ends, ream and file to remove burrs before installation and shall properly clean and cap all empty conduits. Rigid metal conduit shall be reamed after threads are cut. Joints shall be cut square and shall but solidly into couplings. Running threads will not be permitted. Immediately after cutting or threading, coat bare threads with cold galvanizing spray.

3.4.9. Exposed runs in any accessible areas shall be installed parallel to walls and ceilings. Conduits shall not be run horizontally in masonry walls or partitions.

3.4.10. Install conduit system in a continuous fashion both mechanically and electrically, from outlet to outlet and to equipment cabinets, junction boxes or pull boxes. Conduit shall enter and be secured to cabinets and boxes in such a manner that all parts of the system will have electrical continuity.

3.4.11. Raceways which penetrate roof: Raceways through roof seals shall be installed in curbs provided for mechanical equipment. Penetration of roof seal may also utilize suitable pitch pockets of a type approved by the Engineer. Submit method to be used with roofing and W. Expansion Joints.

3.4.12. Provide expansion fittings for embedded raceways or rigidly supported raceways exceeding 100 feet or cross building expansion joints, O-Z Type AX with bonding jumper for rigid conduit, and O-Z Type TX with bonding jumper for electric metallic tubing. Where embedded raceways cross expansion joints, provide combination deflection/expansion fittings, O-Z Type AXDX, installed in accordance with manufacturer's recommendations.

3.4.13. Use care in locating raceways in close proximity to heating duct, steam and hot water lines. Where such crossings are unavoidable, clear covering of line by at least 8". All raceways runs shown are diagrammatic. Determine exact locations in the field except where dimensions are specified on the drawings.

3.5. CONDUIT FITTINGS AND RACEWAY INSTALLATION

3.5.1. All conduit fittings must be compatible with the conduit material and be supplied with covers and neoprene gaskets. Condulets are to be of Crouse-Hinds or Appleton manufacture. Condulets are to be of the threaded type. Use obround series for up to 1 inch conduit and a mogul series for conduit 1-1/4 inch and larger.

3.5.2. In protected areas (MCC Room substation, instrument control room and offices) conduits terminating in boxes, enclosures and apparatus cases must be fitted with double lock-nuts and T & B, Appleton or O.Z. Electrical Manufacturing Co.'s Type "BL" insulated bushings. The accessible locknut or bushing shall be of the grounding type properly bonded. In all other areas conduits shall terminate in threaded hubs. T & B 370 Series bullet hubs shall be provided where required.

3.5.3. Where conduits pass from areas of high humidity to cooler ambients, they must be sealed with Crouse-Hinds Type EYS or EZS or Appleton sealing condulets with suitable drain fittings on the humid side.

3.5.4. Vertical loops in conduit must be avoided, if possible, but when they are necessary must be drained with ECD drain installed in proper condulet.

3.5.5. Long vertical run of conduit shall have cable supports installed as required by the National Electrical Code. This shall be done by using O.Z. Electrical Manufacturing Co.'s Type "M" cable support or an approved equal, at the bottom of a "C" condulet or pull box.

3.5.6. Conduits shall terminate at motors, transformers, and other vibrating equipment with Liquid-tight PVC-jacketed, flexible conduit and liquid-tight connectors as manufactured by “Sealtite”, Carlon “PV-Flex” or Pittsburgh Standard “Robroy Flex”. Provide sufficient length of conduit to avoid transmission vibration. At motors have cast iron conduit boxes of one size larger than manufacturer’s standard size. Contractor shall provide reducers or drill and tap larger where required.

3.5.7. Conduit racks must be neatly arranged, properly aligned and supported. Before installation, contractor shall carefully check all routings indicated or proposed, for interferences with building structure, piping and equipment. Where any interference is encountered, the Contractor shall obtain the approval of the Owner's Representative for a routing to clear the interferences. After such approval has been obtained, the Contractor shall proceed with the installation.

3.5.8. Conduit shall be supported at spacings not to exceed 10 feet for 1-1/2 inch and larger; not exceeding 6 feet for smaller conduit. Conduit shall be attached to supports at this same maximum spacing by Unistrut (or equal) conduit straps. Perforated strap hangers are not permitted. One hole straps, U-bolts, Korn clamps may be used where desirable.

3.5.9. All unused openings shall be plugged with T 7 B Series 121 metallic busing and Series 815 steel pennies or other equivalent means. Secure conduits in place and protect where necessary to prevent damage to work during construction. Plug ends of conduit with cork, oakum, or “Push-Pennies” to avoid filling with mortar, debris, etc. Use O.Z. Type “CA” caps or “B” bushings.

3.5.10. Conduit and other raceways shall be installed in a manner that will prohibit the entrance of water, moisture, or any other contaminants at all instruments, push-button, switches, panels, boxes or any other equipment or device. Satisfactory and adequate methods of installation , materials, sealing, breathing, drainage means, and physical aspects of installation shall be used to insure the proper minimum maintenance installation.

3.6. CONDUIT AND TRAY OPENINGS

3.6.1. The Contractor shall be responsible for all sleeves and openings through walls and floors necessary for passage of electrical conduits and raceways. Where Contractor must provide openings and/or drill concrete floors and/or walls, he shall be responsible for the repair of these openings. All openings in walls and floors and under switchgear and panels where electrical cables, and conduits are installed shall be closed up by the Contractor to prevent dust, dirt, and water from entering. Provide drip leg on all cable tray penetrations in electrical room. Curbing shall be provided where necessary to accomplish the above.

3.6.2. The Contractor shall be responsible for sealing all wall and floor openings and all floor and wall sleeves openings utilized by the Contractor whether furnished by Others or by the Contractor.

3.6.3. Sleeves and openings shall be sealed with materials that will withstand fire and heat to the same rating as the wall, floor, or ceiling through which the conduit or tray passes and shall not be less than a 30 minute barrier. One of the following methods shall be used:

3.6.3.1.1. Method "A": Seal with SF20 Fire Retardant Silicone Foam, made with DC 3-6548 components, as supplied by Brand Industrial Services, Inc or approved equal.

3.6.3.1.2. Method "B": Seal with CTC PR-855 fire retardant "Silicone RTV Foam" made with DC 3-6548 components, as supplied by Chase Technology Corporation, or approved equal.

3.6.3.1.3. Method "C": Use mineral wool batts to fill opening around conduits, cables, and trays. On both sides of the mineral wool, apply and face the opening with either "Pyrocrete" Fire proofing (Carboline Co.,) to the thickness shown below:

3.6.3.1.3.1. For 2-Hr. rated walls, floors, or ceilings are 1 inch LWP cement plaster.

3.6.3.1.3.2. For 3-Hr. rated, use 1-1/2 inch LWP cement plaster.

3.6.3.1.3.3. For 4-Hr. rated, use 2 inch LWP cement plaster.

3.6.4. Forming material when using Method "A" and/or "B" above, which may remain in place to meet flame ratings of 2, 3, and 4 hours shall be non-flammable and shall be "Cerafiber type as manufactured by the Johns-Manville Corp., or approved equal. Waxed cardboard may be used for forming when the forming material is not left in place. Foam and forming material thicknesses will vary depending upon fire rating.

3.6.5. All opening seals shall meet the requirements of the Owner's insurance carrier.

3.7. CABLE TRAY

3.7.1. Installation:

3.7.1.1. Install cable trays as indicated; in accordance with equipment manufacturer's instructions, and with recognized industry practices (NEMA VE 2), to ensure that the cable tray equipment complies with requirements of NEC, and applicable portions of NFPA 70B and NECA's "Standards of Installation" pertaining to general electrical installation practices.

3.7.1.2. Coordinate cable tray with other electrical work as necessary to properly interface installation of cable tray work with other work.

3.7.1.3. Provide sufficient space encompassing cable trays to permit access for installing and maintaining cables.

3.7.2. At cable tray openings, Contractor shall provide Crouse-Hinds type TW series thru-wall barrier, cable/conduit sealing device providing a fire rated seal and also seal against water and dust entrance. Provide plugs and blank sealing block assemblies filling unused spaces. Seal all conduit/cable tray penetrations through all outside wails and all interior cable tray opening through Motor Control Center room, I/O room and Control room. Do not seal cable tray chase/openings between floor levels in plant.

3.7.3. Testing:

3.7.3.1. Test cable trays to ensure electrical continuity of bonding and grounding connections.

3.7.3.2. Manufacturer shall provide test reports witnessed by an independent testing laboratory of the "worst case" loading conditions outlined in this specification and performed in accordance with the latest revision of NEMA VE 1.

END OF SECTION

SECTION 260553 – IDENTIFICATION FOR ELECTRICAL SYSTEMS

1. PART 1 -GENERAL

1. DESCRIPTION OF WORK

1.1.1. Extent of electrical equipment identification is as indicated on the drawings, schedules and by this specification. Labeling of equipment with voltage, equipment function, such as air handling unit, etc. may be abbreviated or shown as indicated by schedule marking, i.e. AHU-1, etc.

1.1.2. Types of electrical identification specified in this section include the following:

1.1.2.1. Buried cable warnings.

1.1.2.2. Cable conductor identification.

1.1.2.3. Operational instruction and warnings.

1.1.2.4. Danger Signs.

1.1.2.5. Equipment marking.

1.1.2.6. System identification.

1.1.2.7. Electrical cabinets and enclosures.

1.1.2.8. Access panels and doors to electrical facilities.

1.1.2.9. Electrical switchboards and switchgear.

1.1.2.10. Electrical substations.

1.1.2.11. Motor control centers.

1.1.2.12. Transformers.

1.1.2.13. Panelboards.

1.1.2.14. Separately mounted starters.

1.1.2.15. Disconnects.

1.1.2.16. Time clocks.

1.1.2.17. Example of panelboard label/marking:

1.1.2.17.1. “Panel ABC”

1.1.2.17.2. 120/208 Volt, 3 phase, 4 wire

1.1.2.17.3. Served from circuit

1.1.2.17.4. MDP-15

1.1.3. Coordinate names, abbreviations and other designations used in electrical identification work, with corresponding designations shown, specified or scheduled. Provide numbers, lettering and working as indicated or, if not otherwise indicated, as recommended by manufacturers or as required for proper identification and operation/maintenance of electrical systems and equipment. Comply with ANSI A13.1 pertaining to minimum sizes for letters and numbers.

1.2. QUALITY CONTROL

1.2.1. Manufacturers: Firms regularly engaged in manufacture of electrical identification products of types required, whose products have been in satisfactory use in similar service for not less than 3 years.

1.2.2. Installer: A firm with not less than 3 years of successful experience in installation of raceways similar to those required for this product.

1.2.3. UL Compliance: Comply with applicable portions of UL safety standards pertaining to electrical marking and labeling identification systems.

1.2.4. NEC Compliance: Comply with NEC as applicable to installation of identifying labels and markers for wiring and equipment.

1.3. QUALITY ASSURANCE

1.3.1. Identification materials shall comply with all Environmental Protection Agency Criteria shall be applied in accordance with manufacturer's instructions.

1.3.2. All warning signs required by local, state and federal agencies regarding safety and all OSHA guidelines shall be furnished and installed as required.

1.4. SUBMITTALS

1.4.1. Product Data: Submit manufacturer's product specification and installation instructions for each identification material and device required. Include data substantiating that materials comply with specification requirements.

1.4.2. Samples: Submit samples of each color, lettering style and other graphic representation required for each identification material or system.

2. PART 2 - PRODUCTS

2.1. ACCEPTABLE MANUFACTURERS

2.1.1. Manufacturers: Subject to compliance with requirements, provide products of one of the following (for each type of marker):

2.1.1.1. W.H. Brady Company

2.1.1.2. Ideal Industries, Inc.

2.1.1.3. LEM Products, Inc.

2.1.1.4. Seton Name Plate Company

2.1.1.5. 3M Corp.

2.2. ELECTRICAL IDENTIFICATION MATERIALS

2.2.1. General: Except as otherwise indicated, provide manufacturer's standard products of categories and types required for each application.

2.2.2. Color-Coded Plastic Tape:

2.2.2.1. General: Provide manufacturer's standard self-adhesive vinyl tape not less than 3 mils thick by 1/2" wide.

2.2.2.2. Colors: Unless otherwise indicated or required by governing regulations, provide orange tape.

2.2.2.3. All line marker tape shall have a metallic insert to aid in location by cable finding equipment.

2.2.3. Underground-Type Plastic Line Marker:

2.2.3.1. General: Manufacturer's standard permanent, bright-colored, continuous-printed plastic tape, intended for direct-burial service; not less than 6" wide x 4 mils thick. Provide tape with printing which most accurately indicates type of service of buried cable.

2.2.4. Cable/Conductor Identification Bands:

2.2.4.1. Tag with W.H. Brady Company Special (761) self-laminating B-292 vinyl film or Owner approved equal. Fasten wire makers to wire or cable at all terminal ends. Type or machine print wire numbers on the markers.

2.2.5. Plasticized Tags:

2.2.5.1. General:

2.2.5.1.1. Manufacturer's standard pre-printed or partially pre-printed accident-prevention and operational tags, of plasticized card stock with matt finish suitable for writing, approximately 3-1/4" x 5-5/8", with brass grommets and wire fasteners, and with appropriate pre-printed wording including large-size primary wording (as examples: DANGER, CAUTION, DO NOT OPERATE).

2.2.5.1.2. Metal-Backed Butyrate Warning and Caution Signs: Weather-resistant, non fading, preprinted cellulose acetate butyrate signs with 20-gauge, galvanized steel backing, with colors, legend, and size appropriate to the location, shall be installed in interior plant locations subject to hosedown/washdown and in interior locations. Provide ¼ inch grommets in corners for mounting.

2.2.6. Baked Enamel Danger Signs:

2.2.6.1. General:

2.2.6.1.1. Provide manufacturer's standard "DANGER" signs of baked enamel finish on 20-gage steel; of standard red, black and white graphics; 14" x 10" size except where 10" x 7" is the largest size which can be applied where needed, and except where larger size is needed for adequate vision; with recognized standard explanation wording (as examples: HIGH VOLTAGE, KEEP AWAY, BURIED CABLE, DO NOT TOUCH SWITCH).

2.2.7. Engraved Plastic-Laminate Signs:

2.2.7.1. General:

2.2.7.1.1. Provide engraving stock melamine plastic laminate, complying with FS L-P-387, in sizes and thicknesses indicated, engraved with engraver's standard letter style of sizes and wording indicated, black and white core (letter color) except as otherwise indicated, punched for mechanical fastening except where adhesive mounting is necessary because of substrate.

2.2.7.1.2. Engraved, Plastic-Laminated Labels, Signs, and Instruction Plates: Engraving stock melamine plastic laminate, 1/16" minimum thick for signs up to 20 square inches, or 8 inches in length; 1/8" thick for larger sizes. Engraved legend in white letters on black face for non-emergency/interruptible loads, and in white letters on red face for emergency/non-interruptible loads. Legends shall be punched for mechanical fasteners.

2.2.7.1.3. Fasteners: Self-tapping stainless steel screws, except contact-type permanent adhesive where screws cannot or should not penetrate substrate, or number 10/32 stainless steel machine screws with nuts and flat and lock washers.

2.2.8. Circuit, conduit, box and panelboard identification:

2.2.8.1. General:

2.2.8.1.1. Junction boxes and panel covers for normal power systems shall be labeled with circuit number and panel number using black numbers on white background, j-boxes may be painted. Panel cover identification shall be plastic engraved plates bolted to the cover.

3. PART 3 - EXECUTION

3.1. APPLICATION AND INSTALLATION

3.1.1. General Installation Requirements:

3.1.1.1.1. Install electrical identification products as indicated, in accordance with manufacturer's written instructions, and requirements of NEC.

3.1.1.1.2. Coordination: Where identification is to be applied to surfaces which require finish, install identification after completion of painting.

3.1.1.1.3. Regulations: Comply with governing regulations and requests of governing authorities for identification of electrical work.

3.1.2. Underground Cable Identification:

3.1.2.1. General:

3.1.2.1.1. During back-filling/top-soiling of each exterior underground electrical, signal or communication cable, install continuous underground-type plastic line marker, located directly over buried line at 6" to 8" below finished grade. Where multiple small lines are buried in a common trench and do not exceed an overall width of 16", install a single line marker.

3.1.2.1.2. Install line markers regardless of whether the cable is direct buried or protected in conduit.

3.1.3. Cable/Conductor Identification:

3.1.3.1. General:

3.1.3.1.1. Apply cable/conductor identification on each cable and conductor in each box/enclosure/cabinet where wires of more than one circuit or communication/signal system are present, except where another form of identification with marking system used in panelboards, shop drawings, contract documents, and similar previously established identification for project electrical work.

3.1.3.1.2. Use conductors with color factory-applied the entire length of the conductors except as follows:

3.1.3.1.3. The following field-applied color-coding methods may be used in lieu of factory-coded wire for sizes No. 6 AWG and larger.

3.1.3.1.4. Apply colored, pressure-sensitive plastic tape in half-lapped turns for a distance of 6 inches from terminal points and in boxes where splices or taps are made. Apply the last two laps of tape with no tension to prevent possible unwinding. Use ½" wide tape in colors as specified. Do not obliterate cable identification markings by taping. Tape locations may be adjusted slightly to prevent such obliteration.

3.1.3.1.5. Multiple Circuits: Where multiple branch circuits or control wiring or communications/signal conductors are present in the same box or enclosure (except for three-circuit, four-wire home runs), label each

conductor or cable. Provide legend indicating source, voltage, circuit number, and phase for branch circuit wiring. Phase and voltage of branch circuit wiring may be indicated by means of coded color of conductor insulation. For control and communications/signal wiring, use color coding or wire/cable marking tape at terminations and at intermediate locations where conductors appear in wiring boxes, troughs, and control cabinets. Use consistent letter/number conductor designations throughout on wire/cable marking tapes.

3.1.3.1.6. Match identification markings with designations used in panelboard shop drawings, Contract Documents, and similar previously established identification schemes for the facility's electrical installations.

3.1.3.1.7. Tag or label conductors in every pullbox, outlet box or junction box as follows:

3.1.3.1.7.1. Future Connections: Conductors indicated to be for future connection or connection under another contract with identification indicating source and circuit numbers.

3.1.3.1.8. Cable Ties: Fungus-inert, self-extinguishing, sunlight resistant one-piece, self-locking nylon cable ties, 0.18" minimum width, 50 lb. minimum tensile strength, and suitable for a temperature range from minus 50°F to 350°F.

3.1.4. Operational Identification and Warnings:

3.1.4.1. General:

3.1.4.1.1. Wherever required to ensure safe and efficient operation and maintenance of electrical systems, and electrically connected mechanical systems, including prevention of misuse of electrical facilities by unauthorized personnel, install self-adhesive plastic signs or similar equivalent identification enclosures. Where detailed instructions or explanations are needed, provide plasticized tags with clearly written messages adequate for intended purposes.

3.1.5. Danger Signs:

3.1.5.1. General:

3.1.5.1.1. In addition to installation of danger signs required by governing regulations and authorities, install appropriate danger signs at locations indicated and at locations subsequently identified by Installer of electrical work as constituting similar danger for persons in or about project.

3.1.5.1.2. High Voltage: Install danger signs wherever it is possible, under any circumstances, for persons to come into contact with electrical power of voltages higher that 250 volts.

3.1.5.1.3. Critical Switches/Controls: Install danger signs on switches and similar controls, regardless of whether concealed or locked up, where untimely or inadvertent operation could result in significant danger to persons, or damage to or loss of property.

3.1.6. Equipment/System Identification:

3.1.6.1.1. General: Install fan engraved plastic-laminate sign on each major unit of electrical equipment unless it is specified with its own identification tag. This includes communication/signal/alarm systems.

3.1.6.1.2. Except as otherwise indicated, provide single line of text, with ½" high lettering on 1 ½" high label (2" high where two lines are required), black lettering in white field. Text shall match terminology and numbering of the Contract Documents and shop drawings. Apply labels for each unit of the following categories of electrical equipment:

3.1.6.1.2.1. Main switchboard, panelboards, electrical cabinets, motor control centers and enclosures, communications, signal and alarm systems, control stations, pushbutton enclosures, transformers, substations etc.

3.1.6.1.3. Motors shall be identified with a label affixed on a plate or other flat surface close to the motor.

3.1.6.1.4. Due care shall be exercised to insure that the markers are clearly visible from nearby walkways.

3.1.6.1.5. Install signs at locations indicated or, where not otherwise indicated, at location for best convenience of viewing without interference with operation and maintenance of equipment. Secure to substrate with fasteners, or use adhesive where fasteners should not or cannot penetrate the substrate.

3.1.6.1.6. Apply warning, caution, and instruction signs and stencils as follows:

3.1.6.1.6.1. Install warning, caution, or instruction signs where required by NEC, where indicated, or where reasonably required to assure safe operation and maintenance of electrical systems and of the items to which they connect. Install engraved plastic-laminated instruction signs with approved legend where instructions or explanations are needed for system or equipment operation.

3.1.6.1.7. Provide warning signs on all equipment or devices operating at 300 volts or more, reading "DANGER – 480 VOLTS," etc., with white letter on red background, of standard Code size. Signs may be decals, stencils, or nameplates.

3.1.6.1.8. Control wires shall be red and marked at all terminations with Class J, B-320, Brady Sleeve, PVC (vinyl) computer printable wire markers.

3.1.6.1.9. All terminals of a control connection between or among control devices shall bear the same designation.

3.1.6.1.10. All spare conductors shall be uniquely marked and tagged to identify the location of the opposite end.

3.1.6.1.11. Control Devices:

3.1.6.1.11.1. All internally mounted control devices such as relays, timers, transformers, and control amplifiers shall be identified with labels affixed on or near the device.

3.1.6.1.11.2. Externally mounted control devices such as limit and pressure switches, transmitters, sensing heads, solenoid valves, control valves etc., shall be identified with markers supplied by the Owner. Vibrators and remotely mounted electric eyes and electric eye sensing heads shall be identified with markers supplied by the Owner.

3.1.6.1.11.3. Externally mounted control amplifiers for remote sensors such as manufactured by Micro-Switch and Peco, shall be identified by markers supplied by the Owner.

3.1.6.1.12. Immediately after each cable installation, the cable must be permanently tagged at both ends and at location where cable exits tray into conduits. Tagging shall indicate the assigned circuit number, MCC number or panel number and motor/equipment ID No.

END OF SECTION

SECTION 260800 – COMMISSIONING OF ELECTRICAL SYSTEMS

1. PART 1 - GENERAL

1.1. SCOPE

1.1.1. Intent. The intent of the inspection, testing and check out work specified herein, or required, is to ensure that all electrical workmanship and equipment, whether Owner furnished or Contractor furnished, is installed and performs in accordance with the design specifications, drawings, manufacturer's instructions and all applicable codes and requirements. Also, it is intended to provide the following:

1.1.2. To determine if the equipment has been damaged during shipment or installation.

1.1.3. To verify if the equipment is in accordance with the specifications.

1.1.4. To provide initial acceptance tests and recorded data that can be used as a benchmark for future routine maintenance and trouble shooting.

1.1.5. To ensure a successful start-up with a minimum of last-minute problems.

1.1.6. Determine the suitability of the equipment and systems for energization and placing into operating service.

1.1.7. Provide assurance that each system component is not only installed satisfactorily but performs, and will continue to perform, its function in the system with reasonable reliability throughout the life of the plant.

1.1.8. Contractor Responsibility. The Contractor shall provide all necessary supervision and labor, materials, tools, test instruments or other equipment or services and expenses required to inspect, test, adjust, set, calibrate, functionally and operationally check all work and components of the various electrical systems and circuitry throughout the installation. Also, the Contractor shall include the furnishing of sufficient personnel to assist operating forces in any additional checks they may require for acceptance, start-up, run-in and placing the equipment and systems into continuous service.

1.2. GENERAL REQUIREMENTS FOR ALL INSPECTION, TESTING, AND START-UP

1.2.1. Preparation of Procedures and Schedules. The Contractor is responsible for the preparation of proposed procedures and schedules for all inspections, tests, settings and calibrations specified or otherwise required prior to or during the check-out for start-up and acceptance of all the electrical components, equipment and systems. This work shall be coordinated and shall be compatible with both the work of other crafts and the project schedule. The above must be organized and submitted with all proposed testing and check out forms for the review and approval of the Owner's Representative. The procedures must provide specific instructions for the checking and testing of each component in addition to the system functional checks. Tests and inspections shall also be scheduled as the job progresses and

may require repetition in greater detail at a later stage of construction. All procedures submitted shall include job safety rules proposed.

1.2.2. Inspection of Purchased Equipment. Equipment purchased by the Contractor or purchased by the Owner for Contractor installation shall be inspected and tested to determine its condition. See other applicable sections of the specifications and contract documents for required Contractor checks and responsibilities.

1.2.3. Report of Damage. At any stage of construction and when observed, any electrical equipment or system determined to be damaged, faulty, or requiring repairs shall be reported to the Owner's Representative. Corrective action may require prior approval.

1.2.4. Identification of Equipment and Wiring and Miscellaneous Checks.

1.2.4.1. Prior to check out and testing for start-up, the Contractor shall ensure that all equipment and wiring is properly and permanently identified with nameplates and other identification as specified elsewhere. The Contractor shall also check and tighten all terminals and connection points, remove all shipping blocks and hardware, thoroughly clean all equipment, repair all damaged or scratched finishes, inspect for broken and missing parts and review and collect manufacturer's drawings and instructions for delivery to the Owner's Representative. The Contractor shall also make routine checks and tests as the job progresses and as necessary to insure that all wiring and equipment is properly installed and wired.

1.2.5. Notifying Owner of Test and Checks and Witnessing. The Owner or the Owner's Representative, at their own discretion, may witness any and all inspections, tests, settings and calibrations as they are being conducted. The Contractor shall notify the Owner's Representative at least 24 hours in advance of all tests and checks to be performed.

1.2.6. Approval for Energizing, De-Energizing or Tying-In a System.

1.2.6.1. Equipment or circuitry shall not be energized, de-energized or tied-in to a system without prior review and approval of the Owner's Representative. The Contractor shall be responsible for obtaining the proper permits for this type work.

1.2.7. Notification of Manufacturer's Representative. The testing and check out of important electrical equipment such as drive equipment and motor control centers may require notification of the proposed checks to the equipment manufacturer's representative. It is important that equipment warranties or guarantees shall not be voided by the Contractor's testing and check out work. The checks and tests shall normally be supplemental to and compatible with the manufacturer's installation instruction leaflets and literature. Where deviations are apparent, the manufacturer's review shall be obtained prior to testing. Reasonable cooperation shall be extended to permit witnessing by the manufacturer's representative if so requested. Where any questionable repairs, modifications, significant adjustments,

tests or checks are to be made, the Contractor shall contact the Owner's Representative to determine if the work should be performed by or with the manufacturer's representative.

1.2.8. Test Instruments and Materials. The Contractor shall furnish all the material, test equipment, and power sources required for testing, calibrating and check out. Test equipment shall have calibrations traceable to the National Bureau of Standards. Contractor and/or testing laboratory dated calibration labels shall be visible on the test equipment. Calibrations over 6 months old shall not be acceptable on field test instruments shall be at least twice that of the accuracy of the equipment, device, relay or meter under test. All testing instruments shall be checked to insure satisfactory operation prior to proceeding with the tests. Serial and model numbers of the instruments used shall be recorded on the test forms.

1.2.9. Personnel Qualifications. All testing and check out work shall be performed with qualified personnel skilled in the particular tests being conducted. This is essential for obtaining and properly evaluating data while the tests are in progress and for insuring that important facts and questionable data are reported.

1.2.10. Safety Precautions. The contractor shall ensure that all testing and checkouts work is conducted in a safe manner. Special safety precautions such as the following shall be utilized where appropriate: (1) locking and tagging procedures; (2) barricades; (3) deenergization and/or isolation of equipment prior to testing; (4) review of procedures with the Owner's safety personnel; (5) erection of warning signs; (6) stationing of guards and watchmen; (7) maintenance of voice communications; and (8) personnel orientation.

1.2.11. Inspection, Test, and Calibration Reports. All inspections, test, and calibrations shall be reported in writing on Owner-approved report forms. The recorded data form shall have the signatures of the persons conducting the tests, authorized witnesses, and the Owner's Representative. The forms shall be designed to serve as the test and inspection checklist for inspection requirements. A minimum of 4 copies of all test and check out data shall be given to the Owner's Representative. This shall also include any data taken prior to adjustments, repairs, drying out, or similar work prior to final testing and acceptance. "As-found" and "as-left" test data shall be recorded and reported in writing. The Owner's Representative shall forward one complete site of all forms and data to the design engineer for review and for filing with project set. Two of the 4 sets shall be delivered to the Owner for record and maintenance purposes.

1.2.12. Quality and Workmanship. The inspections of all the electrical systems shall include methods, workmanship, and quality of materials.

1.2.13. Test Sequence. The sequence of all tests and checks shall be such that the equipment can be energized immediately after the completion of applicable tests.

1.2.14. Notification of Substantial Electrical Completion. When applicable electrical tests and inspections specified herein, or otherwise required are completed and results reported, reviewed and approved, then the Contractor may consider that portion of

the electrical equipment system or installations electrically complete. The Contractor shall then affix appropriate and approved dated completion or calibration labels to the tested equipment. The Contractor shall notify the Owner's Representative of electrical completion. If the Owner's Representative finds completed work unacceptable, he will notify the Contractor in writing of the unfinished or deficient work to be corrected by the Contractor. The Contractor shall notify the Owner's Representative in writing when all exceptions have been corrected. A "Notification of Substantial Electrical Completion" shall be prepared and approved following satisfactory compliance. If later in-service operation or further testing determines problems attributable to the Contractor, these shall be corrected by the Contractor or suitable arrangements shall be made to suit operating circumstances.

1.2.15. Applicable Documents and Standards. Except as specified otherwise and where applicable, all inspections and tests shall be based upon and shall utilize the following documents and references:

1.2.15.1. All Standard, Special and Supplemental Conditions of the Contract.

1.2.15.2. The project specifications.

1.2.15.3. The Contract drawings.

1.2.15.4. Manufacturer's instruction manuals, drawings and information applicable to the apparatus, equipment and systems.

1.2.15.5. National Fire Protection Association (NFPA) No. 70 and No. 70B Recommended Practice for Electrical Equipment Maintenance.

1.2.15.6. National Electrical Testing Association (NETA).

1.2.15.7. National Electrical Manufacture's Association (NEMA).

1.2.15.8. American Society for Testing and Materials (ASTM).

1.2.15.9. Institute of Electrical and Electronic Engineers (IEEE).

1.2.15.10. American National Standards Institute (ANSI).

1.2.15.11. Insulated Power Cable Engineers Association (IPCEA).

1.2.15.12. National Electrical Safety Code (ANSI C2).

1.2.15.13. OSHA Part 1910, Sub-Part S-1910.308.

1.3. SPECIFIC TEST AND INSPECTION BY THE CONTRACTOR

1.3.1. General.

1.3.1.1. The following specific items of work shall be performed by the Contractor. The list is not all inclusive, nor does it define how the tests and checks are to be made. It is merely to define the minimum and type of tests that are required. The equipment and cable shall be deenergized and isolated as necessary to perform the tests.

1.3.1.2. It should be stated again that the engagement of the independent testing firm in no way relieves Contractor of the responsibility for, or diminishes the importance of, performance tests, check outs, and inspections during the various stages of construction. The specific work of the independent test firm is defined in Paragraph 5.0.

1.3.1.3. The Contractor shall perform tests and inspections as defined in the other sections of this specification, in particular,

1.3.1.4. All equipment received for the job and for which the Contractor is responsible shall be stored and cared for per the manufacturer's instructions. It shall be the Contractor's responsibility to obtain such information even where the instructions are not shipped with the equipment.

1.3.2. Equipment Test and Inspection During Construction and Prior to Acceptance Testing.

1.3.2.1. Test run all motors preferably uncoupled or unloaded, before placing into regular service. A check on the motor for rotation, speed, current and temperature rise shall be made and results recorded. The proper color codes for phase identifications shall be maintained. This may require lead swaps at the motor for proper rotation. A motor phase rotation meter should be used prior to connection at motor to prevent later swaps.

1.3.3. Grounding Systems:

1.3.3.1. All main plant building loops and major equipment grounds shall be tested to remote earth or directly referenced to an extremely low resistance (approximately 5 ohm) reference ground benchmark. Visual inspection of all systems, raceway and equipment grounds shall be made to determine the adequacy and integrity of the grounding. All ground testing results shall be properly recorded, witnessed, and reported to the Owner's Representative.

1.3.3.2. Ground tests shall be performed using a J. G. Biddle Company or equivalent low resistance, Null balance type, ground testing ohmmeter, and test lead resistance compensated for, measure the resistance of the ground under test and remote earth or a reference ground. The test instrument shall be the type which compensates for potential and current rod resistances.

1.3.3.3. Where ground test results indicate the need for additional grounding conductors or rods that are not indicated on drawings or specified, design changes will be initiated to obtain the acceptable values. The Contractor is responsible for the proper installation of the grounding shown on drawings or specified and for the correction of improper installations as determined by inspections and tests.

1.3.4. Wire and Cable:

1.3.4.1. Before energizing, the continuity and insulation resistance of every circuit external to equipment shall be measured with a megger from each wire to all

others and ground. Tests shall normally be conducted at voltages 500 volts or lower.

1.3.4.2. Insulation resistance measurements shall be taken of the following:

1.3.4.2.1. Motor Feeders: With motors disconnected, measure insulation resistance from load side of contactors or circuit breakers. Repeat this test after motors are connected.

1.3.4.2.2. Motor Control Circuits (600 Volts): With push buttons and over-current devices connected, measure insulation resistance from phase to ground only. It will be necessary to lift the neutral ground on the control transformers to perform this test. Also, isolate any control elements that should not be meggered.

1.3.4.2.3. Lighting Panel Feeders: Measure insulation resistance with circuit breakers, lighting transformers and panel boards connected, but with lighting branch circuit breakers or switches open.

1.3.4.2.4. Lighting Branch Circuits: Measure insulation resistance after all lamp holders, receptacles, fixtures, etc., are connected but before lamping.

1.3.4.2.5. Feeder Circuits: Measure insulation resistance with connections to circuit breakers made up but with breakers open and loads not connected.

1.3.4.3. All cables and wires shall be checked for proper identification numbering and/or color coding.

1.3.5. Sealing of Openings: The Contractor shall inspect the entire job with the Owner's Representative to insure that all openings and holes that the Contractor provided or utilized are properly sealed as specified elsewhere.

1.3.6. Record Drawings: The Contractor shall maintain a master set of record drawings that shows changes and any other deviations from the base drawing. The markups must be made as the changes are done. At the conclusions of the job, these master record drawings must be complete and delivered to the Owner's Representative for forwarding to the design group.

END OF SECTION

SECTION 262200 - LOW-VOLTAGE TRANSFORMERS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this section and all other sections of Division 26.

1.2 SUMMARY

A. This Section includes the following types of dry-type transformers rated 600 V and less, with capacities up to 1000 kVA:

1. Distribution transformers.

1.3 SUBMITTALS

A. Product Data: Include rated nameplate data, capacities, weights, dimensions, minimum clearances, installed devices and features, and performance for each type and size of transformer indicated.

B. Shop Drawings: Detail equipment assemblies and indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection.

1. Wiring Diagrams: Power, signal, and control wiring.

C. Qualification Data: For testing agency.

D. Source quality-control test reports.

E. Field quality-control test reports.

F. Operation and Maintenance Data: For transformers to include in emergency, operation, and maintenance manuals.

1.4 QUALITY ASSURANCE

A. Testing Agency Qualifications: An independent agency, or manufacturer's trained and qualified personnel, with the experience and capability to conduct the testing indicated, that is a nationally recognized testing laboratory (NRTL) as defined by OSHA in 29 CFR 1910.7.

B. Source Limitations: Obtain each transformer type through one source from a single manufacturer.

C. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.

D. Comply with IEEE C57.12.91, "Test Code for Dry-Type Distribution and Power Transformers."

1.5 DELIVERY, STORAGE, AND HANDLING

A. Temporary Heating: Apply temporary heat according to manufacturer's written instructions within the enclosure of each ventilated-type unit, throughout periods during which equipment is not energized and when transformer is not in a space that is continuously under normal control of temperature and humidity.

1.6 COORDINATION

A. Coordinate size and location of concrete housekeeping pads with actual transformer provided. Cast anchor-bolt inserts into pads. Concrete, reinforcement, and formwork requirements are specified in Division 03 or on the drawings.

B. Coordinate installation of wall-mounting and structure-hanging supports with actual transformer provided.

1.7 WARRANTY/GUARANTEE

A. Warranty Period: Two years from the date of shipment or one year from date of project Substantial Completion, whichever is later.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Manufacturers: Subject to compliance with requirements, provide products by one (1) of the following:

1. ACME Electric Corporation; Power Distribution Products Division.
2. Challenger Electrical Equipment Corp.; a division of Eaton Corp.
3. Eaton Electrical Inc.; Cutler-Hammer Products.
4. Sola/Hevi-Duty.
5. Square D; Schneider Electric.
6. Siemens

2.2 GENERAL TRANSFORMER REQUIREMENTS

A. Description: Factory-assembled and -tested, air-cooled units for 60-Hz service.

B. Cores: Grain-oriented, non-aging silicon steel.

C. Coils: Continuous windings without splices except for taps.

1. Internal Coil Connections: Brazed or pressure type.
2. Coil Material: Copper or aluminum.

2.3 DISTRIBUTION TRANSFORMERS

A. Comply with NEMA ST 20, and list and label as complying with UL 1561.

B. Cores: One leg per phase.

C. Enclosure: Ventilated NEMA 250, Type 1.

1. Core and coil shall be encapsulated within resin compound, sealing out moisture and air.

D. Transformer Enclosure Finish: Comply with NEMA 250.

1. Finish Color: ANSI 49 gray.

E. Taps for Transformers 7.5 to 24 kVA: One (1) 5 % tap above and one (1) 5 % tap below normal full capacity.

F. Taps for Transformers 25 kVA and Larger: Two (2) 2.5 % taps above and two (2) 2.5 % taps below normal full capacity.

G. Insulation Class: 220^{0}C, UL-component-recognized insulation system with a maximum of 115^{0}C rise above 40^{0}C ambient temperature.

H. Transformers shall be of the KVA size, primary and secondary voltages and configuration(s) (delta-wye, delta-delta etc.) indicated on the contract documents.

I. Energy Efficiency for Transformers Rated 15 kVA and Larger:

1. Complying with NEMA TP 1, Class 1 efficiency levels.
2. Tested according to NEMA TP 2.

J. Wall Brackets: Manufacturer's standard brackets.

K. Low-Sound-Level Requirements: Minimum of 3 dBA less than NEMA ST 20 standard sound levels when factory tested according to IEEE C57.12.91.

L. Low-Sound-Level Requirements: Maximum sound levels, when factory tested according to IEEE C57.12.91, as follows:

1. 9 kVA and Less: 37 dBA.
2. 30 to 50 kVA: 42 dBA.
3. 51 to 150 kVA: 47 dBA.
4. 151 to 300 kVA: 52 dBA.
5. 301 to 500 kVA: 57 dBA.
6. 501 to 750 kVA: 59 dBA.
7. 751 to 1000 kVA: 61 dBA.

2.4 IDENTIFICATION DEVICES

A. Nameplates: Engraved, laminated-plastic or metal nameplate for each distribution transformer, mounted with corrosion-resistant screws. Nameplates and label products are specified in Division 26 Section "Identification for Electrical Systems."

2.5 SOURCE QUALITY CONTROL

A. Test and inspect transformers according to IEEE C57.12.91.

B. Factory Sound-Level Tests: Conduct sound-level tests on equipment for this Project.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine conditions for compliance with enclosure- and ambient-temperature requirements for each transformer.

B. Verify that field measurements are as needed to maintain working clearances required by NFPA 70 and manufacturer's written instructions.

C. Examine walls, floors, roofs, and concrete bases for suitable mounting conditions where transformers will be installed.

D. Verify that ground connections are in place and requirements in Division 26 Section "Grounding and Bonding for Electrical Systems" have been met.

E. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. Install wall-mounting transformers level and plumb with wall brackets fabricated by transformer manufacturer.

3.3 CONNECTIONS

A. Ground equipment according to Division 26 Section "Grounding and Bonding for Electrical Systems."

B. Connect wiring according to Division 26 Section "Low-Voltage Electrical Power Conductors and Cables."

3.4 FIELD QUALITY CONTROL

A. Tests and Inspections:

1. Perform each visual and mechanical inspection and electrical test stated in NETA Acceptance Testing Specification. Certify compliance with test parameters.

B. Remove and replace units that do not pass tests or inspections and retest as specified above.

C. Test Labeling: On completion of satisfactory testing of each unit, attach a dated and signed "Satisfactory Test" label to tested component.

3.5 ADJUSTING

A. Record transformer secondary voltage at each unit for at least 48 hours of typical occupancy period. Adjust transformer taps to provide optimum voltage conditions at secondary terminals. Optimum is defined as not exceeding nameplate voltage plus 10 % and not being lower than nameplate voltage minus 3% at maximum load conditions. Submit recording and tap settings as test results.

B. Output Settings Report: Prepare written report recording output voltages and tap settings.

3.6 CLEANING

A. Vacuum dirt and debris; do not use compressed air to assist in cleaning.

END OF SECTION 262200

SECTION 262413 - SWITCHBOARDS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this section and all other sections of Division 26.

1.2 SUMMARY

A. Section Includes:

1. Service and distribution switchboards rated 600 V and less.
2. Transient voltage suppression devices.
3. Disconnecting and overcurrent protective devices.
4. Instrumentation.
5. Control power.
6. Accessory components and features.
7. Identification.

1.3 SUBMITTALS

A. Product Data: For each type of switchboard, overcurrent protective device, transient voltage suppression device, ground-fault protector, accessory, and component indicated. Include dimensions and manufacturers' technical data on features, performance, electrical characteristics, ratings, accessories, and finishes.

B. Shop Drawings: For each switchboard and related equipment.

1. Include dimensioned plans, elevations, sections, and details, including required clearances and service space around equipment. Show tabulations of installed devices, equipment features, and ratings.
2. Detail enclosure types for types other than NEMA 250, Type 1.
3. Detail bus configuration, current, and voltage ratings.
4. Detail short-circuit current rating of switchboards and overcurrent protective devices.
5. Include descriptive documentation of optional barriers specified for electrical insulation and isolation.

6. Detail utility company's metering provisions with indication of approval by utility company.
7. Detail features, characteristics, ratings, and factory settings of individual overcurrent protective devices and auxiliary components.
8. Include time-current coordination curves for each type and rating of overcurrent protective device included in switchboards. Include selectable ranges for each type of overcurrent protective device.

9. Include schematic and wiring diagrams for power, signal, and control wiring.

C. Qualification Data: For qualified testing agency.

D. Field Quality-Control Reports:

1. Test procedures used.
2. Test results that comply with requirements.
3. Results of failed tests and corrective action taken to achieve test results that comply with requirements.

E. Operation and Maintenance Data: For switchboards and components to include in emergency, operation, and maintenance manuals. In addition to items specified in Division 01 Section "Operation and Maintenance Data," include the following:

1. Routine maintenance requirements for switchboards and all installed components.
2. Manufacturer's written instructions for testing, maintaining and adjusting overcurrent protective devices.
3. Time-current coordination curves for each type and rating of overcurrent protective device included in switchboards. Include selectable ranges for each type of overcurrent protective device.

1.4 QUALITY ASSURANCE

A. Installer Qualifications: An employer of workers qualified as defined in NEMA PB 2.1 and trained in electrical safety as required by NFPA 70E, or manufacturer’s factory trained and qualified technician.

B. Testing Agency Qualifications: Member Company of NETA or an NRTL.

1. Testing Agency's Field Supervisor: Currently certified by NETA to supervise on-site testing.

C. Source Limitations: Obtain switchboards, overcurrent protective devices, components, and accessories from single source from single manufacturer.

D. Product Selection for Restricted Space: Drawings indicate maximum dimensions for switchboards including clearances between switchboards and adjacent surfaces and other items. Comply with indicated maximum dimensions.

E. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

F. Comply with NEMA PB 2.

G. Comply with NFPA 70.

H. Comply with UL 891.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Deliver switchboards in sections or lengths that can be moved past obstructions in delivery path.

B. Remove loose packing and flammable materials from inside switchboards and [install temporary electric heating (250 W per section) to prevent condensation.

C. Handle and prepare switchboards for installation according to NEMA PB 2.1.

1.6 PROJECT CONDITIONS

A. Installation Pathway: Remove and replace access fencing, doors, lift-out panels, and structures to provide pathway for moving switchboards into place.

B. Environmental Limitations:

1. Do not deliver or install switchboards until spaces are enclosed and weather-tight, wet work in spaces is complete and dry, work above switchboards is complete, and temporary HVAC system is operating and maintaining ambient temperature and humidity conditions at occupancy levels during the remainder of the construction period.
2. Rate equipment for continuous operation under the following conditions unless otherwise indicated:

   a. Ambient Temperature: Not exceeding 104^0F (40^0C).
   b. Altitude: Not exceeding 6600 feet (2000 m).

C. Interruption of Existing Electric Service: Do not interrupt electric service to facilities occupied by Owner or others unless permitted under the following conditions and then

only after arranging to provide temporary electric service according to requirements indicated:

1. Notify the Owner's representative, in writing, no fewer than 10 days in advance of proposed interruption of electric service.
2. Indicate method of providing temporary electric service.
3. Do not proceed with interruption of electric service without the Owner's written permission.
4. Comply with NFPA 70E.

1.7 COORDINATION

A. Coordinate layout and installation of switchboards and components with other construction that penetrates walls or is supported by them, including electrical and other types of equipment, raceways, piping, encumbrances to workspace clearance requirements, and adjacent surfaces. Maintain required workspace clearances and required clearances for equipment access doors and panels.

B. Coordinate sizes and locations of concrete bases with actual equipment provided. Cast anchor-bolt inserts into bases. Concrete, reinforcement, and formwork requirements are specified in Division 03 or on the contract drawings.

1.8 TVSS WARRANTY

A. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace transient voltage suppression devices that fail in materials or workmanship within specified warranty period.

1. TVSS Warranty Period: Five years from date of Substantial Completion.

1.9 EXTRA MATERIALS

A. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Potential Transformer Fuses: Equal to 10% of quantity installed for each size and type, but no fewer than two (2) of each size and type.
2. Control-Power Fuses: Equal to 10 % of quantity installed for each size and type, but no fewer than two (2) of each size and type.
3. Fuses and Fusible Devices for Fused Circuit Breakers: Equal to 10 % of quantity installed for each size and type, but no fewer than three (3) of each size and type.
4. Fuses for Fused Switches: Equal to 10 % of quantity installed for each size and type, but no fewer than three (3) of each size and type.

5. Fuses for Fused Power-Circuit Devices: Equal to 10 % of quantity installed for each size and type, but no fewer than three (3) of each size and type.
6. Indicating Lights: Equal to 10 % of quantity installed for each size and type, but no less than one of each size and type.
7. Provide Fuse cabinet, location coordinated with Owner.

1.10 SWITCHBOARD WARRANTY/GUARANTEE

A. Provide manufacturer's standard form in which manufacturer agrees to repair or replace any devices that fail in materials or workmanship within specified warranty period.

B. Warranty Period: Two (2) years from date of shipment or one (1) year from the date of project Substantial Completion, whichever is later.

PART 2 - PRODUCTS

2.1 MANUFACTURED UNITS

A. Basis-of-Design Product: Square D QED-2 style switchboard. Subject to compliance with requirements, provide product by one (1) of the following:

1. Square D; a brand of Schneider Electric, QED-2.
2. Eaton Electrical Inc.; Cutler-Hammer Business Unit.
3. ABB, Reliagear SB

B. Front- and Side-Accessible Switchboards:

1. Main Devices: Fixed, individually mounted.
2. Branch Devices: Panel mounted.
3. Sections front and rear aligned.

C. Nominal System Voltage: 480Y/277 V or as indicated on the contract drawings.

D. Main-Bus Continuous: 2000A or as indicated on the contract drawings.

E. Indoor Enclosures: Steel, NEMA 250, Type 1.

F. Enclosure Finish for Indoor Units: Factory-applied finish in manufacturer's ANSI 49 gray finish over a rust-inhibiting primer on treated metal surface.

G. Customer Metering Compartment: A separate customer metering compartment and section with front hinged door, for indicated metering, and current transformers for each

meter. Current transformer secondary wiring shall be terminated on shorting-type terminal blocks. Include potential transformers having primary and secondary fuses with disconnecting means and secondary wiring terminated on terminal blocks.

H. Bus Transition and Incoming Pull Sections: Matched and aligned with basic switchboard.

I. Hinged Front Panels: Allow access to circuit breaker, metering, accessory, and blank compartments.

J. Pull Box on Top of Switchboard. Provided only where indicated on the contract drawings:

1. Adequate ventilation to maintain temperature in pull box within same limits as switchboard.
2. Set back from front to clear circuit-breaker removal mechanism.
3. Removable covers shall form top, front, and sides. Top covers at rear shall be easily removable for drilling and cutting.
4. Bottom shall be insulating, fire-resistive material with separate holes for cable drops into switchboard.
5. Cable supports shall be arranged to facilitate cabling and adequate to support cables indicated, including those for future installation.

K. Buses and Connections: Three-phase, four-wire, unless otherwise indicated.

1. Phase- and Neutral-Bus Material: Hard-drawn copper of 98 percent conductivity, with copper feeder circuit-breaker line connections.
2. Ground Bus: 1/4-by-2-inch hard-drawn copper of 98 % conductivity, equipped with mechanical connectors for feeder and branch-circuit ground conductors. For busway feeders, extend insulated equipment grounding cable to busway ground connection and support cable at intervals in vertical run.
3. Main Phase Buses and Equipment Ground Buses: Uniform capacity for entire length of switchboard's main and distribution sections. Provide for future extensions.
4. Neutral Buses: 100 % of the ampacity of phase buses unless otherwise indicated, equipped with mechanical connectors for outgoing circuit neutral cables. Brace bus extensions for busway feeder neutral bus.
5. Main bonding jumper: Provide main bonding jumper for service entrance or separately derived systems per NEC requirements.
6. Braced for 65,000A, symmetrical at 480V.

2.2 TRANSIENT VOLTAGE SUPPRESSION DEVICES

A. Basis-of-Design Product: Subject to compliance with requirements, provide product by one (1) of the following:

1. Eaton Electrical Inc.; Cutler-Hammer Business Unit.

2. Siemens Energy & Automation, Inc.
3. Square D; a brand of Schneider Electric.

B. Surge Protection Device Description: IEEE C62.41-compliant, exterior mounted units, wired-in, solid-state, parallel-connected, modular (with field-replaceable modules) type, with sine-wave tracking suppression and filtering modules, ANSI Rev 2 and UL 1449, second edition, short-circuit current rating matching or exceeding the switchboard short-circuit rating, and with the following features and accessories:

1. Fuses, rated at 200-kA interrupting capacity.
2. Fabrication using bolted compression lugs for internal wiring.
3. Redundant suppression circuits.
4. Redundant replaceable modules.
5. Arrangement with wire connections to phase buses, neutral bus, and ground bus.
6. LED indicator lights for power and protection status.
7. Audible alarm, with silencing switch, to indicate when protection has failed.
8. Form-C contacts rated at 5 A and 250-V ac, one normally open and one normally closed, for remote monitoring of system operation. Contacts shall reverse position on failure of any surge diversion module or on opening of any current-limiting device. Coordinate with building power monitoring and control system.
9. Four-digit, transient-event counter set to totalize transient surges.

C. Peak Single-Impulse Surge Current Rating: 120 kA per mode/240 kA per phase.

D. Withstand Capabilities: 12,000 IEEE C62.41, Category C3 (10 kA), 8-by-20-micro-second surges with less than 5% change in clamping voltage.

E. Protection modes and UL 1449 SVR for grounded wye circuits with 480Y/277 and 208Y/120 V, three-phase, four-wire circuits shall be as follows:

1. Line to Neutral: 600 V for 480Y/277; 330 V for 208Y/120.
2. Line to Ground: 600 V for 480Y/277; 330 V for 208Y/120.
3. Neutral to Ground: 600 V for 480Y/277; 330 V for 208Y/120.

F. Protection modes and UL 1449 SVR for 480 V, three-phase, three-wire, delta circuits shall be as follows:

1. Line to Line: 2000 V for 480 V.
2. Line to Ground: 1500 V for 480 V.

2.3 DISCONNECTING AND OVERCURRENT PROTECTIVE DEVICES

A. Insulated-Case Circuit Breaker (ICCB): 100% rated, sealed, insulated-case power circuit breaker with interrupting capacity rating to meet available fault current. Provide ICCB's

for all frame sizes 1000A and larger. Basis of Design: Square D MasterPact NW. Interrupting rating of 65kA minimum at 480V.

1. Fixed circuit-breaker mounting.
2. Two-step, stored-energy closing.
3. Standard-function, microprocessor-based trip units with interchangeable rating plug, trip indicators, and the following field-adjustable settings:

    a. Long time, short time and instantaneous time adjustments.
    b. Ground-fault pickup level, time delay, and I^2t response for circuit breakers 1000A and larger.
4. Remote trip indication and control.
5. Communication Capability: Integral communication module with functions and features compatible with power monitoring and control system specified in Division 26 Section "Electrical Power Monitoring and Control."
6. Control Voltage: 120-V ac.

B. Molded-Case Circuit Breakers (MCCB): Comply with UL 489, with interrupting capacity to meet available fault currents. Provide for frame sizes 800A and lower. 100% rated for group mounting. Interrupting rating of 65kA minimum at 480V. Basis of Design: Square D PowerPacT type P-Frame and L-Frame with LSI electronic trip.

1. Electronic trip circuit breakers with rms sensing; field-replaceable rating plug or field-replicable electronic trip; and the following field-adjustable settings:

    a. Instantaneous trip.
    b. Long- and short-time pickup levels.
    c. Long- and short-time time adjustments.

2. Molded-Case Circuit-Breaker (MCCB) Features and Accessories:

    a. Standard frame sizes, trip ratings, and number of poles.
    b. Lugs: Mechanical style, suitable for number, size, trip ratings, and conductor material.
    c. Lock-out hasp for padlock.

2.4 INSTRUMENTATION – Refer to Section 262713 “Electricity Metering” & 260913 “Electrical Power Monitoring & Control”

A. Provide metering compartment with Square D model PM8244 meter or equivalent. Factory install meter complete including all required potential and current transformers, control power transformer, finger safe fused switch or breakers for control and potential transformers circuits, current transformer shorting block and any other required accessories. Meter shall have front panel display with ability for user view and scroll through all available voltage, current and power readings as well as historical peaks.

B. Provide Ethernet connectivity and software to allow for user to remotely monitor.

C. Provide factory technician to program and startup meter prior to substantial completion, including installation and setup of monitoring software on Owner's desktop and mobile devices.

2.5 CONTROL POWER

A. Control Circuits: 120-V ac, supplied through secondary disconnecting devices from control-power transformer.

B. Control-Power Fuses: Primary and secondary fuses for current-limiting and overload protection of transformer and fuses for protection of control circuits.

C. Control Wiring: Factory installed, with bundling, lacing, and protection included. Provide flexible conductors for No. 8 AWG and smaller, for conductors across hinges, and for conductors for interconnections between shipping units.

2.6 ACCESSORY COMPONENTS AND FEATURES

A. Portable Test Set: For testing functions of solid-state trip devices without removing from switchboard. Include relay and meter test plugs suitable for testing switchboard meters and switchboard class relays.

2.7 IDENTIFICATION

A. Service Equipment Label: NRTL labeled for use as service equipment for switchboards with one or more service disconnecting and overcurrent protective devices.

B. Permanent Engraved Phenolic Nameplates: Provide white with black lettering ID plates for each circuit.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Receive, inspect, handle, and store switchboards according to NEMA PB 2.1.

B. Examine switchboards before installation. Reject switchboards that are moisture damaged or physically damaged.

C. Examine elements and surfaces to receive switchboards for compliance with installation tolerances and other conditions affecting performance of the Work.

D. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. Install switchboards and accessories according to NEMA PB 2.1.

B. Anchor switchboards to floor slabs or equipment pads and brace to structural wall or ceiling to meet IBC Seismic Design Category D and ASCE 7-16 Ch. 13 requirements.

C. Equipment Mounting: Install switchboards on concrete base, four (4) inch (100-mm) nominal thickness. Comply with requirements for concrete base specified in Division 03 Section Cast-in-Place Concrete or as indicated on the drawings.

1. Install dowel rods to connect concrete base to concrete floor. Unless otherwise indicated, install dowel rods on eighteen (18) inch (450-mm) centers around the full perimeter of concrete base.
2. For supported equipment, install epoxy-coated anchor bolts that extend through concrete base and anchor into structural concrete floor.
3. Place and secure anchorage devices. Use setting drawings, templates, diagrams, instructions, and directions furnished with items to be embedded.
4. Install anchor bolts to elevations required for proper attachment to switchboards.

D. Temporary Lifting Provisions: Remove temporary lifting eyes, channels, and brackets and temporary blocking of moving parts from switchboard units and components.

E. Operating Instructions: Frame and mount the printed basic operating instructions for switchboards, including control and key interlocking sequences and emergency procedures. Fabricate frame of finished wood or metal and cover instructions with clear acrylic plastic. Mount on front of switchboards.

F. Install filler plates in unused spaces of panel-mounted sections.

G. Install overcurrent protective devices, transient voltage suppression devices, and instrumentation.

H. Set field-adjustable switches and circuit-breaker trip ranges as specified in Division 26 Section "Overcurrent Protective Device Coordination Study."

I. Comply with NECA 1.

3.3 CONNECTIONS

A. Comply with requirements for terminating feeder bus specified in Division 26 Section "Enclosed Bus Assemblies." Drawings indicate general arrangement of bus, fittings, and specialties.

B. Comply with requirements for terminating cable trays specified in Division 26 Section "Cable Trays for Electrical Systems." Drawings indicate general arrangement of cable trays, fittings, and specialties.

3.4 IDENTIFICATION

A. Identify field-installed conductors, interconnecting wiring, and components; provide warning signs complying with requirements for identification specified in Division 26 Section "Identification for Electrical Systems."

B. Switchboard Nameplates: Label each switchboard compartment with a nameplate complying with requirements for identification specified in Division 26 Section "Identification for Electrical Systems."

C. Device Nameplates: Label each disconnecting and overcurrent protective device and each meter and control device mounted in compartment doors with a nameplate complying with requirements for identification specified in Division 26 Section "Identification for Electrical Systems."

3.5 FIELD QUALITY CONTROL

A. Testing Agency: Engage a qualified testing agency or factory trained and qualified personnel to perform tests and inspections.

B. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect, test, and adjust components, assemblies, and equipment installations, including connections.

1. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect components, assemblies, and equipment installations, including connections, and to assist in testing.

C. Acceptance Testing Preparation:

1. Test insulation resistance for each switchboard bus, component, connecting supply, feeder, and control circuit.
2. Test continuity of each circuit.

D. Tests and Inspections:

1. Perform each visual and mechanical inspection and electrical test stated in NETA Acceptance Testing Specification. Certify compliance with test parameters.
2. Correct malfunctioning units on-site, where possible, and retest to demonstrate compliance; otherwise, replace with new units and retest.
3. Test and adjust controls, remote monitoring, and safeties. Replace damaged and malfunctioning controls and equipment.

E. Switchboard will be considered defective if it does not pass tests and inspections.

F. Prepare test and inspection reports including a certified report that identifies switchboards included and that describes scanning results. Include notation of deficiencies detected, remedial action taken and observations after remedial action.

3.6 ADJUSTING

A. Adjust moving parts and operable components to function smoothly and lubricate as recommended by manufacturer.

B. Set field-adjustable circuit-breaker trip ranges as specified in Division 26 Section "Overcurrent Protective Device Coordination Study."

3.7 PROTECTION

A. Temporary Heating: Apply temporary heat, to maintain temperature according to manufacturer's written instructions, until switchboard is ready to be energized and placed into service.

3.8 DEMONSTRATION

A. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain switchboards, overcurrent protective devices, instrumentation, and accessories, and to use and reprogram microprocessor-based trip, monitoring, and communication units.

B. END OF SECTION 262413

SECTION 262413.1 – OUTDOOR SWITCHBOARDS WITH UTILITY METERING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this section and all other sections of Division 26.

1.2 SUMMARY

A. Section Includes:

1. Service and distribution switchboards rated 600 V and less.
2. Disconnecting and overcurrent protective devices.
3. Instrumentation.
4. Control power.
5. Accessory components and features.
6. Identification.

1.3 SUBMITTALS

A. Product Data: For each type of switchboard, overcurrent protective device, transient voltage suppression device, ground-fault protector, accessory, and component indicated. Include dimensions and manufacturers' technical data on features, performance, electrical characteristics, ratings, accessories, and finishes.

B. Shop Drawings: For each switchboard and related equipment.

1. Include dimensioned plans, elevations, sections, and details, including required clearances and service space around equipment. Show tabulations of installed devices, equipment features, and ratings.
2. Detail enclosure types for types other than NEMA 250, Type 1.
3. Detail bus configuration, current, and voltage ratings.
4. Detail short-circuit current rating of switchboards and overcurrent protective devices.
5. Include descriptive documentation of optional barriers specified for electrical insulation and isolation.
6. Detail utility company's metering provisions with indication of approval by utility company.

7. Detail features, characteristics, ratings, and factory settings of individual overcurrent protective devices and auxiliary components.
8. Include time-current coordination curves for each type and rating of overcurrent protective device included in switchboards. Include selectable ranges for each type of overcurrent protective device.

9. Include schematic and wiring diagrams for power, signal, and control wiring.

C. Qualification Data: For qualified testing agency.

D. Field Quality-Control Reports:

1. Test procedures used.
2. Test results that comply with requirements.
3. Results of failed tests and corrective action taken to achieve test results that comply with requirements.

E. Operation and Maintenance Data: For switchboards and components to include in emergency, operation, and maintenance manuals. In addition to items specified in Division 01 Section "Operation and Maintenance Data," include the following:

1. Routine maintenance requirements for switchboards and all installed components.
2. Manufacturer's written instructions for testing, maintaining and adjusting overcurrent protective devices.
3. Time-current coordination curves for each type and rating of overcurrent protective device included in switchboards. Include selectable ranges for each type of overcurrent protective device.

1.4 QUALITY ASSURANCE

A. Installer Qualifications: An employer of workers qualified as defined in NEMA PB 2.1 and trained in electrical safety as required by NFPA 70E, or manufacturer’s factory trained and qualified technician.

B. Testing Agency Qualifications: Member Company of NETA or an NRTL.

1. Testing Agency's Field Supervisor: Currently certified by NETA to supervise on-site testing.

C. Source Limitations: Obtain switchboards, overcurrent protective devices, components, and accessories from single source from single manufacturer.

D. Product Selection for Restricted Space: Drawings indicate maximum dimensions for switchboards including clearances between switchboards and adjacent surfaces and other items. Comply with indicated maximum dimensions.

E. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

F. Comply with NEMA PB 2.

G. Comply with NFPA 70.

H. Comply with UL 891.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Deliver switchboards in sections or lengths that can be moved past obstructions in delivery path.

B. Remove loose packing and flammable materials from inside switchboards.

C. Handle and prepare switchboards for installation according to NEMA PB 2.1.

1.6 PROJECT CONDITIONS

A. Installation Pathway: Remove and replace access fencing, doors, lift-out panels, and structures to provide pathway for moving switchboards into place.

B. Environmental Limitations:

1. Switchboards shall be suitable for installation in an outdoor environment.
2. Rate equipment for continuous operation under the following conditions unless otherwise indicated:

   a. Ambient Temperature: Between -10^{0}F (-23^{0}C) and 104^{0}F (40^{0}C).
   b. Altitude: Not exceeding 6600 feet (2000 m).

C. Interruption of Existing Electric Service: Do not interrupt electric service to facilities occupied by Owner or others unless permitted under the following conditions and then only after arranging to provide temporary electric service according to requirements indicated:

1. Notify the Owner's representative, in writing, no fewer than 10 days in advance of proposed interruption of electric service.
2. Indicate method of providing temporary electric service.
3. Do not proceed with interruption of electric service without the Owner's written permission.
4. Comply with NFPA 70E.

1.7 COORDINATION

A. Coordinate layout and installation of switchboards and components with other construction that penetrates walls or is supported by them, including electrical and other types of equipment, raceways, piping, encumbrances to workspace clearance requirements, and adjacent surfaces. Maintain required workspace clearances and required clearances for equipment access doors and panels.

B. Coordinate sizes and locations of concrete pads with actual equipment provided.

1.8 EXTRA MATERIALS

A. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Potential Transformer Fuses: Equal to 10% of quantity installed for each size and type, but no fewer than two (2) of each size and type.
2. Control-Power Fuses: Equal to 10 % of quantity installed for each size and type, but no fewer than two (2) of each size and type.
3. Fuses and Fusible Devices for Fused Circuit Breakers: Equal to 10 % of quantity installed for each size and type, but no fewer than three (3) of each size and type.
4. Fuses for Fused Switches: Equal to 10 % of quantity installed for each size and type, but no fewer than three (3) of each size and type.
5. Fuses for Fused Power-Circuit Devices: Equal to 10 % of quantity installed for each size and type, but no fewer than three (3) of each size and type.
6. Indicating Lights: Equal to 10 % of quantity installed for each size and type, but no less than one of each size and type.
7. Provide Fuse cabinet, location coordinated with Owner.

1.9 SWITCHBOARD WARRANTY/GUARANTEE

A. Provide manufacturer's standard form in which manufacturer agrees to repair or replace any devices that fail in materials or workmanship within specified warranty period.

B. Warranty Period: Two (2) years from date of shipment or one (1) year from the date of project Substantial Completion, whichever is later.

PART 2 - PRODUCTS

2.1 MANUFACTURED UNITS

A. Basis-of-Design Product: Square D QED-2 style two section, outdoor, pad mounted switchboard. Subject to compliance with requirements, provide product by one (1) of the following:

1. Square D; a brand of Schneider Electric, QED-2.
2. Eaton Electrical Inc.; Cutler-Hammer Business Unit.
3. ABB, Reliagear SB

B. Front- and Side-Accessible Outdoor, Pad Mounted, Two Section Switchboard:

1. Incoming Utility Section: Bottom Feed. Prepped for Power Company CT's and PT's.
2. Main Circuit Breaker Section: Bottom Feed.
3. Main Device in Main Circuit Breaker Section: Fixed, individually mounted.
4. Branch Devices: None.
5. Sections front and rear aligned.

C. Nominal System Voltage: 480Y/277 V or as indicated on the contract drawings.

D. Main-Bus Continuous: 2000A or as indicated on the contract drawings.

E. Outdoor Enclosures: Steel, NEMA Type 3R.

F. Enclosure Finish for Outdoor Units: Factory-applied finish in manufacturer's ANSI 49 gray finish over a rust-inhibiting primer on treated metal surface.

G. Customer Metering Compartment: None.

H. Utility Section and Metering Compartment: Utility Section shall be on the left when facing the front of the switchboard. Refer to Appendix A of this specification, excerpts from PP&L's 2022 Electric Service Manual Requirements (ESR), as well as the entire ESR document available at this link: https://www.pacificpower.net/content/dam/pcorp/documents/en/pp-rmp/electric-service-requirements/ESR_FULL.pdf and comply with the requirements for 2000A Switchboard Metering.

I. Bus Transition and Incoming Pull Sections: Matched and aligned with basic switchboard.

J. Hinged Front Panels: Allow access to circuit breaker, metering, accessory, and blank compartments.

K. Buses and Connections: Three-phase, four-wire, unless otherwise indicated.

1. Phase- and Neutral-Bus Material: Hard-drawn copper of 98 percent conductivity, with copper feeder circuit-breaker line connections.
2. Ground Bus: 1/4-by-2-inch hard-drawn copper of 98 % conductivity, equipped with mechanical connectors for feeder and branch-circuit ground conductors. For busway feeders, extend insulated equipment grounding cable to busway ground connection and support cable at intervals in vertical run.
3. Main Phase Buses and Equipment Ground Buses: Uniform capacity for entire length of switchboard's main and distribution sections. Provide for future extensions.
4. Neutral Buses: 100 % of the ampacity of phase buses unless otherwise indicated, equipped with mechanical connectors for outgoing circuit neutral cables. Brace bus extensions for busway feeder neutral bus.
5. Main bonding jumper: Provide main bonding jumper for service entrance or separately derived systems per NEC requirements.
6. Braced for 65,000A, symmetrical at 480V.

2.2 DISCONNECTING AND OVERCURRENT PROTECTIVE DEVICES

A. Main Insulated-Case Circuit Breaker (ICCB): 100% rated, sealed, insulated-case power circuit breaker with interrupting capacity rating to meet available fault current. Provide ICCB's for all frame sizes 1000A and larger. Basis of Design: Square D MasterPact NW. Interrupting rating of 65kA minimum at 480V. The Main Circuit Breaker section shall be on the right when facing the front of the switchboard.

1. Fixed circuit-breaker mounting.

2. Line side of main breaker bussed from incoming Utility Section. Load side of main circuit breaker prepped for feeder cabling to downstream equipment. Load side feeder will be comprised of six (6) parallel sets of #500 kcmil copper per phase and neutral plus equipment grounding conductors.
3. Two-step, stored-energy closing.
4. Standard-function, microprocessor-based trip units with interchangeable rating plug, trip indicators, and the following field-adjustable settings:

   a. Long time, short time and instantaneous time adjustments.
   b. Ground-fault pickup level, time delay, and I^2t response for circuit breakers 1000A and larger.
5. Remote trip indication and control.
6. Communication Capability: Integral communication module.
7. Control Voltage: 120-V ac.

2.3 INSTRUMENTATION

A. Prep incoming Utility and metering equipment section in accordance with Pacific Power and Light (PP&L) requirements.

B. Provide removable buss section for each phase to allow for field installation of PP&L furnished current transformers. Make provisions for potential transformers per PP&L requirements.

C. Refer to Appendix A of this specification, excerpts from PP&L's 2022 Electric Service Manual Requirements (ESR), as well as the entire ESR document available at this link: https://www.pacificpower.net/content/dam/pcorp/documents/en/pp-rmp/electric-service-requirements/ESR_FULL.pdf and comply with the requirements for 2000A Switchboard Metering.

2.4 CONTROL POWER

A. Control Circuits: 120-V ac, supplied through secondary disconnecting devices from control-power transformer.

B. Control-Power Fuses: Primary and secondary fuses for current-limiting and overload protection of transformer and fuses for protection of control circuits.

C. Control Wiring: Factory installed, with bundling, lacing, and protection included. Provide flexible conductors for No. 8 AWG and smaller, for conductors across hinges, and for conductors for interconnections between shipping units.

2.5 IDENTIFICATION

A. Service Equipment Label: NRTL labeled for use as service equipment for switchboards.

B. Permanent Engraved Phenolic Nameplates: Provide white with black lettering ID plates for each circuit.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Receive, inspect, handle, and store switchboards according to NEMA PB 2.1.

B. Examine switchboards before installation. Reject switchboards that are moisture damaged or physically damaged.

C. Examine elements and surfaces to receive switchboards for compliance with installation tolerances and other conditions affecting performance of the Work.

D. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. Install switchboard and accessories according to NEMA PB 2.1.

B. Equipment Mounting: Install switchboard on concrete base, four (4) inch (100-mm) nominal thickness. Comply with PP&L requirements for switchboard metering equipment..

1. Install dowel rods to connect concrete base to concrete floor. Unless otherwise indicated, install dowel rods on eighteen (18) inch (450-mm) centers around the full perimeter of concrete base.
2. For supported equipment, install epoxy-coated anchor bolts that extend through concrete base and anchor into structural concrete floor.
3. Place and secure anchorage devices. Use setting drawings, templates, diagrams, instructions, and directions furnished with items to be embedded.
4. Install anchor bolts to elevations required for proper attachment to switchboards.

C. Temporary Lifting Provisions: Remove temporary lifting eyes, channels, and brackets and temporary blocking of moving parts from switchboard units and components.

D. Operating Instructions: Mount the printed basic operating instructions for switchboards, including control and key interlocking sequences and emergency procedures on the front of the switchboard. Material shall be impervious to weather.

E. Install filler plates in unused spaces of panel-mounted sections.

F. Install overcurrent protective devices, transient voltage suppression devices, and instrumentation.

G. Set field-adjustable switches and circuit-breaker trip ranges as specified in Division 26 Section "Overcurrent Protective Device Coordination Study."

H. Comply with NECA 1.

3.3 IDENTIFICATION

A. Identify field-installed conductors, interconnecting wiring, and components; provide warning signs complying with requirements for identification specified in Division 26 Section "Identification for Electrical Systems."

B. Switchboard Nameplates: Label each switchboard compartment with a nameplate complying with requirements for identification specified in Division 26 Section "Identification for Electrical Systems."

C. Device Nameplates: Label each disconnecting and overcurrent protective device and each meter and control device mounted in compartment doors with a nameplate complying with requirements for identification specified in Division 26 Section "Identification for Electrical Systems."

3.4 FIELD QUALITY CONTROL

A. Testing Agency: Engage a qualified testing agency or factory trained and qualified personnel to perform tests and inspections.

B. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect, test, and adjust components, assemblies, and equipment installations, including connections.

1. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect components, assemblies, and equipment installations, including connections, and to assist in testing.

C. Acceptance Testing Preparation:

1. Test insulation resistance for each switchboard bus, component, connecting supply, feeder, and control circuit.
2. Test continuity of each circuit.

D. Tests and Inspections:

1. Perform each visual and mechanical inspection and electrical test stated in NETA Acceptance Testing Specification. Certify compliance with test parameters.
2. Correct malfunctioning units on-site, where possible, and retest to demonstrate compliance; otherwise, replace with new units and retest.
3. Test and adjust controls, remote monitoring, and safeties. Replace damaged and malfunctioning controls and equipment.

E. Switchboard will be considered defective if it does not pass tests and inspections.

F. Prepare test and inspection reports including a certified report that identifies switchboards included and that describes scanning results. Include notation of deficiencies detected, remedial action taken and observations after remedial action.

3.5 ADJUSTING

A. Adjust moving parts and operable components to function smoothly and lubricate as recommended by manufacturer.

B. Set field-adjustable circuit-breaker trip ranges as specified in Division 26 Section "Overcurrent Protective Device Coordination Study."

3.6 PROTECTION

A. Temporary Heating: Apply temporary heat, to maintain temperature according to manufacturer's written instructions, until switchboard is ready to be energized and placed into service.

3.7 DEMONSTRATION

A. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain switchboards, overcurrent protective devices, instrumentation, and accessories, and to use and reprogram microprocessor-based trip, monitoring, and communication units.

B. END OF SECTION 262413.1

SECTION 262416 - PANELBOARDS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this section and all other sections of Division 26.

1.2 SUMMARY

A. Section Includes:

1. Distribution panelboards.
2. Lighting and appliance branch-circuit panelboards.

1.3 DEFINITIONS

A. VPR: Voltage Protection Rating.

B. SPD: Surge Protective Device.

1.4 SUBMITTALS

A. Product Data: For each type of panelboard, switching and overcurrent protective device, surge protective device, accessory, and component indicated. Include dimensions and manufacturers' technical data on features, performance, electrical characteristics, ratings, and finishes.

B. Shop Drawings: For each panelboard and related equipment.

1. Include dimensioned plans, elevations, sections, and details. Show tabulations of installed devices, equipment features, and ratings.
2. Detail enclosure types and details for types other than NEMA 250, Type 1.
3. Detail bus configuration, current, and voltage ratings.

4. Short-circuit current rating of panelboards and overcurrent protective devices.
5. Detail features, characteristics, ratings, and factory settings of individual overcurrent protective devices and auxiliary components.
6. Include wiring diagrams for power, signal, and control wiring.
7. Include time-current coordination curves for each type and rating of overcurrent protective device included in panelboards. Also include selectable ranges for each type of overcurrent protective device.

C. Qualification Data: For qualified testing agency.

D. Field Quality-Control Reports:

1. Test procedures used.
2. Test results that comply with requirements.
3. Results of failed tests and corrective action taken to achieve test results that comply with requirements.

E. Panelboard Schedules: For installation in panel boards.

F. Operation and Maintenance Data: For panel boards, and components to include in emergency, operation, and maintenance manuals. In addition to items specified in Division 01 Section "Operation and Maintenance Data," include the following:

1. Manufacturer's written instructions for maintaining, testing and adjusting overcurrent protective devices.
2. Time-current curves, including selectable ranges for each type of overcurrent protective device that allows adjustments.

1.5 QUALITY ASSURANCE

A. Testing Agency Qualifications: Member company of NETA or an NRTL.

1. Testing Agency's Field Supervisor: Currently certified by NETA to supervise on-site testing.

B. Source Limitations: Obtain panelboards, overcurrent protective devices, components, and accessories from single source from single manufacturer.

C. Product Selection for Restricted Space: Drawings indicate maximum dimensions for panelboards including clearances between panelboards and adjacent surfaces and other items. Comply with indicated maximum dimensions.

D. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

E. Comply with NEMA PB 1.

F. Comply with NFPA 70.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Remove loose packing and flammable materials from inside panelboards; Store in a clean, dry location.

B. Handle and prepare panelboards for installation according to NEMA PB 1.

1.7 PROJECT CONDITIONS

A. Environmental Limitations:

1. Do not deliver or install panelboards until spaces are enclosed and weathertight, wet work in spaces is complete and dry, work above panelboards is complete, and temporary HVAC system is operating and maintaining ambient temperature and humidity conditions at occupancy levels during the remainder of the construction period.
2. Rate equipment for continuous operation under the following conditions unless otherwise indicated:

   a. Ambient Temperature: Not exceeding 23^0F to plus 104^0F.
   b. Altitude: Not exceeding 6600 feet.

B. Service Conditions: NEMA PB 1, usual service conditions, as follows:

1. Ambient temperatures within limits specified.
2. Altitude not exceeding 6600 feet.

C. Interruption of Existing Electric Service (when applicable): Do not interrupt electric service to facilities occupied by Owner or others unless permitted under

the following conditions and then only after arranging to provide temporary electric service according to requirements indicated:

1. Notify Owner, in writing, no fewer than 10 days in advance of proposed interruption of electric service.
2. Do not proceed with interruption of electric service without Owner's written permission.
3. Comply with NFPA 70E.
4. In existing facilities, the operation of electrical panels or power switches; required to achieve an outage must be accomplished by Owner's personnel only. Unauthorized operation of electric panels, power switches, by contractors their personnel will result in extremely serious consequences for which the contractor will be held accountable.

5. Mandatory Requirements: The following requirements are mandatory:

   a. Protective Equipment: Electricians must use full protective equipment (i.e., certified and tested insulating material to cover exposed energized electrical components, certified and tested insulated tools, etc.) while working on energized systems in accordance with NFPA 70E.

   b. Owner Energized Work Permit: An Owner Energized Work Permit is required for any work on energized circuits or equipment. Permit must be approved by OWNER Department of Operations and Maintenance prior to performing energized work. Submit the work permit with the outage request.

6. On applicable projects, electrical contractor shall identify existing circuits and existing panels for the renovation area and trace and identify existing circuits. Identifying and tracing of the circuits shall be done with machinery and appropriate safety gear. Should an outage become necessary, it will need to be requested a minimum of ten (10) working days in advance through the Owner's Project Manager.

1.8 COORDINATION

A. Coordinate layout and installation of panelboards and components with other construction that penetrates walls or is supported by them, including electrical and other types of equipment, raceways, piping, encumbrances to workspace

clearance requirements, and adjacent surfaces. Maintain required workspace clearances and required clearances for equipment access doors and panels.

1.9 WARRANTY

A. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace devices that fail in materials or workmanship within specified warranty period.

1. Warranty Period: Two (2) years from date of shipment or one (1) year from the date of project Substantial Completion, whichever is later.

1.10 EXTRA MATERIALS

A. Coordinate with Division 26 Section "Fuses" for quantities of spare fuses and spare-fuse cabinet to be provided.

B. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Keys: Two spares for each type of panelboard cabinet lock.
2. Circuit Breakers Including GFCI and Ground Fault Equipment Protection (GFEP) Types: Two spares for each panelboard.
3. Fuses for Fused Switches: Equal to 10 % of quantity installed for each size and type, but no fewer than three (3) of each size and type.
4. Fuses for Fused Power-Circuit Devices: Equal to 10 % of quantity installed for each size and type, but no fewer than three (3) of each size and type.
5. Furnish and install fuse cabinet.

1.11 WARRANTY/GUARANTEE

A. See Division 26 Specification Section “Basic Electrical Requirements’ for warranty and guarantee requirements.

PART 2 - PRODUCTS

2.1 GENERAL REQUIREMENTS FOR PANELBOARDS

A. Enclosures: Flush- and surface-mounted cabinets.

1. Rated for environmental conditions at installed location.

   a. Indoor Dry and Clean Locations: NEMA 250, Type 1.
   b. Outdoor Locations: NEMA 250, Type 3R.
   c. Wash-Down Areas: NEMA 250, Type 4X stainless steel.
   d. Other Wet or Damp Indoor Locations: NEMA 250, Type 4X non-metallic.
   e. Indoor Locations Subject to Dust, Falling Dirt, and Dripping Noncorrosive Liquids: NEMA 250, Type 12.

2. Lockable Hinged Front Cover - Entire front trim hinged to box and with standard door within hinged trim cover i.e. “hinged door-in-door” style.
3. Gutter Extension and Barrier: Same gage and finish as panelboard enclosure; integral with enclosure body. Arrange to isolate individual panel sections.
4. Finishes:

   a. Panels and Trim: Steel, factory finished immediately after cleaning and pretreating with manufacturer's standard two (2) coat, baked-on finish consisting of prime coat and thermosetting topcoat.
   b. Back Boxes: Galvanized steel.

5. Directory Card: Inside panelboard door, mounted in transparent card holder.

B. Incoming Mains Location: Top or bottom fed as needed.

C. Phase, Neutral, and Ground Buses:

1. Material: Copper or aluminum.
2. Equipment Ground Bus: Adequate for feeder and branch-circuit equipment grounding conductors; bonded to box.
3. Isolated Ground Bus: Adequate for branch-circuit isolated ground conductors; insulated from box where indicated on the panel schedule.

4. Extra-Capacity Neutral Bus: Neutral bus rated 200 % of phase bus and UL listed as suitable for nonlinear loads where indicated on the panel schedule.

D. Conductor Connectors: Suitable for use with conductor material and sizes.

1. Material: Suitable for aluminum or copper conductors.
2. Main and Neutral Lugs: Mechanical type.
3. Ground Lugs and Bus-Configured Terminators: Mechanical type.
4. Feed-Through Lugs: Mechanical type, suitable for use with conductor material. Locate at opposite end of bus from incoming lugs or main device.
5. Subfeed (Double) Lugs: Mechanical type suitable for use with conductor material. Locate at same end of bus as incoming lugs or main device as needed.
6. Gutter-Tap Lugs: Mechanical type suitable for use with conductor material. Locate at same end of bus as incoming lugs or main device.
7. Extra-Capacity Neutral Lugs: Rated 200 % of phase lugs mounted on extra-capacity neutral bus where indicated on the contract documents.

E. Service Equipment Label (where applicable): NRTL labeled for use as service equipment for panelboards or load centers with one or more main service disconnecting and overcurrent protective devices.

F. Future Devices: Mounting brackets, bus connections, filler plates, and necessary appurtenances required for future installation of devices.

G. Panelboard Short-Circuit Current Rating: Fully rated to interrupt symmetrical short-circuit current available at terminals.

H. Branch Circuit Panelboard Sizing and Capacity: As indicated on the drawings.

I. Wall-mounted Distribution Panels/Boards: Provide distribution panels with a minimum 63” of breaker mounting space, (i.e. sufficient space to hold fourteen 225A breakers) and with the minimum breaker capacity or prepared spaces for installing 400A branch circuit breakers in the future. Provide greater than 63” of breaker mounting space if required to accommodate all the breakers and prepared spaces indicated on the one-line diagrams.

2.2 DISTRIBUTION PANELBOARDS

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
   1. Square D; a brand of Schneider Electric., I-Line series (Basis of Design)
   2. Eaton Electrical Inc.; Cutler-Hammer Business Unit – Molded case drawout type.
   3. ABB/General Electric Company; GE Consumer & Industrial - Electrical Distribution. neXT series.

B. Panelboards: NEMA PB 1, power and feeder distribution type.

C. Mains: Circuit breaker or main lugs only as indicated on the contract documents.

D. 65kAIC minimum at 480V or as indicated on the drawings.

E. Square D PowerPact B frame (Basis of Design), fixed thermal magnetic feeder breakers for 15 through 90A frame unless indicated to be electronic trip LI or LSI on the drawings or approved equal by listed manufacturer.

F. Square D PowerPact H frame (Basis of Design), electronic trip LSI feeder breakers for 100 through 150A frame or approved equal by listed manufacturer.

G. Square D PowerPact J frame (Basis of Design), electronic trip LSI feeder breakers for over 150A through 250A frame or approved equal by listed manufacturer.

H. Square D PowerPact L frame (Basis of Design), electronic trip LSI feeder breakers for over 250A through 400A frame or approved equal by listed manufacturer.

I. Mounting: Panels shall be constructed such that all feeder breakers may be removed or added without tools making contact with the line side bussing/connectors. Plug on or draw-out connection shall be used to support minimal installation/downtime and fewer points of continued maintenance.

J. Future spaces: To minimize future costs and simplify procurement, all future spaces called out on the single line diagram or panel schedule shall be fully bussed and contain all mounting assemblies and trim assemblies such that filling those future spaces shall only require purchasing the breaker itself.

2.3 LIGHTING AND APPLIANCE BRANCH-CIRCUIT PANELBOARDS

A. Manufacturers: Subject to compliance with requirements, provide products by one (1) of the following:
   1. Square D; a brand of Schneider Electric (Basis of Design). NQ for 240 and 208V; NF for 480V applications.
   2. Eaton Electrical Inc.; Cutler-Hammer Business Unit.
   3. ABB/General Electric Company; GE Consumer & Industrial - Electrical Distribution.

B. Panelboards: NEMA PB 1, lighting and appliance branch-circuit type.

C. Mains: Circuit breaker or main lugs only as designated on contract drawings.

D. Branch Overcurrent Protective Devices: Bolt-on circuit breakers, replaceable without disturbing adjacent units.

E. Doors: Lockable Hinged Front Cover: Entire front trim hinged to box and with standard door within hinged trim cover i.e. “hinged door-in-door” style

2.4 DISCONNECTING AND OVERCURRENT PROTECTIVE DEVICES

A. Manufacturers: Subject to compliance with requirements, provide products by one (1) of the following:

   1. Eaton Electrical Inc.; Cutler-Hammer Business Unit.
   2. Square D; a brand of Schneider Electric.
   3. General Electric Company; GE Consumer & Industrial - Electrical Distribution.

B. Molded-Case Circuit Breaker (MCCB): Comply with UL 489, with interrupting capacity to meet available fault currents and as indicated on the contract documents.
   1. As indicated in section 2.2 for Power Distribution Panelboards and as specified below for Lighting and Appliance panelboards.
   2. Thermal-Magnetic Circuit Breakers: Inverse time-current element for low-level overloads, and instantaneous magnetic trip element for short circuits.
   3. Adjustable Instantaneous-Trip Circuit Breakers: Magnetic trip element with front-mounted, field-adjustable trip setting.
   4. Electronic trip circuit breakers: Frame sizes 100 through 400 A main breakers in Lighting and Appliance panelboard, rms sensing; field-

replaceable rating plug or field-replicable electronic trip; and the following field-adjustable settings:

a. Instantaneous trip.
b. Long- and short-time pickup levels.
c. Long- and short-time time delay adjustments.

5. GFCI Circuit Breakers: Single- and two-pole configurations with Class A ground-fault protection (6-mA trip).
6. Molded-Case Circuit-Breaker (MCCB) Features and Accessories:

a. Standard frame sizes, trip ratings, and number of poles.
b. Lugs: Mechanical style, suitable for number, size, trip ratings, and conductor materials.
c. Application Listing: Appropriate for application; Type SWD for switching fluorescent lighting loads; Type HID for feeding fluorescent and high-intensity discharge (HID) lighting circuits.
d. Ground-Fault Protection: Integrally mounted relay and trip unit with adjustable pickup and time-delay settings, push-to-test feature, and ground-fault indicator.
e. Multipole units enclosed in a single housing or factory assembled to operate as a single unit.

C. Fused Switch: NEMA KS 1, Type HD; clips to accommodate specified fuses; lockable handle with interrupting capacity to meet available fault currents and as indicated on the contract documents

1. Fuses, and Spare-Fuse Cabinet: Comply with requirements specified in Division 26 Section "Fuses."
2. Fused Switch Features and Accessories: Standard ampere ratings and number of poles.
   Auxiliary Contacts: One normally open and normally closed contact(s) that operate with switch handle operation.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Receive, inspect, handle, and store panelboards according to NEMA PB 1.1.

B. Examine panelboards before installation. Reject panelboards that are damaged or rusted or have been subjected to water saturation.

C. Examine elements and surfaces to receive panelboards for compliance with installation tolerances and other conditions affecting performance of the Work.

D. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. Install panelboards and accessories according to NEMA PB 1.1.

B. Temporary Lifting Provisions: Remove temporary lifting eyes, channels, and brackets and temporary blocking of moving parts from panelboards.

C. Mount top of trim 72 inches above finished floor unless otherwise indicated or as required due to overall tub dimensions.

D. Mount panelboard cabinet plumb and rigid without distortion of box. Mount recessed panelboards with fronts uniformly flush with wall finish and mating with back box.

E. Install overcurrent protective devices and controllers not already factory installed.

1. Set field-adjustable, circuit-breaker trip ranges.

2. Breakers settings shall be furnished to the Contractor by the Engineer.

F. Install filler plates in unused spaces.

G. On recessed panel installations, stub four (4) one (1) inch empty conduits from panelboard into accessible ceiling space or space designated to be ceiling space in the future. Stub four (4) one (1) inch empty conduits into raised floor space or below slab not on grade.

H. Arrange conductors in gutters into groups and bundle and wrap with wire ties after completing load balancing.

I. Comply with NECA 1.

3.3 IDENTIFICATION

A. Identify field-installed conductors, interconnecting wiring, and components; provide warning signs complying with Division 26 Section "Identification for Electrical Systems."

B. Create a directory to indicate installed circuit loads after balancing panelboard loads; incorporate Owner's final room designations. Obtain approval before installing. Use a computer or typewriter to create directory; handwritten directories are not acceptable. Contractor shall also provide an electronic copy of new and/or revised schedule in excel or word format to Operations & Maintenance work management system thru Director of Operations & Maintenance.

C. Panelboard Nameplates: Label each panelboard with a nameplate complying with requirements for identification specified in Division 26 Section "Identification for Electrical Systems."

D. Device Nameplates: Label each branch circuit device in distribution panelboards with a nameplate complying with requirements for identification specified in Division 26 Section "Identification for Electrical Systems."

3.4 FIELD QUALITY CONTROL

A. Acceptance Testing:

1. Test insulation resistance for each panelboard bus, component, connecting supply, feeder, and control circuit.
2. Test continuity of each circuit.

B. Tests and Inspections:

1. Perform each visual and mechanical inspection and electrical test stated in NETA Acceptance Testing Specification. Certify compliance with test parameters.
2. Correct malfunctioning units on-site, where possible, and retest to demonstrate compliance; otherwise, replace with new units and retest.
3. Comply with NFPA 70E.

C. Panelboards will be considered defective if they do not pass tests and inspections.

D. Prepare test and inspection reports, including a certified report that identifies panelboards included and that describes scanning results. Include notation of deficiencies detected, remedial action taken and observations after remedial action.

3.5 ADJUSTING

A. Adjust moving parts and operable component to function smoothly and lubricate as recommended by manufacturer.

B. Set field-adjustable circuit-breaker trip ranges as specified in Division 26 Section "Overcurrent Protective Device Coordination Study."

3.6 PROTECTION

A. Temporary Heating: Apply temporary heat to maintain temperature according to manufacturer's written instructions.

END OF SECTION 262416

SECTION 26 24 19

INTELLIGENT MOTOR CONTROL CENTERS

PART 1 - GENERAL

1.1 SUMMARY

A. Scope: Provide labor, material, equipment, related services, and supervision required, including, but not limited to, manufacturing, fabrication, configuration and installation for low voltage motor control centers as required for the complete performance of the Work, as shown on the Drawings, as specified herein.

B. Related Sections: Related sections include, but shall not be limited to, the following:

1. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.
2. Applicable general requirements for electrical Work specified within Division 26 Specification Sections apply to this Section.
3. The following MCC information is typically depicted on the Drawings: bus configuration, bus ratings, interrupting ratings, component size and type, power line and feeder connections, application specific unit control wiring, elevation and footprint, etc. Where not shown on or able to be derived from the Drawings, the minimum requirements specified herein shall be provided.
4. Refer to the following specifications for additional requirements:
   a. Section 26 29 23 Variable Frequency Motor Controllers

1.2 REFERENCES

A. General Publications: The publications listed below form a part of this Specification to the extent referenced. The publications are referred to in the text by the basic designation only. The edition/revision of the referenced publications shall be the latest date as of the date of the Contract Documents, unless otherwise specified.

1. American National Standards Institute (ANSI)
   a. ANSI Z55.1, “Gray Finishes for Industrial Apparatus and Equipment”
2. ASTM International (ASTM)
   a. ASTM B117, “Standard Practice for Operating Salt Spray (Fog) Apparatus”
3. Institute of Electrical and Electronics Engineers, Inc. (IEEE)
   a. IEEE 519, “Guide for Harmonic Control and Reactive Compensation of Static Power Converters

b. IEEE 1584, “Guide for Performing Arc Flash Hazard Calculations”

c. IEEE C37-20-7, “Guide for testing metal-enclosed switchgear rated up to 52kV for internal arcing faults"

4. National Fire Protection Agency (NFPA)
   a. NFPA 70, “National Electrical Code (NEC)”
   b. NFPA 70E, “Standard for Electrical Safety in the Workplace”
   c. NFPA 79, “Electrical Standard for Industrial Machinery”
5. National Electrical Manufacturer Association (NEMA)
   a. NEMA 250, “Enclosures for Electrical Equipment”
   b. NEMA ICS 18, “Motor Control Centers”
6. Underwriters Laboratories, Inc. (UL)
   a. UL 50, “Enclosures for Electrical Equipment, Non-Environmental Considerations”
   b. UL 498, “Standard for Attachment Plugs and Receptacles”
   c. UL 508, “Standard for Industrial Control Equipment”
   d. UL 508A, “Standard for Industrial Control Panels”
   e. UL 508C, “Standard for Power Conversion Equipment”
   f. UL 845, “Motor Control Centers”

1.3 DEFINITIONS

A. Unless specifically defined within the Contract Documents, the words or acronyms contained within this specification shall be as defined within, or by the references listed within this specification, the Contract Documents, or, if not listed by either, by common industry practice.

1. AFIE: Arc Flash Incident Energy
2. iMCC: Intelligent Motor Control Center
3. IMPR: Intelligent Motor Protection Relay
4. MCC: Motor Control Center
5. MCCB: Molded Case Circuit Breaker
6. MOV: Metal Oxide Varistors
7. OIT: Operator Interface Terminal
8. PIV: Peak Inverse Voltage
9. RVSS: Reduced Voltage Solid State motor controller
10. SCR: Silicon-Controlled Rectifier
11. SNMP: Simple Network Management Protocol
12. VFD, AFD: Variable Frequency Drive

1.4 SUBMITTALS

A. General: Submittals shall be in accordance with this section.

1. Submit sufficient information to determine compliance with the Contract Documents. Identify submittal data with the specific equipment tags and/or service descriptions to which they pertain. Submittal data shall be clearly marked to identify the specific model numbers, options, and features of equipment and work proposed.

2. Submit required product data and shop drawings specific to each product and accessory proposed. In addition, include the following information:

a. Equipment assembly. Indicate dimensions, shipping section dimensions, weights, foundation requirements, required clearances, location and size of each field connection, and mounting and installation instructions.

b. Include elementary and interconnection diagrams for power, signal, control, and communications wiring. Diagrams shall provide the minimum detail as shown for drawings in the appendix of NFPA 79.

B. Harmonic Analysis Report **NOT REQUIRED ON THIS PROJECT**: Through a manufacturer selected combination of the acceptable and specified AFE and passively filtered type VFD's as well as standard VFD's without any integral to the VFD filtering plus, as required and determined by the manufacturer, active harmonic filtering in each MCC, manufacturers are required to meet the total harmonic distortion and power factor requirements of this specification for each MCC. Provide project-specific calculations and manufacturer's statement of compliance with IEEE 519, latest revision and this specification with the initial for-review submittal package. Refer to Section 26 24 29 Variable Frequency Motor Controllers for additional information and requirements.

1. Harmonic Calculations: The MCC manufacturer shall supply harmonic calculations made in accordance with the latest version of IEEE 519 showing the specified THVD, line notching and the specified THCD limits are met. Calculations shall assume worst case system conditions. The calculations shall include:

a. All input data and assumptions

b. Explanation of method used to perform the analysis.

c. All calculations and computer printouts used in the analysis, including input documentation.

d. A system impedance diagram based on Electrical single-line diagram.

e. All calculations shall be made in accordance with IEEE 519 with all drives at 100% speed. The point of common coupling shall be the line side each MCC. These calculations shall also be done with all drives running.

f. A detailed description of the tests, procedures and supporting calculations required to substantiate the installed systems compliance with the THD limits.

1) The description shall include information on the proposed test equipment and test conditions.

g. Each point of common coupling shall be defined as the line side of each MCC. At the point of common coupling, the following numbers shall meet with the maximum load on the MCC.

h. Submittals without calculations will not be reviewed.

C. Operation & Maintenance (O&M) manuals shall be provided in accordance with the minimum requirements specified in this Section. Operation and Maintenance Data, Electrical Requirements and additional requirements specified herein.

1. Submit required Operations & Maintenance data specific to each product and accessory proposed. In addition, include the following information:

a. Manufacturer, supplier, support, and repair center specific contact information.

b. Manufacturer's standard operation and maintenance data assembled for each size and type of equipment furnished.

c. All construction, installation, schematic, and wiring diagrams updated to an as-installed and commissioned state.

d. All configured settings/parameters for adjustable components updated to an as-installed and commissioned stated if different from the factory default. Electronic copies of configuration files shall be provided, on media acceptable to the Owner (e.g. CD, USB stick, etc.), where these configurations can be saved as an electronic file for future upload into replaced or repaired components.

e. List of furnished and recommended spare parts.

f. Statement of standard Warranty.

1.5 QUALITY ASSURANCE

A. Manufacturer Qualifications: Manufacturer shall be a firm engaged in the manufacture of specified products of types and sizes required, and whose products have been in satisfactory use in similar service for a minimum of ten years.

B. All work performed and all materials used shall be in accordance with the National Electrical Code and with applicable local regulations and ordinances. Equipment assemblies, materials, and equipment shall be listed and labeled by Underwriter's Laboratories or by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.

1.6 DELIVERY, STORAGE AND HANDLING

A. Prior to delivery to the Project site, ensure that suitable storage space is available to store materials in a well-ventilated area protected from weather, moisture, soiling, extreme temperatures, humidity, and corrosive atmospheres.

B. Deliver materials to the Project site in supplier's or manufacturer's original wrappings and containers, labeled with supplier's or manufacturer's name, material or product brand name, and equipment tag number or service name as identified within the Contract Documents.

C. Coordinate delivery and receiving requirements with Owner's authorized representative.

1.7 WARRANTY

A. Warranty Period: Two (2) years from date of shipment or one (1) year from the date of project Substantial Completion, whichever is later.

1.8 SPECIAL TOOLS AND SPARE PARTS

A. Spare parts shall be provided for each type and size of unit installed. At a minimum, the following shall be provided:

1. Provide the minimum spare parts recommended by the manufacturer.
2. 1 set of each type of power and control fuse installed within equipment
3. 1 control power transformer of each size installed
4. 2 control relays of each size and type installed. Provide an additional spare for every 25 of a specific type installed.
5. 2 indicating pilot lights, if not LED type, of each color and type installed
6. 1 control operators (e.g. selector switch, pushbutton, etc.) complete with contact blocks for each type, color, size, installed
7. 1 intelligent motor protection relay of each type and size installed.

B. Any manufacturer specific special tool, not normally found in an electrician's toolbox, required to remove and install recommended or furnished spare parts shall be furnished. At a minimum the following shall be provided:

1. Provide PC-based configuration software tool and a minimum of one communication interface cable for each type of cable required to connect a PC-based computer to the devices specified herein for configuration and programming.
2. Electronic configuration files, in a media format acceptable by the Owner (e.g. CD, USB stick, etc.), updated to an as-installed and commissioned state.

C. Spare parts shall be properly marked and packaged for long term storage. Printed circuit boards shall be provided in separate anti-static containers.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Products: Intelligent Motor Control Centers specified herein shall be the product of a single manufacturer. Products and manufacturers specified are to establish a standard of quality for design, function, materials, and appearance. Products shall be modified as necessary by the manufacturer for compliance with requirements.:

1. Square D Model 6 Intelligent Motor Control Center by Schneider Electric
2. Eaton Freedom 2100 Motor Control Center
3. Allen Bradley by Rockwell Automation Centerline 2100 IntelliCenter Motor Control Center
4. ABB – ReliaGear MCC with Intelligent MCC features and functions

2.2 GENERAL REQUIREMENTS

A. The following MCC information is typically depicted on the Drawings: bus configuration, bus ratings, interrupting ratings, component size and type, power line and feeder connections, application specific unit control wiring, elevation and footprint, etc. Where not shown on or able to be derived from the Drawings, the minimum requirements specified herein shall be provided.

B. Motor Control Centers (MCCs) shall be 600 VAC class suitable for operation on three-phase, 60 Hz system. MCCs and their components shall conform to the applicable requirements of NEMA ICS 18-2001 and UL-845.

C. The system operating voltage and number of wires shall be as indicated on the contract documents.

D. MCCs shall be rated for an available short circuit capacity of 65,000 amperes rms, or as indicated on the contract documents, whichever is greater.

E. All motor control centers furnished shall be equipped with automatically operated shutters over vertical bus connection openings.

F. All motor control centers shall be Intelligent Motor Control Centers.

G. All MCC's and their components shall be built to NEMA standards and be UL listed.

2.3 CONSTRUCTION

A. MATERIALS

1. Steel material shall comply with UL 845 and CSA requirements.

2. Each MCC shall consist of one or more vertical sections of heavy gauge steel bolted together to form a rigid, freestanding assembly. The entire assembly shall be constructed and packaged to withstand normal stresses included in transit and during installation.

B. STRUCTURES

1. Structures shall be type NEMA 12 or NEMA 1A.
2. Structures shall be totally enclosed, dead-front, freestanding assemblies. Structures shall be capable of being bolted together to form a single assembly.
3. The overall height of the MCC shall not exceed 90 in (2286 mm), not including base channel or lifting angle, baffle, or plenum. Base channels shall be provided. Lifting angles shall be provided.
4. Each standard section for drawout type units shall have all the necessary hardware and bussing to add or move units within section. All unused space shall be covered by hinged blank doors. All spaces equipped to accept future units and include hinged blank doors. Vertical bus openings shall be covered by automatic bus shutters.
5. Each section shall include a top plate (single piece or two-piece). These plates shall be removable for ease in cutting conduit entry openings.
6. For motor control centers with a main breaker or switch, the main structure or the one immediately next to the main structure shall include an Absence of Voltage tester. The Absence of Voltage tester shall provide positive indication that voltage is not present on the motor control center load bus. For motor control centers without a main breaker or main switch, the incoming structure or the one immediately next to the incoming structure shall include an Absence of Voltage tester. The Absence of Voltage tester shall provide positive indication that voltage is not present on the motor control center bus.

C. WIREWAYS

1. Structures shall contain a minimum 12 inch (305 mm) high horizontal wireway at the top of each section and a minimum 6 inch (152 mm) high horizontal wireway at the bottom of each section. These wireways shall run the full length of MCC to allow room for power and control cable to connect between units in different sections.
2. A full-depth vertical wireway shall be provided in each MCC section that accepts modular drawout units. The vertical wireway shall connect with both the top and bottom horizontal wireway. The vertical wireway shall be 4 in (102 mm) wide minimum with a separate hinged door.
3. Access to the wireways shall not require the opening of control unit doors.
4. Structures that house a single, full section control unit shall not be required to have vertical wireways. Those control units shall open directly into the MCC horizontal wireways.
5. A steel barrier for communication cabling and I/O wiring shall be provided in each vertical wireway to provide separation from motor cables.

D. INSULATION AND ISOLATION

1. All power bussing and splice connections shall be isolated from the unit compartments and the wireways. The horizontal bus shall be mounted onto an insulating support assembly that braces the bus against the forces generated during a short circuit. The horizontal bus shall be isolated from the top horizontal wireway by a rigid non-conductive barrier. The barrier design shall allow qualified personnel to slide the barriers both left and right, to allow access to the bus and connections for maintenance without using tools or having to remove the barrier.

2. The vertical bus shall be housed in a molded and insulated support that provides bus insulation and shall brace the bus against the forces generated during a short circuit. These supports shall have openings for unit stab-on connections. Each opening shall be provided with an automatic shutter to close off the stab opening.

3. Barriers shall be provided in the vertical structure and unit designs to prevent the contact of any energized bus or terminal by a fishtape inserted through the conduit or wireway areas.

2.4 BUSSING

A. All bussing and connectors shall be tin or silver plated copper, with ratings as shown on the contract documents..

B. The main horizontal bus shall be rated at the amperage shown on the contract documents, continuous, and shall extend the full length of the MCC. Bus ratings shall be based on 65°C maximum temperature rise in a 40°C ambient. Provisions shall be provided for splicing additional sections onto either end of the MCC.

C. Horizontal bus splice bars stack shall be installed to the end of the MCC power bus to allow the installation of additional sections.

D. Each section that accepts plug-in units shall be provided with a vertical bus for distributing power from the main bus to the individual plug-in starter units. This bus shall be of the same material and plating as the main bus and shall be rated at the amperage shown on the contract documents, continuous current.

E. A tin or silver-plated copper ground bus shall be provided that shall run the entire length of the MCC. A mechanical lug shall be provided in the MCC for a #8-250 kcmil ground cable. The ground bus shall be provided with holes for each vertical section to accept user-supplied ground lugs for any loads requiring a ground conductor.

F. Each vertical section shall have a tin or silver plated copper vertical ground bus that shall be connected to the horizontal ground bus. This vertical ground bus shall be installed so that the plug-in units engage the ground bus prior to engagement of the power stabs and shall disengage only after the power stabs are disconnected upon removal of the plug-in unit.

G. Automatic shutters shall be installed on the vertical bus that will isolate the vertical bus when the unit is removed. These shutters shall be included on all installed units and provisions.

2.5 UNIT CONSTRUCTION

A. Units with circuit breaker disconnects through 400 ampere frame, and fusible switch disconnects through 400 A, shall connect to the vertical bus through a spring reinforced stab-on connector. Units with larger disconnects shall be connected directly to the main horizontal bus with appropriately sized cable or riser bus.

B. All conducting parts on the line side of the unit disconnect shall be shrouded by a suitable insulating material to prevent accidental contact with those parts.

C. A lever handle operator shall be provided on each disconnect. With the unit stabs engaged onto the vertical phase bus and the unit door closed, the handle mechanism shall allow complete ON/OFF control of the unit. All circuit breaker operators shall include, but shall not be limited, a separate TRIPPED position to clearly indicate a circuit breaker trip condition. It shall be possible to reset a tripped circuit breaker without opening the control unit door. Clear indication of disconnect status shall be provided.

D. A mechanical interlock shall prevent the operator from opening the unit door when the disconnect is in the ON position. Another mechanical interlock shall prevent the operator from placing the disconnect in the ON position while the unit door is open. It shall be possible for authorized personnel to defeat these interlocks.

E. A non-defeatable interlock shall be provided to prevent installing or removing a plug-on unit unless the disconnect is in the OFF position.

F. The plug-in unit shall have a grounded stab-on connector which engages the vertical ground bus prior to, and releases after, the power bus stab-on connectors.

G. Provisions shall be provided for locking all disconnects in the OFF position.

H. Unit construction shall combine with the vertical wireway isolation barrier to provide a fully compartmentalized design.

2.6 MAIN AND FEEDER MOLDED CASE CIRCUIT BREAKERS

A. Circuit breakers shall have voltage and interrupting ratings that meet the application requirements. Interrupting rating shall be available up to 65 kAIR without fuses.

B. MCCBs shall be designed to trip the circuit breaker in the event of high-level short-circuit currents. This design shall be independent of the thermal-magnetic or electronic trip unit.

C. MCCB's off all frame sizes shall be equipped with electronic trip units.

D. Electronic Trip Circuit Breakers

1. Electronic trip unit shall be true RMS sensing.
2. Trip units shall have the capability to electronically adjust the settings.

2.7 COMPONENTS FOR TYPICAL UNITS

A. Combination Starters:

1. All combination starters shall use a unit disconnect as described in Typical Unit Construction article above. Magnetic starters shall be furnished in all combination starter units.
2. All starters shall utilize NEMA/EEMAC rated contactors.
3. All starters shall be minimum of size 1.
4. When provided, control circuit transformers shall include two primary protection fuses and one secondary fuse (in the non-ground secondary conductor). The transformer shall be sized to accommodate the contactor(s) and all connected control circuit loads. The transformer rating shall be fully visible from the front when the unit door is opened.
5. When a unit control circuit transformer is not provided, the disconnect shall include an electrical interlock for disconnection of externally powered control circuits.
6. Auxiliary control circuit interlocks shall be provided where indicated. Auxiliary interlocks shall be field convertible to normally open or normally closed operation.
7. Hand-Off-Auto Selector Switch: Not required.

B. Provide engraved phenolic nameplates for each MCC and unit compartment. Nameplates shall be white background with black letters, measuring a minimum of 1.5 in (38 mm) high by 6.25 in (159 mm) wide total outside dimensions.

C. Pilot Device Panel: Each combination starter unit shall be provided with a hinged/removable control station plate. Provide a red LED "RUN" pilot light for each combination starter.

D. Motor Overloads: Motor overload protection shall be provided by Intelligent Motor Protection Relays.

E. Intelligent Motor Protection Relays (IMPR)

1. Overloads shall be fully programmable electronic relays.
2. Relay communications shall be compatible with the MCC software over Ethernet IP.

3. Overloads shall provide a 4.5:1 adjustment range matched to NEMA contactor sizes 1-6. Overload relays shall monitor all 3 phases for current. Parameter setting of the overload via network communications shall be supported.
4. The overload shall feature a Test/Reset button.
5. The overload shall feature General Purpose I/O – 6 Inputs, 3 Outputs.
6. Inputs powered by 120VAC.
7. Outputs shall be 120VAC.
8. Automatic, network, electronic and manual reset functions shall be supported.
9. The following protective functions shall be provided based on user configurable parameters:
    a. Overload / Underload / Jam
    b. Current Unbalance
    c. Ground fault
    d. Rapid cycling
    e. Motor Thermal Capacity
10. The overload shall provide the following user configurable parameters:
    a. Trip Class (5, 10, 15, 20, 25, 30)
    b. Command (Trip, Reset/Run, Display Lock, Network Configuration Enable)
    c. Rapid Cycle Timer
    d. Overload Restart Delay
    e. Underload Restart Delay (Dry Well Recovery Timer)
    f. Number of restarts after faults (Automatic Reset Mode)
    g. Number of restarts after Under Load fault (Automatic Reset Mode)
    h. Underload Trip Delay
11. The overload shall provide the following diagnostic and operational information:
    a. Error and Trip Indicators
    b. Ground Fault, L1, L2, L3 and 3 Phase Average Currents
    c. Current Unbalance
    d. Thermal Capacity Remaining
    e. Fault History (Last 4 faults)
    f. Product Identifier / Scale Factor
    g. Restart delay timer values
    h. Run time accumulator

2.8 VARIABLE FREQUENCY AC DRIVE UNIT CONSTRUCTION

A. Refer to specification section 262923 Variable Frequency Drive for additional requirements.

B. The AC drive controller unit shall be a combination disconnect-drive MCC style unit. The input circuit breaker or fusible switch shall provide NEC required branch circuit protection. The branch circuit protection shall have an external operator.

C. Output voltage and current ratings shall match the adjustable frequency operating requirements of standard NEMA design A or NEMA design B motors.

1. The normal duty overload current capacity shall be 110% of rated current for one (1) minute out of ten (10) minutes.

2. The heavy-duty overload current capacity shall be 150% of rated current for one (1) minute out of ten (10) minutes.

D. Units shall be of modular construction so that it shall be possible to readily interchange units of the same size without modifications to the MCC structure.

E. Conducting parts on the line side of the unit disconnect shall be isolated to prevent accidental contact with those parts.

F. AC drive controller units up to 50 horsepower variable torque shall be plug-on units which shall connect to the vertical bus through a spring-reinforced, stab-on connector. Units larger than 50 horsepower variable torque shall be connected directly to the main horizontal bus with appropriately sized cable or riser bus.

G. A disconnect operator shall be provided per Typical Unit Construction Article above.

H. The AC drive controller unit shall incorporate a self-contained, air-based cooling system. Air exhaust vents shall be louvered to help direct air flow away from personnel operating the AC drive controller unit. Fans, ductwork, or filters shall be easily accessible for maintenance.

I. The AC drive controller unit cooling system shall be sized to cool the drive regardless of mounting location within the MCC. The AC drive controller unit shall not be restricted to a specific location in the MCC.

J. An internal overtemperature trip shall be provided to detect cooling system failure or blockage. Upon occurrence of an overtemperature trip, the cooling system fans shall continue running to provide a rapid cool down.

K. Power for the cooling system shall be provided internal to the AC drive controller unit by use of a control power transformer that shall include, but shall not be limited to, two primary fuses and one secondary fuse (in the non-ground secondary conductor).

L. Graphic User Interface

1. A detachable backlit graphical user interface terminal with keypad shall be provided for monitoring, annunciation, and configuration.
2. The keypad shall be capable of storing the configuration for the motor overload to ease replacement and service continuity
3. The keypad shall be capable of providing password protection.
4. The user interface shall be capable of saving and downloading configurations of the VFDs, as well as porting them to other VFDs.
5. The user interface shall offer a Mini-USB port for mass storage or PC device connection.
6. The VFD shall have self-diagnostic capabilities to display alarms, errors, and warnings as they occur and be able to store at least 15 last messages into the memory. These shall be accessible by PC maintenance tools or web server with flash record for data logging.
7. The user interface shall be identical throughout the power range to avoid confusion amongst the users and need for training in several different units.

M. Control:

1. VFD shall interface with the process control system via an Ethernet IP network to monitor, control, display, and record data for use in processing reports. VFD settings shall be retained within VFD's nonvolatile memory.

N. Communications

1. Refer to specification section 262923 Variable Frequency Drive for additional requirements
2. VFD’s shall be equipped with at a minimum of two Ethernet IP RJ-45 communications ports. In addition the following communications options shall be provided as necessary for communications. Refer to communication requirements specified elsewhere within the Contract Documents.
3. VFD Ethernet ports shall be IPv6 compliant, allow for web server access and provide network management via SNMP and clock synchronization.
4. VFD communications modules shall be self-powered by the drive.
5. The VFD shall provide integration connectivity via
   a. DHCP protocol for Fast Device Replacement
   b. DTM library in compliance with standard FDT technology

O. Diagnostics and Configuration

1. The VFD Supplier shall have Windows based PC software for configuring and diagnosing the VFD. It shall be possible to set and modify parameters, control the drive, read actual

values and make trend analysis using the software. The PC-tools may be connected to the VFD by wired or wireless connection.

2. The VFD shall display all faults in plain text and help screens shall be available to guide the user in the troubleshooting.

3. The VFD shall provide a Real Time Clock management for time stamping of detected errors.

4. The VFD must have the ability to dynamically display I/O status.

P. Harmonic Study – NOT REQUIRED ON THIS PROJECT.

1. A harmonic distortion analysis shall be performed by the MCC manufacturer based upon system documentation supplied with the bid documents. The documentation shall consist of one-line diagrams and these specifications. The harmonic distortion analysis report shall be part of the shop drawing process, submitted to the Engineer for review and acceptance with the initial submittal of the MCC's. The report shall be on a per-MCC basis. It is the responsibility of the manufacturer to select and provide the combination of the acceptable types of specified VFD's in conjunction with a harmonic filtering section built into the MCC's as required to meet the requirements of the most current edition of IEEE 519 for THD as well as a minimum power factor of 95% per MCC. Total current harmonic distortion for each MCC shall not be more than 5%.

   For clarification and for example, Manufacturer "A" may elect to supply all VFD's with AFE filtering technology and in combination with the specified number and size of FVNR and Soft Start type controllers for the quantity and size of motors indicated in the contract documents, determine that this approach alone will meet the requirements of the most current edition of IEEE 519 and this specification for each MCC. On the other hand, Manufacturer "B" may elect to supply VFD's equipped with varying degrees of integral harmonic filtering for the same MCC and motors and variety of motor controller types, but also supply an active harmonic filter and power factor correction section in the MCC to achieve the same end result as Manufacturer "A" for each MCC.

   It is anticipated that an active harmonic filter section of at least 200A will be required to meet the requirements of these specifications for both THD and Power Factor.

Q. Harmonic Equipment with power factor correction installed in MCC's NOT REQUIRED ON THIS PROJECT :

1. All Refer to specification section 26 29 23 for VFD harmonic requirements.

2. Manufacturers

   a. Square D Accusine

   b. Eaton HCU2

   c. TCI Harmonic Guard Active filter in MCC construction

   d. Mesta DPM / Hammond Power Solutions HPS TruWave in MCC construction

3. Where a manufacturer determines that based upon their selection of VFD types in conjunction with the other types of motor controllers in each MCC the best approach for them to meet the requirements of this specification is to include an active harmonic filter section built into the MCC, the filter shall be in accordance with the following:

   Active harmonic filters shall be provided to perform electronic cancellation of load produced harmonic currents such that the upstream power harmonic current and voltage shall be reduced below the most current edition of IEEE 519 guidelines for load demand and voltage distortion limits. Performance of the filter shall be independent of the impedance of the power source, AC lines or back-up generator. Necessary current transducers, reactors, and operator interfaces shall be supplied with the MCC. A factory certified start-up technician shall be used to start up each harmonic filter to achieve optimum system performance. Filter shall have Ethernet IP communications compatible with the MCC and connected to the MCC Ethernet switch. Filter shall be sized by and provided by the manufacturer such that current and voltage THD is less than allowed by IEEE 519, latest revision or less than 5%, whichever is lower, as well as correct the power factor at the line side of the MCC to a minimum of 95% for each individual MCC.

2.9 SOLID STATE REDUCED VOLTAGE STARTER UNIT CONSTRUCTION. PROVIDE WHEN SOLID STATE REDUCED VOLTAGE STARTERS ARE SHOWN IN THE MCCS ON THE CONTRACT DOCUMENTS. NONE REQUIRED ON THIS PROJECT

A. The soft start controller unit shall be a combination disconnect/soft starter, MCC-style unit. The input circuit breaker or fusible switch shall provide NEC required branch circuit protection. The branch circuit protection shall have an external operator. Wiring between the soft starter and disconnect shall not be disturbed when removing or installing the soft start controller unit from the MCC. Units shall be of modular construction so that units of the same size can be interchanged without modifications to the MCC structure.

B. In addition to the requirement of this section, solid state reduced voltage starters shall be of one of the manufacturers specified in and meet the performance requirements of specification section 262913 Enclosed Controllers.

C. Conducting parts on the line side of the unit disconnect shall be isolated to prevent accidental contact with those parts.

D. Soft start controller units rated for standard duty (up to 156 A) shall be plug-on units which connect to the vertical bus through a spring-reinforced stab-on connector. Units rated higher than 156 A shall be connected directly to the main horizontal bus with appropriately sized cable or riser bus.

E. A disconnect operator shall be provided per Typical Unit Construction Article.

F. Plug-on soft start controller units shall have a grounded stab-on connector which shall engage the vertical ground bus prior to, and shall release after, the power bus stab-on connectors engage/release.

G. The enclosure shall include, but shall not be limited to, a door-mounted digital keypad for adjusting the soft start parameters and viewing the motor, soft start, and fault status without opening the enclosure door.

H. A shorting contactor shall be supplied in each soft start controller unit to reduce temperature rise within the unit and MCC enclosure. The shorting contactor shall be rated to carry the motor full load current during steady state after full voltage has been applied to the motor by the soft starter.

I. Electrical Ratings, Adjustments and Configurations:

1. Refer to specification section 262913 Enclosed Controllers.

J. Controls

1. Controls shall be over the process control systems Ethernet IP network.

2.10 POWER METERING

A. NOT REQUIRED ON THIS PROJECT

2.11 ETHERNET COMMUNICATIONS

A. General Requirements

1. MCCs shall be provided with industrial quality managed Ethernet switch(s) as required to communicate and perform the functions necessary for connection to an industrial communications network. In conjunction with the Ethernet switches a communications network shall be provided within the MCC to connect all Ethernet communications capable equipment within the MCC. In addition, the following MCC components shall be provided with Ethernet connections for communications over the network:

a. Feeder Circuit Breakers

b. Motor Starters or associated Intelligent Motor Protection Relays

c. Variable Frequency Drives

d. Solid State Reduced Voltage Soft Starters

2. The MCC Ethernet network communications architecture shall form a star topology to optimize network availability and reliability. The star topology shall be created by way of

home run patch cables individually run from each communicating motor starter or circuit breaker unit back to a centrally located Ethernet Switch unit.

3. A minimum of 2 managed Ethernet switches shall be provided per MCC. Ethernet switch connections to MCC equipment for multiple loads of a single process train shall be evenly split between switches so that a switch failure will not disable all communication to all equipment within that process train.[For example, Switch A would be connected to Pumps 1, 3, and 5 while Switch B would be connected to Pumps 2, 4, and 6.]

4. Vendor shall provide add-on instructions or EDS files that can be utilized in a Rockwell/Allen-Bradley Process Automation Controller for all intelligent components and motor controllers of each MCC.

B. Managed Ethernet Switches

1. Subject to compliance with requirements, provide Ethernet TCP/IP Managed Switch.

2. Managed switches shall provide capability for prioritization of network traffic, Rapid Spanning Tree Protocol (RSTP), network redundancy, and Simple Network Management Protocol (SNMP).

3. Each MCC shall contain two or more Ethernet switch devices with a sufficient quantity of RJ-45 ports for all communicating units, management ports, uplink ports, and approximately 10% spares.

4. All Ethernet switches shall be centrally mounted in (a) compartmentalized unit(s) for easy and ready access for troubleshooting without the presence of control power voltages greater than 24Vdc.

   a. Ethernet Switches shall be unit mounted and centrally located near the bottom of the MCC to optimize cable runs across shipping splits. Individually mounting Ethernet switches in horizontal or vertical wireways or behind wireway barriers that require maintenance personnel to work around or near exposed power cables is strictly prohibited. Ethernet Switch units shall be provided with cable supports which help to prevent stress and damage to Ethernet switch ports.

5. Switch compartments shall have no voltage over 24V DC

C. Communications Equipment Power Supply: A separately mounted, 24Vdc power supply unit shall provide power to Ethernet switch unit(s) in order to prevent personnel exposure unsafe voltages. All communicating devices, such as Ethernet switches, overloads, meters, etc., shall be supplied with 24Vdc power externally from the MCC.

D. Communication Cabling

1. The full-depth vertical wireway shall contain a communications barrier that will serve to separate communications from power cabling to prevent noise interference and mechanical damage on the network cables. Intermediate Ethernet connectors used to

extend CAT5e cables through metal barriers may create additional points of failure, and are strictly prohibited.

2. The MCC shall employ industrial grade, shielded 600V rated Category 5e communication patch cables to interconnect communicating units with Ethernet switch units.

E. Configuration protocol

1. The MCC shall communicate Ethernet/IP.

F. A startup/commissioning Network Validation Service shall be provided with each Intelligent MCC line item to reduce installation, commissioning, and setup time on site. Training shall be provided as part of the service. This service shall be performed by a qualified Intelligent MCC services engineer.

2.12 MARKINGS AND LABELING

A. All identification and warning labels and nameplates exterior to the MCC shall be resistant to weather, UV, and their intended installation environment.

B. Each MCC shall be provided with an engraved nameplate identifying the project specific equipment tag and service description.

C. Warning labels and nameplates shall be present at access locations to advise personnel of possible hazards. The MCC shall be marked in accordance with UL, NFPA 70 NEC, NFPA 70E, and other applicable standards.

D. Each MCC and unit compartment shall be identified with an engraved nameplate using 1/2-inch high lettering. Each enclosure mounted control and pilot device shall be identified with an engraved nameplate using 3/8-inch high lettering. Nameplate lettering shall be black on a white background.

PART 3 - EXECUTION

3.1 GENERAL

A. In addition to the requirements specified herein, execution shall be in accordance with the requirements of the contract Drawings.

B. The installation contractor shall examine equipment exterior and interior prior to installation. Report any damage and do not install any equipment that is structurally, moisture, or mildew damaged.

C. Verification of Conditions: The installation contractor shall examine areas and conditions under which the work is to be installed, and notify the Owner's representative in writing, with a copy to

the Engineer, of any conditions detrimental to the proper and timely completion of the work. Do not proceed with the work until unsatisfactory conditions have been corrected.

D. Pre-Installation Conference: Prior to commencing the installation, an onsite pre-installation conference shall review the material selections, installation procedures, and coordination with other trades. Attendees shall include, but shall not be limited to, the Contractor, the Installer, manufacturer's representatives, and any trade that requires coordination with the work. Date and time of the pre-installation conference shall be acceptable to the Owner and the Engineer.

E. Beginning of the work shall indicate acceptance of the areas and conditions as satisfactory by the Installer.

F. Install equipment in accordance with reviewed product data, final shop drawings, manufacturer's written instructions and recommendations, and as indicated on the Drawings.

G. Functional testing, commissioning, and parameter adjusting shall be carried out by the Engineer. Manufacturer shall replace damaged or malfunctioning equipment.

3.2 FACTORY ACCEPTANCE TESTING (FAT)

A. The entire MCC shall go through an unwitnessed factory acceptance test prior to shipment. This FAT shall include, but shall not be limited to, the following:
   1. Physical Inspection of the following:
      a. Structure.
      b. Electrical conductors, including, but not limited to, the following:
      c. Bussing.
      d. General wiring.
      e. Units.
   2. Electrical Tests:
      a. General electrical tests shall include, but shall not be limited to, the following:
         1) Power circuit phasing.
         2) Control circuit wiring.
         3) Instrument transformers.
         4) Meters.
         5) Ground fault system.
         6) Device electrical operation.
      b. AC dielectric tests shall be performed on the power circuit.
   3. Markings/labels include, but shall not be limited to, the following:
      a. Instructional type.
      b. UL/CSA.
      c. Inspector's stamps.
   4. Each device shall be configured and addressed to correspond with software settings.

5. A read/write test shall be performed prior to shipment on network devices, including, but not limited to, overloads, drives, and soft starters.
6. Testing shall be designed to verify system operation and shall include, but shall not be limited to, these verifications as a minimum:
   a. Drawings and bill of materials.
   b. I/O addressing.
   c. Correct device operation by I/O address.
   d. Host communications.
   e. Control network interface.
7. The manufacturer shall use integral quality control checks throughout the manufacturing process to ensure that the MCC meets operating specifications.

3.3 FIELD QUALITY CONTROL

A. Functional testing, commissioning, and first parameter adjusting shall be carried out by the Engineer

3.4 HARMONIC ANALYSIS STUDY – NOT REQUIRED ON THIS PROJECT

A. Within 30 Days of Substantial Completion, factory qualified and trained technician shall perform a harmonic analysis study to identify the effects of nonlinear loads and their associated harmonic contributions on the voltages and currents at each MCC. Analyze possible operating scenarios, including recommendations for VFD input filtering to limit TDD and THD (V) at each VFD at each MCC to specified levels. Testing shall comply with IEEE519.

B. Prepare a harmonic analysis study and report complying with IEEE 399 and NETA Acceptance Testing Specification.

C. Manufacturer shall provide additional harmonic filtering and power factor correction equipment or measures at any MCC that does not meet the minimum requirements of IEEE 519 and these specifications.

3.5 FIELD TESTING AND COMMISSIONING

A. MCCs and related equipment shall be commissioned by the Engineer.

B. Operational Readiness Testing – Performed by Installation Contractor

1. The Installation Contractor shall inspect and test furnished equipment and associated systems for conformance to the contract documents, including equipment manufacturer's recommendations, and readiness for operation. The test shall include the following as a minimum:

   a. Visually inspect for physical damage and proper installation

b. Perform tests in accordance with manufacturer's instructions

c. Perform tests to ensure compliance with Contract Documents

d. Perform tests that equipment is ready for operation

e. Touch-up paint all chips and scratches with manufacturer-supplied paint and transfer remaining paint to Owner

2. Installation Contractor shall submit an operational readiness test report documenting all test results, including all assumptions, conditions, allowances and corrections made during the test. The report shall provide a listing of all modifications and adjustments made onsite to include any settings / parameters not identified as factory defaults within the equipment's O&M documentation. The test report shall include a signed statement from the Contractor, installer(s) and the factory-trained manufacturer's representative(s) certifying that the furnished equipment and associated system have been installed, configured, and tested in accordance with the manufacturer's recommendations, completely conforms to the requirements of the Contract Documents and is ready for operation.

3.6 TRAINING

A. O&M Training: The services of a factory-trained instructor shall be provided for training the Owner's staff in the proper operation and maintenance of the MCC and components. Training shall consist of not less than 2 repeated session(s) with 6 hours of onsite classroom and hands-on instruction for a minimum of 4 attendees per session. The instructor shall provide sufficient time and detail in each session to cover the following as a minimum:

1. Theory of operation
2. Operation of MCC and components furnished
3. Maintenance and configuration
4. Configurations of MCCs and components furnished
5. Troubleshooting and repair
6. Replacement of component level parts

B. Manuals and documentation shall be provided to each participant for training.

END OF SECTION 26 24 19

SECTION 262726 – WIRING DEVICES

1. PART 1 - GENERAL

1.1. DESCRIPTION OF WORK

1.1.1. The types of the electrical wiring devices required for this project include the following:

1.1.1.1. Receptacles.

1.1.1.2. Switches.

1.1.1.3. Covers.

1.1.1.4. Cord plugs.

1.1.1.5. Cord plug connectors.

1.1.1.6. Telephone and power poles.

1.2. QUALITY ASSURANCE

1.2.1. Manufacturers: Firms regularly engaged in manufacture of wiring devices, of types, sizes and ratings required, whose products have been in satisfactory use in similar service for not less than 3 years.

1.2.2. Installer: A firm with at least 2 years of successful installation experience on projects with electrical installation work similar to that required for the project.

1.2.3. NEC Compliance: Comply with National Electrical Code (NFPA No. 70) Article 410 and 380 as applicable to construction and installation of electrical wiring devices.

1.2.4. UL Labels: Provide electrical wiring devices which have been tested, listed and labeled by Underwriters Laboratories.

1.2.5. NEMA Compliance: Comply with National Electrical Manufacturers Association standard for general- and specific-purpose wiring devices.

1.3. SUBMITTALS

1.3.1. Manufacturer's Data, Wiring Devices: Submit manufacturer's data on electrical wiring devices and plates with sufficient information that allows for evaluation of equipment for compliance with contract documents.

2. PART 2 - PRODUCTS

2.1. ACCEPTABLE MANUFACTURERS:

2.1.1. Manufacturer: Subject to compliance with requirements, provide products of one of the following (for each type of wiring device):

2.1.1.1. Leviton Co.,

2.1.1.2. Pass & Seymour / Legrand

2.1.1.3. Hubbell.

2.2. CONVENIENCE OUTLETS

2.2.1. Shall be industrial grade, 125V, 20A, 2P-3W, with provisions for back-wiring from eight separate wiring openings with screw activated clamp type terminals and side wiring with captively held binding screws.

2.2.2. They shall be listed by Underwriters Laboratories and verified as meeting Federal Specifications W-C-596F.

2.2.3. Receptacles shall be by Leviton, Pass and Seymour, Hubbell, or approved equal.

2.2.3.1. Duplex Receptacles

2.2.3.1.1. Hubbell 5352

2.2.3.1.2. Leviton 5352

2.2.3.1.3. Pass and Seymour 5362

2.2.3.1.4. Device and plate color shall be white in office areas. Devices shall gray in mechanical and electrical rooms with industrial or FS covers where surface mounted. Covers in process/clean rooms shall be weatherproof while-in-use suitable for washdown areas as specified below.

2.2.3.1.5. Ground fault interrupter receptacles Industrial Grade, 125V, 20A Leviton G5362 or approved equal by Hubbell or Pass and Seymour. Color shall be white in office areas and gray in mechanical and electrical rooms as well as process/clean rooms. Covers in process/clean rooms shall be weatherproof while-in-use suitable for washdown areas as specified below.

2.2.3.1.6. All 125-volt receptacles in the plant/clean room/process room areas shall be rated for NEMA4 environments and shall be in a weatherproof enclosure, the integrity of which is not affected when the receptacle is in use (attachment plug cap inserted). Receptacles shall use Crouse-Hinds type FS or FD box with No. WLRS-1 or No. WLRD-1 coverplate for standard duplex receptacles and with type WLGF for flush device boxes. (WLFG spring door cover and gasket only) or type WLGF-FS for FS and FD cast device boxes (WLGF-FS spring door cover and gasket only) for use with ground fault circuit

interrupter type receptacles. Receptacles shall be ground fault circuit interrupter (GFCI) type.

2.3. TOGGLE SWITCHES

2.3.1. Shall be 120/277V, 20A quiet AC-type, heavy duty, industrial grade with provisions for back-wiring with screw actuated clamp type terminals, side-wiring with adaptively-held binding screws, and separate screw terminal for the grounding wire. They shall be color coded for current rating and certified by UL as meeting Federal Specifications W-S-896E.

2.3.2. Switches shall be by Hubbell 1223, Pass & Seymour CSB20AC3 or Leviton 1223-2 or approved equal. Toggle switches in office areas shall be white. Toggle switches in mechanical, electrical and process rooms shall be gray.

2.4. COVER PLATES

2.4.1. Plates in office areas shall be white.

2.4.2. Plates for outlets in mechanical and electrical rooms shall be galvanized steel/raised industrial cover type where surface mounted, stainless steel where flush mounted in mechanical and electrical rooms. Covers in process/clean room areas shall be NEMA 4 rated for FS/FD type boxes, suitable for washdown areas by Hubbell/Killark or approved equal.

2.4.3. Plates shall be set plumb and parallel with the wall. Care shall be taken to center all switch plates on paneling, trim, etc. All switch plates shall be located equal distance from all doors where possible and on the lock side of the door regardless of notations on drawings.

2.5. PLUG LOAD TIMER RECEPTACLES

2.5.1 50% of receptacles in office areas are required to be automatically controlled per the Energy Code. These receptacles shall be 125V, 20A with LCD display on top half and simplex receptacle on the bottom half and override button on the face of the receptacle. Receptacles shall be Pass & Seymour/Legrand PLT26351W or approved equal. Wire these receptacles in a “feed-through” configuration to control the downstream receptacles within the same room as shown on the drawings. There will typically be one Plug Load Timer receptacle per office and conference room. Refer to the drawings for additional information.

2.6. SPECIAL WIRING DEVICES

2.6.1. Outlets installed outdoors or in wet locations: Use Crouse-Hinds Type FS or FD box with No. WLRS-1 or No. WLRD-1 cover plate for standard receptacles, and No. DS185 cover plates for switches and with type WLGF for flush device boxes. (WLFG spring door cover and gasket only) or type WLGF-

FS for FS and FD cast device boxes (WLGF-FS spring door cover and gasket only) for use with ground fault circuit interrupter type receptacles. Receptacles shall be ground fault circuit interrupter (GFCI) type.

2.7 AUTOMATIC AND DIMMING LIGHTING CONTROLS

2.7.1 Refer to specification section 262519 LED LIGHTING and the project drawings for requirements on automatic lighting controls and dimmer switches (occupancy and vacancy sensors, photocells, dimmer switches etc.).

3. PART 3 - EXECUTION

3.1. Switches and receptacles shall be installed and located as follows. Dimensions are AFF to center unless otherwise noted:

3.1.1. Switches 48" above finished floors.

3.1.2. Receptacles: 18" above finished floors unless otherwise noted; 4" above counters and work benches except where otherwise noted; and 44" above finished floors in electrical and mechanical equipment rooms, process, warehouse and similar areas.

3.1.3. Where light switches are located adjacent to doors, they shall be installed on "knob" side of door, unless indicated otherwise.

3.1.4. Prior to roughing-in outlet boxes, verify door swings, type of wall finishes and locations above counters.

3.1.5. Install wiring devices plumb and flush with walls so that device plates are tight to finished surfaces.

3.1.6. Install single pole switches to turn lights on in the up position.

3.1.7. Mount wiring devices as heights above finished floor as shown on the Mounting Height Schedule, unless shown otherwise.

3.1.8. Where more than one switch is shown at a location, switches shall be ganged in one box.

3.1.9. Receptacles shall be installed with the "U" grounding contact at the top. Where receptacles are mounted horizontally they shall be installed with the neutral contact at the top.

END OF SECTION

SECTION 262816.16 – ENCLOSED SWITCHES

1. PART 1 - GENERAL

1.1. DESCRIPTION OF WORK

1.1.1. Local in-sight disconnect switches for all equipment and motors indicated on the drawings whether furnished by this Contractor or by others.

1.1.2. Provide local in-sight disconnect switches for HVAC equipment only as shown on the drawings. Local disconnect switches are not required for process equipment motors.

1.1.3. Provide fractional HP thermal element switches at all unit heaters and motor operated overhead doors.

1.2. SUBMITTALS

1.2.1. Shop drawings submittals shall include catalog cuts, fabrication details, ratings and other pertinent data required to evaluate the proposed equipment.

1.3. QUALITY ASSURANCE

1.3.1. Manufacturers: Firms regularly engaged in manufacture of motor and circuit disconnect switches of types and capacities required, whose products have been in satisfactory use in similar service for not less than 3 years.

1.3.2. Installer: A firm with at least 3 years of successful installation experience on projects with electrical installation work similar to that required for the project.

1.4. STANDARDS AND CODES

1.4.1. Except where otherwise required by this section, the following standards and codes shall govern.

1.4.1.1. UL 98 - Enclosed and Dead Front Switches

1.4.1.2. NEMA KS 1 - Enclosed Switches

1.4.1.3. NEMA 250 - Enclosures for Electrical Equipment

1.5. PRODUCT DELIVERY, STORAGE AND HANDLING

1.5.1. Deliver switches individually wrapped in factory-fabricated fiberboard type containers.

1.5.2. Handle switches carefully to avoid damage to material components, enclosure and finish. Do not install damaged switches; remove from project site.

1.5.3. Store switches in a clean dry space. Protect switches from dirt, fumes, water and physical damage.

2. PART 2 - PRODUCTS

2.1. ACCEPTABLE MANUFACTURERS

2.1.1. Disconnect switches for integral horsepower motors, 1/2 horsepower and larger, by:

2.1.1.1. Schneider Electric/Square D

2.1.1.2. ABB/General Electric

2.1.1.3. Siemens

2.1.1.4. Eaton-Cutler Hammer.

2.2. EQUIPMENT

2.2.1. Disconnects for fractional horsepower motors, less than 1/2 horsepower, at 125 volts, and for equipment of similar capacity and voltage shall be Bussman types as follows:

2.2.1.1. Type SRY mounted on 4" square box cover, for plug-in equipment in unfinished areas.

2.2.1.2. Type SRW mounted on 2-3/4" switch box, for plug-in equipment in finished areas, installed in flush switch box.

2.2.1.3. Type SSY mounted on 4" square box cover, for direct connected equipment in unfinished area.

2.2.2. Type SSW mounted on 2-3/4" switch box stainless steel plate, for direct connected equipment in unfinished areas, installed in flush switch box.

2.2.3. Disconnects for integral horsepower motors, 1/2 horsepower and larger, and for equipment of similar capacity shall be heavy duty industrial type, 600 V, 3 blades with solid neutrals where required. At interior locations, enclosures shall be NEMA Type 1; at exterior locations enclosures shall be NEMA Type 4.

2.2.4. Disconnects for integral horsepower motors shall have provision of padlocking in the "ON" or "OFF" positions.

2.2.5. Fuses: Provide fuses for safety switches, as recommended by switch manufacturer, of classes, types, and ratings needed to fulfill electrical requirements for service indicated.

2.2.6. Switch Interior:

2.2.6.1. All switches shall have switch blades which are visible when the switch is OFF and the cover is open.

2.2.6.2. Where the type of switch enclosures are not shown, they shall be of NEMA Type: 4-4X-5 stainless steel or 4X polyester: Dusttight, Watertight, Corrosion Resistant.

2.2.6.3. Lugs shall be front removable and UL Listed for 60º C or 75º C conductors (30-100 amperes), 75º C conductors (200-1200 ampere), aluminum or copper conductors (Type 1, 3R or 4X polyester), copper conductors only (Type 12, 12K, Type 4-4X-5 stainless steel).

2.2.6.4. 30 through 100 ampere switches shall be equipped with factory installed fuse pullers, Type 12, 12K, Type 4-4X-5 stainless steel, Type 4X polyester or field installed fuse pullers, Type 1 or Type 3R.

2.2.6.5. Switches required for Type 4-4X-5 stainless steel applications shall have all copper current carrying parts.

2.2.6.6. All current carrying parts shall be plated to resist corrosion.

2.2.6.7. Switches shall have removable arc suppressers to facilitate easy access to line side lugs.

2.2.6.8. Switches shall have provisions for a field installable electrical interlock.

2.2.7. Switch Mechanism:

2.2.7.1. Switch operating mechanism shall be quick-made, quick-break such that, during normal operation of the switch, the operation of the contacts shall not be capable of being restrained by the operating handle after the closing or opening action of the contacts has started.

2.2.7.2. The operating handle shall be an integral part of the box, not the cover.

2.2.7.3. Provisions for padlocking the switch in the OFF position with at least three padlocks shall be provided.

2.2.7.4. The handle position shall travel at least 90 between OFF and ON positions to clearly distinguish and indicate handle position.

2.2.7.5. All switches shall have a dual cover interlock mechanism to prevent unintentional opening of the switch cover when the switch is ON and prevent turning the switch ON when the cover is open. The cover interlock mechanism shall have an externally operated override but the override shall not permanently disable the interlock mechanism. The tool used to override the cover interlock mechanism shall not be required to enter the enclosure in order to override the interlock.

2.2.8. Switch Enclosures:

2.2.8.1. Switch covers shall be attached with welded pin-type, top hinged, attached with removable screws and securable in the open position.

2.2.8.2. The enclosure shall be finished with gray baked enamel paint with a brush finish on Type 304 stainless steel (Type 4-4X-5 stainless steel).

2.2.8.3. The enclosure shall have ON and OFF markings stamped into the cover (Type 4-4X-5 stainless steel, inked on an adhesive label (Type 4X polyester).

2.2.8.4. . The operating handle shall be provided with a dual colored, red/black position indication.

2.2.8.5. All switches shall have provisions to accept up to three 3/8 in hasp padlocks to lock the operating handle in the OFF position.

2.2.8.6. Type 4-4X-5 stainless steel enclosure shall contain no knockouts. Supply watertight hubs as indicated on the plans.

2.2.8.7. Type 4X polyester enclosures shall be provided with polyester conduit hubs for field installation.

2.2.8.8. Cover sealing means for switches rated through 200 amperes shall be quick release trunk latches (Type 4-4X-5 stainless steel), Type 316 stainless steel captive screws (Type 4X polyester).

2.2.8.9. Type 4-4X-5 stainless steel enclosures shall be dual rated as Type 3R to facilitate their use in outdoor applications.

2.2.8.10. Cover viewing window will be incorporated on 30-200 A, 4-4X-5 stainless steel, two or three pole switches.

2.2.9. Switch Ratings:

2.2.9.1. Switches shall be NEMA Standard Type Heavy-Duty “HD” horsepower rated for ac and/or dc as indicated on the plans, 100 percent duty rated.

2.2.9.2. The UL Listed short circuit current rating of the switches shall be: 10,000 rms symmetrical amperes when used with or protected by Class H or K fuses (30-600 ampere), 200,000 rms symmetrical amperes when used with or protected by Class R or Class J fuses (30-600 ampere switches employing appropriate fuse rejection schemes), 200,000 rms symmetrical amperes when used with or protected by Class L fuses (800-1200 ampere).

2.2.9.3. Shall have external operating handle which indicate ON and OFF position and shall have lock-open padlocking provisions. Mechanical interlock shall permit opening of the door only when the switch is in the OFF position, defeatable by a special tool to permit inspection via override screw to permit opening front cover with switch in the ON position. Solid neutral for each switch being installed in a circuit which includes a neutral conductor.

3. PART 3 - EXECUTION

3.1. INSTALLATION OF SAFETY AND DISCONNECT SWITCHES

3.1.1. General: Install safety and disconnect switches where indicated in accordance with the manufacturer's written instructions, the applicable requirements of NEC and the National Electrical Contractors Association's "Standard of Installation," and in accordance with recognized industry practices to ensure that products serve the intended function.

3.1.2. Install disconnect switches used with motor-driven appliances, and motors and controllers within sight of the controller position unless otherwise indicated.

3.1.3. Coordinate motor and circuit disconnect switch installation work with electrical raceway and cable work, as necessary for proper interface.

3.1.4. Install fuses in fusible switches.

3.1.5. Install motor and circuit disconnects on adjacent walls maintaining NEC required access. Where required, due to access requirements, mount motor and circuit disconnects on free standing unistrut stands adjacent to equipment. In utility areas, mount motor and circuit disconnects on adjacent walls maintaining required NEC Clearance in front of disconnects. On roofs, or other exterior locations, mount motor and circuit disconnects on equipment or provide free-standing unistrut stand adjacent to equipment

3.1.6. Furnish and install plastic phenolic nameplate labels as per Section 260553.

3.1.7. All disconnect switches indicated on the Drawings by a symbol to be raintight or weatherproof shall have a NEMA 4 enclosure.

3.1.8. Provide warning labels inside disconnect switches affixed in prominent location indicating type and size of fuse installed and fuse manufacturer's catalog number. Place warning label "Do Not Install Fuses Other Than Those Listed Below: (list fuse types and sizes allowed)

3.1.9. Fuses for feeders, branch circuits, motors and other equipment shall be selected in types of ratings in accordance with NEC to provide a coordinated system of overcurrent protection, whereby if a fault or harmful overload should occur, only the fuses nearest the fault or overload will open. Provide one spare set of three of each size and type of fuses rated at 100 amperes and over.

3.1.10. For disconnect switches serving motors connected to variable frequency drives and where indicated on the drawings, provide a microswitch (emergency stop contacts or proximity switch) designed to accommodate an external signal such that the variable frequency drive will shutdown upon opening of the disconnect switch.

3.1.11. All safety disconnect switches for motors, and heating, ventilating and air conditioning equipment shall be fusible switch type and horsepower rated for dual-element fuses unless protected elsewhere in the circuit by fuses. Type "HACR" breakers may be used for protection where prior approval is obtained from the appropriate authority and written approval is obtained from the Owner's Representative.

3.2. FUSES

3.2.1. Owner Furnished. Where required, fuses shall be included as a part of any fusible device within equipment designated to be Owner furnished. Otherwise, the Contractor will provide fuses.

3.2.2. Contractor Furnished. The Contractor shall furnish and install fuses in all fusible devices and equipment that are furnished by the Contractor.

3.2.3. Manufacturer and Listing. Class J fuses shall normally be used unless otherwise specified or indicated on the drawings.

3.2.4. Near the completion of the work, the Contractor shall confirm that proper fuses are installed in each fusible device (both Owner and Contractor Furnished) and shall furnish the Owner's Representative a complete listing of fuses, indicating the manufacturer, size, voltage, class and type, equipment number, horsepower, and descriptive title of equipment protected.

END OF SECTION

SECTION 262923 – VARIABLE FREQUENCY DRIVES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this section and all other sections of Division 26.

1.2 SUMMARY

A. Section includes combination VFDs, rated 600 V and less, for speed control of three-phase, squirrel-cage induction motors, for both separately enclosed, preassembled units as well as those installed in a Motor Control Center.

B. VFD Application:

1. Enclosed Preassembled Drives for Motors 125 HP and Smaller: Drives shall be equipped with passive harmonic filter technology. Drive shall be heavy duty HD rated at the horsepower supplied.

2. Enclosed Preassembled Drives for Motors 150 HP and Larger: NONE REQUIRED ON THIS PROJECT. Drives shall be low harmonic distortion type equipped with Active Front End (AFE) technology. Drive shall be heavy duty HD rated at the horsepower supplied.

3. Drives installed in Intelligent Motor Control Centers All Motor Sizes: Drives shall be equipped with Active Front End (AFE) or, active or passive harmonic filter technology, or unfiltered but used in conjunction with an active harmonic filtering unit installed in the MCC lineup. NOT APPLICABLE ON THIS PROJECT. Drives shall be heavy duty HD rated at the horsepower supplied. Drives installed in MCC's shall be of a technology(ies) selected by the manufacturer to be used in combination with or without an active harmonic filter as required and with power factor correction to meet the requirements of IEEE 519, latest revision, and these specifications on a per MCC basis. Refer to specification section A26 24 19 Intelligent Motor Control Centers for additional requirements.

4. Refer to the contract documents for VFD application (enclosed or installed in VFD).

1.3 DEFINITIONS

A. BAS: Building automation system.

B. CPT: Control power transformer.

C. EMI: Electromagnetic interference.

D. IGBT: Insulated-gate bipolar transistor.

E. LAN: Local area network.

F. LED: Light-emitting diode.

G. MCP: Motor-circuit protector.

H. NC: Normally closed.

I. NO: Normally open.

J. OCPD: Overcurrent protective device.

K. PCC: Point of common coupling.

L. PID: Control action, proportional plus integral plus derivative.

M. PWM: Pulse-width modulated.

N. RFI: Radio-frequency interference.

O. TDD: Total demand (harmonic current) distortion.

P. THD(V): Total harmonic voltage demand.

Q. VFD: Variable-frequency motor controller.

R. VFD: Variable-frequency drive (synonymous with VFD).

1.4 SUBMITTALS

A. General Submittal Requirements: When issued, comply with the General Conditions.

B. In addition to the requirements identified in Architectural Specification Division 01 Section "Submittal Procedures" the supplier shall also comply with the following:

1. The Variable-Frequency Drive Submittal to be submitted electronically as one (1) complete submission as a "pdf" file for review. Partial Submittals will be rejected.

   a. The complete submittal must be reviewed and approved by the Engineer before installation can take place. The submission shall include the following:

      1) Product data
      2) Shop drawings (See Paragraph 'E' below for requirements)
      3) Harmonic Analysis Study required by specification section A26 24 19 Intelligent Motor Control Centers.

C. Product Data: For each type and rating of VFD indicated. Include features, performance, electrical ratings, operating characteristics, shipping and operating weights, and furnished specialties and accessories.

D. Shop Drawings: For each VFD indicated. Include dimensioned plans, elevations, and sections; and conduit entry locations and sizes, mounting arrangements, and details, including required clearances and service space around equipment.

   1. Show tabulations of installed devices, equipment features, and ratings. Include the following:

      a. Each installed unit's type and details.
      b. Factory-installed devices.
      c. Enclosure types and details.
      d. Nameplate legends.
      e. Short-circuit current (withstand) rating of enclosed unit.
      f. Features, characteristics, ratings, and factory settings of each VFD and installed devices.
      g. Specified modifications.

   2. Schematic and Connection Wiring Diagrams: For power, signal, and control wiring.

E. Coordination Drawings: Floor plans, drawn to scale, showing dimensioned layout, required working clearances, and required area above and around VFDs. Show VFD layout and relationships between electrical components and adjacent structural and mechanical elements including raceway stub-ups and field wiring. Show support locations, type of support, and weight on each support. Indicate field measurements.

F. Qualification Data: For qualified testing agency.

G. Product Certificates: For each VFD, from manufacturer.

H. Source quality-control reports.

I. Field quality-control reports.

J. Operation and Maintenance Data: For VFDs to include in emergency, operation, and maintenance manuals. In addition to items specified in Division 01 Section "Operation and Maintenance Data," include the following:

1. Manufacturer's written instructions for testing and adjusting thermal-magnetic circuit breaker and MCP trip settings.
2. Manufacturer's written instructions for setting field-adjustable overload relays.
3. Manufacturer's written instructions for testing, adjusting, and reprogramming microprocessor control modules.
4. Manufacturer's written instructions for setting field-adjustable timers, controls, and status and alarm points.

K. Load-Current and List of Settings of Adjustable Overload Relays: Compile after motors have been installed and arrange to demonstrate that switch settings for motor-running overload protection suit actual motors to be protected.

1.5 QUALITY ASSURANCE

A. Testing Agency Qualifications: Member company of NETA or an NRTL or qualified and trained manufacture's technician.

1. Testing Agency's Field Supervisor: Currently certified by NETA to supervise on-site testing.

B. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

C. Comply with NFPA 70.

D. VFD's shall be built to NEMA standards and be UL listed.

1.6 DELIVERY, STORAGE, AND HANDLING

A. If stored in space that is not permanently enclosed and air conditioned, remove loose packing and flammable materials from inside controllers and install temporary electric heating.

1.7 PROJECT CONDITIONS

A. Environmental and Operating Limitations: Rate equipment for continuous operation, capable of driving full load without derating, under the following conditions unless otherwise indicated:

1. Ambient Temperature: Not less than 0^0F and not exceeding 120^0F.
2. Ambient Storage Temperature: Not less than minus 4^0F and not exceeding 140^0F
3. Humidity: Less than 95% (noncondensing and non-corrosive).
4. Altitude: Not exceeding 3300 feet.
5. Incoming 3-phase 480 VAC power: +5% or -10% at 60 Hz.
6. A 50% voltage sags for thirty (30) cycles.

B. Interruption of Existing Electrical Systems (when applicable): Do not interrupt electrical systems in facilities occupied by Owner or others unless permitted under the following conditions and then only after arranging to provide temporary electrical service according to requirements indicated:

1. Notify Owner no fewer than ten (10) days in advance of proposed interruption of electrical systems.
2. Indicate method of providing temporary electrical service.
3. Do not proceed with interruption of electrical systems without Owner's written permission.
4. Comply with NFPA 70E.

C. Product Selection for Restricted Space: Drawings indicate maximum dimensions for VFDs, including clearances between VFDs, and adjacent surfaces and other items.

1.8 COORDINATION

A. Coordinate features of motors, load characteristics, installed units, and accessory devices to be compatible with the following:

1. Torque, speed, and horsepower requirements of the load.
2. Ratings and characteristics of supply circuit and required control sequence.
3. Ambient and environmental conditions of installation location.

B. Coordinate sizes and locations of concrete bases with actual equipment provided.

C. Coordinate sizes and locations of roof curbs, equipment supports, and roof penetrations with actual equipment provided.

D. Submit written, signed off coordination statement confirming input control signal is compatible with automatic controls and/or building automation control system.

1.9 EXTRA MATERIALS

A. Furnish extra materials described below that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Power Fuses: Equal to Five (5) percent of quantity installed for each size and type, but no less than three of each size and type.
2. Control Power Fuses: Equal to Five (5) percent of quantity installed for each size and type, but no less than three of each size and type.
3. Indicating Lights: One (1) of each type and color installed.
4. Auxiliary Contacts: Furnish One (1) spare(s) for each size and type of magnetic controller installed.
5. Power Contacts: Furnish One (1) spares for each size and type of magnetic contactor installed.

1.10 WARRANTY/GUARANTEE

A. See Division 26 Specification Section “Basic Electrical Requirements’ for warranty and guarantee requirements.

B. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace VFDs that fails in materials or workmanship within specified warranty period.

1. Warranty Period: Two (2) years from date of shipment or one (1) year from the date of project Substantial Completion, whichever is later.

PART 2 - PRODUCTS

2.1 MANUFACTURED UNITS

A. Acceptable Manufacturers: Subject to compliance with ANSI, IEEE and NEMA requirements, and unless otherwise indicated all VFD’s shall be products manufactured by one (1) of the following:

1. Allen Bradley by Rockwell Automation – PowerFlex 755TL AFE Series for 150 hp and larger not installed in an MCC (NONE REQUIRED ON THIS PROJECT), PowerFlex series for all other applications.
2. Eaton Electrical Inc.; Cutler-Hammer Business Unit – RGX AFE Series for 150 hp and larger not installed in an MCC (NONE REQUIRED ON THIS PROJECT), PowerXL series for all other applications.
3. Square D – Altivar ATV680 AFE Series for 150 hp and larger not installed in an MCC (NONE REQUIRED ON THIS PROJECT), Altivar ATV series for all other applications.

4. ABB – ACS880-37 ULH Series for 150 hp and larger not installed in an MCC (NONE REQUIRED ON THIS PROJECT), ACH580 series for all other applications.

B. General Requirements for VFDs: Comply with NEMA ICS 7, NEMA ICS 61800-2, and UL 508C.

C. Application: Constant torque and variable torque. Match drive characteristics and capabilities to the type of load supplied per the single line diagram.

D. VFD Description: Heavy Duty rated, variable-frequency power converter (PWM inverter) factory packaged in an enclosure, or factory installed in an MCC assembly as indicated on the contract drawings, with integral disconnecting means and overcurrent and overload protection; listed and labeled by an NRTL as a complete unit; arranged to provide self-protection, and protection and variable-speed control of one or more three-phase induction motors by adjusting output voltage and frequency. The VFD shall convert the fixed voltage and frequency available from the utility or facility wide system to a variable voltage and frequency output. Efficiency shall exceed 95% at 100% speed and load. Line side displacement power factor shall exceed 0.95 regardless of speed and load. The VFD shall be rated for 110% current for 1 minute for variable torque loads and 150% current for one (1) minute for constant torque loads.

1. Units suitable for operation of NEMA MG 1, Design A and Design B motors as defined by NEMA MG 1, Section IV, Part 30, "Application Considerations for Constant Speed Motors Used on a Sinusoidal Bus with Harmonic Content and General Purpose Motors Used with Adjustable-Voltage or Adjustable-Frequency Controls or Both."
2. Units suitable for operation of inverter-duty motors as defined by NEMA MG 1, Section IV, Part 31, "Definite-Purpose Inverter-Fed Polyphase Motors."
3. Listed and labeled for integrated short-circuit current (withstand) rating by an NRTL acceptable to authorities having jurisdiction.

E. Design and Rating: Match load type, such as fans, blowers, pumps, conveyors and agitators; and type of connection used between motor and load such as direct or through a power-transmission connection.

F. Output Rating: Three-phase; 10 to 60 Hz, with voltage proportional to frequency throughout voltage range and 66 Hz, with torque constant as speed changes; maximum voltage equals input voltage.

G. Unit Operating Requirements:

1. Input AC Voltage Tolerance: Plus 10% and minus 10% of VFD input voltage rating.
2. Input AC Voltage Unbalance: Not exceeding 5%.
3. Input Frequency Tolerance: Plus or minus 3% of VFD frequency rating.

4. Minimum Efficiency: 96% at 60 Hz, full load.
5. Minimum Displacement Primary-Side Power Factor: 96% or higher under any load or speed condition.
6. Minimum Short-Circuit Current (Withstand) Rating: 65kA.
7. Ambient Temperature Rating: Not less than 14^0F and not exceeding 120^0F.
8. Ambient Storage Temperature Rating: Not less than minus 4^0F and not exceeding 140^0F.
9. Humidity Rating: Less than 95% (noncondensing).
10. Altitude Rating: Not exceeding 3300 feet.
11. Vibration Withstand: Comply with IEC 60068-2-6.
12. Overload Capability: 1.5 times the base load current for sixty (60) seconds; minimum of 1.8 times the base load current for three seconds.
13. Starting Torque: Minimum 100% of rated torque from 3 to 60 Hz.
14. Speed Regulation: Plus or minus 5%.
15. Output Carrier Frequency: Selectable; 0.5 to 15 kHz.
16. Stop Modes: Programmable; includes fast, free-wheel, and dc injection braking.
17. VFD's above 125hp: The input current to the VFD shall have a total harmonic content less than 5% of full rated capability at the input terminals of the VFD on a power system sized according to IEEE 519-2014, at line voltage unbalance up to 3%, and under all motor load conditions.
18. VFD's 125hp and Less: The input current to the VFD shall have a total harmonic content less than 8% of full rated capability at the input terminals of the VFD on a power system sized according to IEEE 519-2014, at line voltage unbalance up to 3%, and under all motor load conditions.
19. On applicable projects, Motor Control Centers shall also be equipped with harmonic filtering and power factor correction equipment as required to meet the requirements of Specification Section A26 24 19 INTELLIGENT MOTOR CONTROL CENTERS. Drives shall be compatible with the supplied filtering equipment. Refer to specification section A26 24 19 INTELLIGENT MOTOR CONTROL CENTERS for additional information and requirements. HARMONIC FILTERING UNITS FOR THE MCC'S ARE NOT REQUIRED ON THIS PROJECT.

H. Inverter Logic: Microprocessor based, thirty two (32) bit, isolated from all power circuits.

I. Isolated Control Interface: Allows VFDs to follow remote-control signal over a minimum 40:1 speed range.

1. Signal: Electrical.

J. Internal Adjustability Capabilities:

1. Minimum Speed: 10% of maximum rpm.
2. Maximum Speed: 100% of maximum rpm.
3. Current Limit: 30% to minimum of 150% of maximum rating.
4. Minimum frequency (5 Hz)

5. Maximum frequency (80 Hz)
6. Two (2) preset speeds
7. Four (4) acceleration times (2-300 seconds)
8. Four (4) de-acceleration times (2-300 seconds)
9. Minimum speed dwell time (0-18 seconds)
10. Voltage boost (0-40V) for starting torque control
11. Adjustable Carrier Frequency 700-10,000 Hz for motor noise reduction or flexible switching technology. This adjustment shall be without derating the drive or motor.
12. Current Limit 70% - 120% Critical Frequency Avoidance (2 bands with 10 Hz adjustable widths

K. Self-Protection and Reliability Features:

1. Input transient protection by means of surge suppressors to provide three (3) phase protection against damage from supply voltage surges 10% or more above nominal line voltage.
2. Loss of Input Signal Protection: Selectable response strategy, including speed default to a percent of the most recent speed, a preset speed, or stop; with alarm.
3. Under- and overvoltage trips.
4. Inverter overcurrent trips.
5. VFD and Motor Overload/Over temperature Protection: Microprocessor-based thermal protection system for monitoring VFDs and motor thermal characteristics, and for providing VFD over temperature and motor overload alarm and trip; settings selectable via the keypad; NRTL approved.
6. Critical frequency rejection, with selectable, adjustable deadbands.
7. Instantaneous line-to-line and line-to-ground overcurrent trips.
8. Loss-of-phase protection.
9. Reverse-phase protection.
10. Short-circuit protection.
11. Motor overtemperature fault.

L. Automatic Reset/Restart: Attempt three restarts after drive fault or on return of power after an interruption and before shutting down for manual reset or fault correction; adjustable delay time between restart attempts.

M. Power-Interruption Protection: To prevent motor from re-energizing after a power interruption until motor has stopped, unless "Bidirectional Autospeed Search" feature is available and engaged.

N. Bidirectional Auto speed Search: Capable of starting VFD into rotating loads spinning in either direction and returning motor to set speed in proper direction, without causing damage to drive, motor, or load.

O. Torque Boost: Automatically varies starting and continuous torque to at least one and one half (1.5) times the minimum torque to ensure high-starting torque and increased torque at slow speeds.

P. Motor Temperature Compensation at Slow Speeds: Adjustable current fall-back based on output frequency for temperature protection of self-cooled, fan-ventilated motors at slow speeds.

Q. Integral Input Disconnecting Means and OCPD: NEMA AB 1, instantaneous-trip circuit breaker, thermal-magnetic circuit breaker with pad-lockable, door-mounted handle mechanism.

1. Disconnect Rating: Not less than 115% of VFD input current rating.
2. Disconnect Rating: Not less than 115% of NFPA 70 motor full-load current rating or VFD input current rating, whichever is larger.
3. Auxiliary Contacts: NO/NC, arranged to activate before switch blades open.
4. Auxiliary contacts "a" and "b" arranged to activate with circuit-breaker handle.
5. NC/NO alarm contact that operates only when circuit breaker has tripped.

2.2 CONTROLS AND INDICATION

A. Status Lights: Door-mounted LED indicators displaying the following conditions:

1. Power on.
2. Run.
3. Overvoltage.
4. Line fault.
5. Overcurrent.
6. External fault.
7. Run/Stop selection and LED indication (keypad or remote)
8. Speed control selection and LED indication
9. Forward/Reverse selection
10. Manual speed adjustment
11. Frequency meter
12. Motor RPM
13. Ammeter
14. Output Voltage
15. Elapse Time Meter

B. Panel-Mounted Operator Station: Manufacturer's standard front-accessible, sealed keypad and plain-English language digital display; allows complete programming, program copying, operating, monitoring, and diagnostic capability.

1. Keypad: In addition to required programming and control keys, include keys for HAND, OFF, and AUTO modes.

2. Security Access: Provide electronic security access to controls through identification and password with at least three levels of access: View only; view and operate; and view, operate, and service.

   a. Control Authority: Supports at least four conditions: Off, local manual control at VFD, local automatic control at VFD, and automatic control through a remote source.

C. Historical Logging Information and Displays:

1. Real-time clock with current time and date.
2. Running log of total power versus time.
3. Total run time.
4. Fault log, maintaining last four (4) faults with time and date stamp for each.

D. Indicating Devices: Digital display mounted flush in VFD door and connected to display VFD parameters including, but not limited to:

1. Output frequency (Hz).
2. Motor speed (rpm).
3. Motor status (running, stop, fault).
4. Motor current (amperes).
5. Motor torque (percent).
6. Fault or alarming status (code).
7. PID feedback signal (percent).
8. DC-link voltage (V dc).
9. Set point frequency (Hz).
10. Motor output voltage (V ac).

E. Control Signal Interfaces:

1. Electric Input Signal Interface:

   a. Manufacturers standard number programmable analog inputs for the model and series specified: 4- to 20-mA dc.
   b. Manufacturers standard number digital inputs for the model and series specified.
   c. Control input via Ethernet IP communications compatible with the Intelligent MCC software package (basis of design).

2. Remote Signal Inputs: Capability to accept any of the following speed-setting input signals from the BAS or other control systems:

   a. Ethernet IP (Basis of design, must be included, not just capable).

b. RS485
c. Keypad display for local hand operation.
d. 0- to 10-V dc.
e. 4- to 20-mA dc.
f. Potentiometer using up/down digital inputs.
g. Fixed frequencies using digital inputs.

3. Output Signal Interface: Manufacturers standard number of programmable analog output signal(s), 0- to 10-V dc and/or 4- to 20-mA dc, for the model and series specified, which can be configured for any of the following:

a. Output frequency (Hz).
b. Output current (load).
c. DC-link voltage (V dc).
d. Motor torque (percent).
e. Motor speed (rpm).
f. Set point frequency (Hz).

4. Remote Indication Interface: Manufacturers standard number of programmable dry-circuit relay outputs for the model and series specified (120-V ac, 1 A) for remote indication of the following:

a. Motor running.
b. Set point speed reached.
c. Fault and warning indication (overtemperature or overcurrent).
d. PID high- or low-speed limits reached.

5. Remote Metering: Provide “kW” and “kWh” values via Ethernet IP communications.

F. Control System Interface: Factory-installed hardware and software to enable the Control or Automation System to monitor, control, and display VFD status, alarm and energy usage. Allows VFD to be used with an external system within a multidrop LAN configuration; settings retained within VFD's nonvolatile memory. The VFD shall provide at a minimum one (1) Modbus and one (1) Ethernet communication port. VFD’s installed as well as not installed in MCC’s shall be fully compatible with the Ethernet network communications and software provided with the Intelligent MCC’s.

1. Embedded TCP/IP over Ethernet communication, IPv6 compliant.
2. Vendor shall provide add-on instructions or EDS files that can be utilized in a Rockwell/Allen-Bradley Process Automation Controller for all VFD’s.
3. Embedded BAS Protocols for Network Communications: Manufacturer shall provide the capability, either standard or with an additional module(s), each of the following protocols, in addition to the required Ethernet IP communications:

a. BACNET MS/TP

b. BACNET IP
c. MODBUS TCP

2.3 LINE CONDITIONING AND FILTERING

A. Input Line Conditioning: Based on the harmonic analysis study and report, provide input filtering, as required, to limit TDD and THD (V) at the each MCC per IEEE 519, latest revision and these specifications. Refer to specification section 262419 Intelligent Motor Control Centers for additional requirements.

B. EMI/RFI Filtering: CE marked; certify compliance with IEC 61800-3.

2.4 ENCLOSURES

A. VFD Enclosures (where not installed in an MCC): NEMA 250, to comply with environmental conditions at installed location.

1. All Dry Indoor Locations: NEMA Type 12.
2. Outdoor Locations: Type 4X non-metallic.
3. Other Wet or Damp Indoor Locations: Type 4X non-metallic.

B. Plenum Rating: UL 1995; NRTL certification label on enclosure, clearly identifying VFD as "Plenum Rated."

2.5 ACCESSORIES Match to Enclosed Controllers

A. General Requirements for Control-Circuit and Pilot Devices: NEMA ICS 5; factory installed in VFD enclosure cover unless otherwise indicated.

1. Push Buttons, Pilot Lights, and Selector Switches: Heavy-duty type.

a. Push Buttons: momentary.
b. Pilot Lights LED types (for VFD's installed in MCC's); Green for de-energized state, red for energized state, amber for warning and white for alarm; push to test. Not required for loose, enclosed VFD's.

B. Control Relays: Auxiliary and adjustable solid-state time-delay relays.

C. Phase-Failure, Phase-Reversal, and Undervoltage and Overvoltage Relays: Solid-state sensing circuit with isolated output contacts for hard-wired connections. Provide adjustable undervoltage, overvoltage, and time-delay settings.

1. Current Transformers: Continuous current rating, basic impulse insulating level (BIL) rating, burden, and accuracy class suitable for connected circuitry. Comply with IEEE C57.13.

D. Cooling Fan and Exhaust System: For NEMA 250, Type 1; UL 508 component recognized: Supply fan, with composite intake and exhaust grills and filters; 120 -V ac; obtained from integral CPT.

E. Door Interlock Circuit Breaker rated at 65,000 AIC with 5% total line impedance.

F. Thermal motor overcurrent relay.

2.6 SOURCE QUALITY CONTROL

A. Testing: Factor test and inspect VFDs according to requirements in NEMA ICS 61800-2.

B. VFDs will be considered defective if they do not pass tests and inspections.

C. Prepare test and inspection reports.

PART 3 - EXECUTION

3.1 EXAMINATION (APPLIES TO INSTALLATION CONTRACTOR)

A. Examine areas, surfaces, and substrates to receive VFDs, with Installer present, for compliance with requirements for installation tolerances, and other conditions affecting performance.

B. Examine VFD before installation. Reject VFDs that are wet, moisture damaged, or mold damaged.

C. Examine roughing-in for conduit systems to verify actual locations of conduit connections before VFD installation.

D. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 HARMONIC ANALYSIS STUDY – NOT REQUIRED ON THIS PROJECT

A. Within 30 Days of Substantial Completion, factory qualified and trained technician shall perform a harmonic analysis study to identify the effects of nonlinear loads and their associated harmonic contributions on the voltages and currents at each MCC. Analyze possible operating scenarios, including recommendations for VFD input filtering to limit

TDD and THD (V) at each VFD at each MCC to specified levels. Testing shall comply with IEEE519.

B. Prepare a harmonic analysis study and report complying with IEEE 399 and NETA Acceptance Testing Specification.

C. Manufacturer shall provide additional harmonic filtering and power factor correction equipment or measures at any MCC that does not meet the minimum requirements of IEEE 519 and these specifications.

3.3 INSTALLATION (APPLIES TO ENCLOSED PREASSEMBLED DRIVES)

A. Coordinate layout and installation of VFDs with other construction including conduit, piping, equipment, and adjacent surfaces. Maintain required workspace clearances and required clearances for equipment access doors and panels.

B. Wall-Mounting Controllers: Install VFDs on walls with tops at uniform height and with disconnect operating handles not higher than seventy eight (78) inches above finished floor unless otherwise indicated, and by bolting units to wall or mounting on lightweight structural-steel channels bolted to wall. For controllers not on walls, provide freestanding racks complying with Division 26 Section "Hangers and Supports for Electrical Systems."

C. Floor-Mounting Controllers: Install VFDs on four (4) inch nominal thickness, 3500 psi minimum concrete housekeeping pad. Comply with requirements for concrete base specified in Division 03 Section Cast-in-Place Concrete.

1. Install dowel rods to connect concrete housekeeping pad to concrete floor. Unless otherwise indicated, install dowel rods on eighteen (18) inch centers around the full perimeter of concrete base.
2. For supported equipment, install epoxy-coated anchor bolts that extend through concrete base and anchor into structural concrete floor.
3. Place and secure anchorage devices. Use setting drawings, templates, diagrams, instructions, and directions furnished with items to be embedded.
4. Install anchor bolts to elevations required for proper attachment to supported equipment.

D. Roof-Mounting Controllers: Install VFD on roofs with tops at uniform height and with disconnect operating handles not higher than seventy eight (78) inches above finished roof surface unless otherwise indicated, and by bolting units to curbs or mounting on freestanding, lightweight, structural-steel channels bolted to curbs. Seal roof penetrations after raceways are installed.

1. Curbs and roof penetrations are specified in Division 07 Section "Roof Accessories."

2. Structural-steel channels are specified in Division 26 Section "Hangers and Supports for Electrical Systems."

E. Temporary Lifting Provisions: Remove temporary lifting eyes, channels, and brackets and temporary blocking of moving parts from enclosures and components.

F. Install fuses in each fusible-switch VFD.

G. Install fuses in control circuits if not factory installed.

H. Install heaters in thermal-overload relays. Select heaters based on actual nameplate full-load amperes after motors have been installed.

I. Install, connect, and fuse thermal-protector monitoring relays furnished with motor-driven equipment.

J. Comply with NECA 1.

3.4 IDENTIFICATION

A. Identify VFDs, components, and control wiring. Comply with requirements for identification specified in Division 26 Section "Identification for Electrical Systems."

1. Identify field-installed conductors, interconnecting wiring, and components; provide warning signs.
2. Label each VFD with engraved nameplate.
3. Label each enclosure-mounted control and pilot device.

B. Operating Instructions: Frame printed operating instructions for VFDs, including control sequences and emergency procedures. Fabricate frame of finished metal, and cover instructions with clear acrylic plastic. Mount on front of VFD units.

3.5 CONTROL WIRING INSTALLATION

A. Install wiring between VFDs and remote devices and the building automation system. Comply with requirements in Division 26 Section "Control-Voltage Electrical Power Cables."

B. Bundle, train, and support wiring in enclosures.

C. Connect selector switches and other automatic control devices where applicable.

1. Connect selector switches to bypass only those manual- and automatic control devices that have no safety functions when switches are in manual-control position.
2. Connect selector switches with control circuit in both manual and automatic positions for safety-type control devices such as low- and high-pressure cutouts, high-temperature cutouts, and motor overload protectors.

3.6 FACTORY TESTS AND CHECKS

A. VFD power semiconductors and diodes shall be 100% inspected and tested, including load testing.

B. Small signal semiconductors, resistors, capacitors and diodes shall be lot sampled. Testing shall include parameter, as well as, functional characteristics.

C. All printed circuit-boards shall be tested under a temperature cycling (0^0C to $+65^0$C) 24-hour load test and then functionally tested via fault finder bench equipment prior to unit installation.

D. All final assemblies shall be tested at full load with application of line-to-line and line-to-ground bolted faults. The VFD shall trip electronically without device failure.

E. After all tests have been performed, each VFD shall undergo a twenty four (24) hour burn-in test. The drive shall be burned-in at 100% inductive or motor load for twenty four (24) hours without an unscheduled shutdown.

F. After the burn-in cycle is complete, each VFD shall be provided by the manufacturer upon request prior to shipment.

3.7 FIELD QUALITY CONTROL – <u>VFD SUPPLIER SHALL INCLUDE THESE SERVICES WITH THE EQUIPMENT BID PACKAGE EL-302.</u>

A. Testing Agency: Engage a qualified testing agency or qualified factory trained technician to perform tests and inspections.

B. Perform tests and inspections.

1. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect components, assemblies, and equipment installations, including connections, and to direct testing.

2. Testing, checkout and startup of the VFD equipment shall be performed under the technical direction of the manufacturer’s service engineer. Under no

circumstances are any portions of the drive system to be energized without authorization from the manufacturer's representative.

3. On applicable projects, the VFD supplier as part of the equipment bid package shall provide independent harmonic testing by an independent testing company or a qualified and trained factory technician. Readings with printouts of the harmonic current at each harmonic, as well as, the total voltage distortion. The following readings shall be provided: HARMONIC TESTING IS NOT REQUIRED ON THIS PROJECT.

   a. At the line side of each MCC:

      1) With all drives running with load
      2) With all drives off

   b. At the power connection to each drive:

      1) With the drive running loaded
      2) With drive off

   c. All the above data shall be submitted to the Owner for review. If these tests show the drives are not in compliance with the specifications, the drive manufacturer shall make all changes required to comply with the specifications at no cost to the Owner. If required, this could mean replacing the drives that are not in compliance.

   d. A copy of all the tests and checks performed in the field, complete with meter readings and recordings, where applicable, shall be submitted to the Owner.

C. Acceptance Testing Preparation:

1. Test insulation resistance for each VFD element, bus, component, connecting supply, feeder, and control circuit.
2. Test continuity of each circuit.

D. Tests and Inspections:

1. Inspect VFD, wiring, components, connections, and equipment installation. Test and adjust controllers, components, and equipment.
2. Test insulation resistance for each VFD element, component, connecting motor supply, feeder, and control circuits.
3. Test continuity of each circuit.
4. Verify that voltages at VFD locations are within 10% of motor nameplate rated voltages. If outside this range for any motor, notify Owner before starting the motor(s).

5. Test each motor for proper phase rotation.
6. Perform each electrical test and visual and mechanical inspection stated in NETA Acceptance Testing Specification. Certify compliance with test parameters.
7. Correct malfunctioning units on-site, where possible, and retest to demonstrate compliance; otherwise, replace with new units and retest.

8. Test and adjust controls, remote monitoring, and safeties. Replace damaged and malfunctioning controls and equipment.

E. VFD's will be considered defective if they do not pass tests and inspections.

F. Prepare test and inspection reports, including a certified report that identifies the VFD and describes scanning results. Include notation of deficiencies detected, remedial action taken, and observations made after remedial action.

3.8 STARTUP SERVICE – VFD SUPPLIER SHALL INCLUDE THESE SERVICES WITH THE EQUIPMENT BID PACKAGE.

A. Engage a factory-authorized service representative to perform startup service.

1. Complete installation and startup checks according to manufacturer's written instructions.

3.9 ADJUSTING – VFD SUPPLIER SHALL INCLUDE THESE SERVICES WITH THE EQUIPMENT BID PACKAGE.

A. Program microprocessors for required operational sequences, status indications, alarms, event recording, and display features. Clear events memory after final acceptance testing and prior to Substantial Completion.

B. Set field-adjustable switches, auxiliary relays, time-delay relays, timers, and overload-relay pickup and trip ranges.

C. Adjust the trip settings of MCPs and thermal-magnetic circuit breakers with adjustable, instantaneous trip elements. Initially adjust to six times the motor nameplate full-load amperes and attempt to start motors several times, allowing for motor cool-down between starts. If tripping occurs on motor inrush, adjust settings in increments until motors start without tripping. Do not exceed eight times the motor full-load amperes or eleven (11) times for NEMA Premium Efficient motors if required. Where these maximum settings do not allow starting of a motor, notify Owner before increasing settings.

D. Set the taps on reduced-voltage autotransformer controllers.

E. Set field-adjustable circuit-breaker trip ranges as specified in Division 26 Section "Overcurrent Protective Device Coordination Study."

F. Set field-adjustable pressure switches.

3.10 PROTECTION (INSTALLING CONTRACTOR)

A. Temporary Heating: Apply temporary heat to maintain temperature according to manufacturer's written instructions until controllers are ready to be energized and placed into service.

B. Replace VFDs whose interiors have been exposed to water or other liquids prior to Substantial Completion.

C. The VFD shall be protected against damage at all times. The drive shall be stored in a clean, dry environment with temperatures and humidity within the range specified by the drive manufacturer. Space heaters shall be energized controlled storage as recommended by the manufacturer. Storage space shall be environmentally controlled.

3.11 DEMONSTRATION – VFD SUPPLIER SHALL INCLUDE THESE SERVICES WITH THE EQUIPMENT BID PACKAGE.

A. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, reprogram, and maintain VFDs.

END OF SECTION 262923

SECTION 265119 – LED LIGHTING

1. **PART 1 - GENERAL**

1.1. DESCRIPTION OF WORK

1.1.1. The extent of interior and exterior lighting fixture work is indicated on the drawings and schedules and by the requirements of this section.

1.1.2. The types of lighting fixtures required for the project include the following:

1.1.2.1. LED type fixtures.

1.1.3. Light Fixtures shall be supplied by the Contractor.

1.2. The application of lighting fixtures required for the project include the following:

1.2.1.1. General lighting.

1.2.1.2. Supplementary lighting.

1.2.1.3. Local lighting.

1.2.1.4. Emergency lighting.

1.2.1.5. Exit lighting.

1.3. REFERENCES

1.3.1. ANSI/NFPA 70 - National Electrical Code.

1.3.2. ANSI/NFPA 101 - Life Safety Code.

1.3.3. NEMA WD 6 - Wiring Devices-Dimensional Requirements

1.4. PRODUCT DELIVERY, STORAGE AND HANDLING

1.4.1. Handle lighting fixtures carefully to prevent breakage, denting and scoring the fixture finish. Do not install lighting fixtures, replace and return damaged units to equipment manufacturer.

1.4.2. Store lighting fixtures in a clean, dry space. Store in original cartons and protect from dirt, physical damage, weather, and construction traffic.

2. **PART 2 - PRODUCTS**

2.1. INTERIOR LIGHTING FIXTURES

2.1.1. General: Install lighting fixtures, of the size, type and rating indicated; complete with, but not necessarily limited to, LED boards, LED drivers, and emergency battery packs. All lighting fixture types shall bear UL label.

2.1.2. All light fixtures shall be factory equipped with an input power quick disconnect internal to the fixture to facilitate changing or replacing of the LED driver or LED board without having to disconnect all the fixtures in the room / on the circuit.

2.2. FIXTURES TYPES

2.2.1. For fixture types see Drawings for Lighting Fixture Schedule.

2.3. EXTERIOR LIGHTING FIXTURES

2.3.1. General: Install lighting fixtures of the size, type and rating indicated U.L. listed for wet locations. Complete with, but not necessarily limited to, housings, LED boards, LED drivers, emergency battery packs and wiring.

2.3.2. Supply fixtures with integral photocells as shown on the fixture schedule.

2.3.3. All light fixtures shall be factory equipped with an input power quick disconnect internal to the fixture to facilitate changing or replacing of the LED driver or LED board without having to disconnect all the fixtures on the circuit.

2.4. EMERGENCY LIGHTING FIXTURES AND BATTERY PACKS

2.4.1. General: Where designated as an emergency fixture on the drawings, furnish emergency fixtures with emergency battery packs. Fixtures on the plans marked with an N/L shall operate as an “always on” night light. Emergency fixtures not marked with an N/L shall also be furnished with an emergency battery pack but shall be controlled with the room lighting controls. These fixtures require a constant voltage supply as well as a switched supply. Upon loss of normal power, whether a night light or a controlled light, the battery pack shall provide power to the fixture at a minimum 50% of full lumen output for a minimum of 90 minutes regardless of whether the room lighting controls are “on” or “off” or “dimmed”.

2.4.2. Exit signs shall be equipped with an emergency battery pack as indicated on the fixtures schedule and shall supply power to the exit light for a minimum of 90 minutes after loss of normal power.

2.5. AUTOMATIC WALL BOX LIGHTING CONTROLS AND DIMMER SWITCHES

2.5.1. General: Furnish and install automatic and/or dimming controls as indicated on the drawings including all sensors, dimming switches and associated wiring as required for a complete system.

2.5.2. Wall mounted vacancy sensor switches (no dimming required) shall be PIR, 120-277V, 6A, Lutron MS-VPS6M2-DV or approved equal. Color shall be white.

2.5.3. Wall mounted combination vacancy/dimmer switches shall be Lutron MS-Z101V or Hubbell LHD-IRS3 or approved equal. Color shall be white. Dimming shall be 0-10V suitable for dimming LED drivers. Sensor shall be PIR. Operation shall be manual on/auto or manual off. Auto-off time delay shall be adjustable for up to 30 minutes.

2.6. AUTOMATIC OCCUPANCY SENSOR LIGHTING CONTROLS AND DIMMER SWITCHES

2.6.1. Manufacturers - Basis of Design – Cooper Controls OAC-P-1500. Hubbell, Leviton and Acuity Brands approved equal.

2.6.2. Ceiling mounted sensors shall be PIR, powered from the power pack and have an adjustable time delay of 1-30 minutes.

2.6.3. Power packs shall be rated 20A, 12-277V rated for LED. Provide an additional relay for HVAC controls. Install power packs above ceiling near a sensor.

2.6.4. Training room lighting controls shall be as indicated on the drawings.

3. PART 3 - EXECUTION

3.1. INSPECTION

3.1.1. Installer must examine the areas and conditions under which lighting fixtures are to be installed and notify the Owner in writing of conditions detrimental to the proper and timely completion of the work. Do not proceed with the work until unsatisfactory conditions have been corrected in a manner acceptable in the Installer. Coordinate installation requirements with the mechanical contractor to avoid conflicts between luminaires, conduit, supports, fittings, and mechanical equipment.

3.2. INSTALLATION

3.2.1. Install lighting fixtures of the types indicated where shown and at the indicated heights; in accordance with the fixture manufacturer's written instructions and with recognized industry practices; to ensure that the fixtures comply with the requirements and serve the intended purposes. Comply with NEMA standards, and requirements of the National Electrical Code pertaining to installation of interior lighting fixtures.

3.2.2. Properly support and align fixture and provide all necessary steel shapes for support of the fixtures. Coordinate complete fixture installation with the building construction. Clean all lighting fixtures immediately prior to final inspection.

3.2.3. Square and rectangular fixtures shall be mounted with sides parallel to building lines, and parallel with ceiling lines or as specifically required by drawings.

3.2.4. Grounding shall be done by connecting the green ground wire to the grounding system or to the pole ground or luminaire grounding on outside fixtures.

3.2.5. Fasten fixtures securely to the structural support members of the building; and check to ensure that solid pendant fixtures are plumb and level all fixtures shall be supported in accordance with National Electric Code (NEC).

3.2.6. Support grid troffers and fixtures recessed in drywall or plaster ceilings independently from the ceiling grid or plaster ring in addition to securing fixtures to the grid or frame. Provide support wires or safety chains at all four corners of the fixture to the structure above. Provide safety chains or wires from all pendant mounted fixtures to structure above. Comply with the seismic requirements of the IBC and any other applicable codes and standards.

3.2.7. For surface mounted fixtures, utilize anchors and support methods that comply with the seismic requirements of the IBC and any other applicable codes and standards.

3.2.8. Provide wire guards, vandal resistant shields for all luminaires as indicated in the Luminaire Schedule shown on the drawings.

3.2.9. Install surface mounted luminaires and exit signs plumb and adjust to align with building lines and with each other. Secure to prohibit movement. Install wall mounted luminaires, emergency lighting units and exit signs at height as indicated on Drawings.

3.2.10. Fixtures should be located with reasonable maintenance criteria in mind. Avoid positioning fixtures where special lifts and scaffolding would be required to service the fixture.

3.2.11. Securely support and anchor all outlet boxes from structure.

3.3. ADJUST AND CLEAN

3.3.1. Clean lighting fixtures of dirt and debris upon completion of installation.

3.3.2. Protect installed fixtures from damage during the remainder of the construction period.

3.4. FIELD QUALITY CONTROL

3.4.1. Upon completion of installation of lighting fixtures, and after building circuitry has been energized, apply electrical energy to demonstrate capacity and compliance with requirements. Correct any malfunctioning units at the site, then retest to demonstrate compliance; remove any fixtures which do not operate properly and replace with new units and proceed with retesting.

3.4.2. Aim and adjust luminaires and clean lenses, diffusers and reflectors at completion of work.

3.4.3. Adjust exit sign directional arrows and aim emergency lighting unit directional lampheads as directed by Owner's Representative.

3.4.4. Clean paint spatters, dirt and debris from installed luminaires at completion of work. Operate each luminaire after installation and connection. Inspect for proper connection and operation.

3.5. GROUNDING

3.5.1. Provide tight equipment grounding connections for each interior lighting fixture installation to the building grounding system.

END OF SECTION